MATH INQUIRERS HOLIDAY PROGRAM

Math Inquirers P2 Tutor Guide

Holiday program for the Primary 2 level

Notice it. Test it. Explain why.

PRIMARY 2 HOLIDAY PROGRAM

Math Inquirers Holiday Program

Notice it. Test it. Explain why.

Program guide for tutors: schedule, lesson routine, ten minute-by-minute lesson scripts with answer keys, a progress tracker and a parent summary. Math Inquirers is online first: every lesson runs live on video; the face-to-face notes are kept for partner centres.

The program at a glance

Level	Children finishing Primary 2 (ages 8 to 9), going into P3, who are ready for more thinking than standard practice
Length	10 lessons of 90 minutes, two a week over the 5 weeks of the holidays
Group size	Up to 8 children per tutor, live online. Pilot with 6. Add a second tutor or helper above 8.
Focus	Thinking skills: making a list in order, guess and check, drawing a model, working backwards, finding patterns, reasoning and explaining. Not speed drills.
The arc of every lesson	Start concrete with objects, record in a table, spot the pattern, state the rule, use it (often backwards), then explain why.
What children take home	A student workbook with an inquiry journal, 10 worksheets and a certificate
Before tutors start	Every tutor signs the tutor confidentiality agreement before receiving any materials.

Schedule

Week	Lesson	Title	Topic	Big idea
Week 1a	1	**Digit Detectives**	Numbers to 1000	Place value
Week 1b	2	**Missing Digit Mysteries**	Addition and Subtraction within 1000	Working backwards

Week 2a	3	**Legs, Wheels and Arrays**	Multiplication and Division	Draw it (arrays)
Week 2b	4	**Pattern Machines**	Patterns with Multiplication	Look for a pattern
Week 3a	5	**Mystery Number Machines**	Working Backwards	Inverse operations
Week 3b	6	**Market Day**	Money	Systematic listing (make a list in order)
Week 4a	7	**Fair Shares**	Fractions	Checking for equal parts
Week 4b	8	**Shape Builders**	2D and 3D Shapes	Spatial reasoning
Week 5a	9	**Measure Masters**	Length, Mass, Volume and Time	Reasoning with equal amounts (swapping)
Week 5b	10	**Challenge Day**	Mixed Puzzles	Choosing a strategy

Suggested pairing: two lessons a week, for example Tuesday and Thursday, or Saturday and Sunday. Keep the same time slot all five weeks.

The 90-minute lesson routine

Every lesson uses the same routine so children know what to expect and tutors can prepare quickly. The times below are the standard timeline; the lesson scripts give the exact minutes for each lesson. Every lesson adds up to exactly 90 minutes, including the named Flex block.

Minutes	Part	What happens	Face to face	Online
0–10.5	**Opening, sneak-peek and warm-up**	Welcome (0–1.5): kit is on the desk from minute 0. Sneak-peek puzzles (1.5–6): two puzzles on ideas not learnt yet, answers shown only to the tutor, nobody marked. A short game (6–8.5). Two quick questions (8.5–10.5). Everyone joins in.	Sit in a circle or horseshoe.	Cameras on. Answers on own paper held to the camera at the same moment, or shown to the tutor only; or thumbs. Never by private chat or message. No whiteboards raised together.
10.5–23.5	**Explore**	Hands-on in pairs. Objects first, no writing yet. One round of breakout rooms (60 seconds each way, about 5.5 minutes of pair work), then the group talks it through (about 4 minutes).	Materials box on each table.	Each child has their own materials at home, out on the desk from minute 0. Pairs go to breakout rooms once, with an auto-timer.
23.5–35.5	**Mini-lesson**	Tutor builds the table on the board, names the rule and asks what stays the same and what changes. Lessons 9 and 10 are shorter (10.5 and 10 minutes).	Whiteboard.	Shared slide and annotation tool.
35.5–38.5	**Brain break**	Stand, stretch, drink water: two minutes of movement, one for water.	A short movement game.	Stand up on camera. Count the stretches together.

38.5–73.5	**Worksheet**	Page 1 in pairs (38.5–52.5), check together (52.5–55), short teaching (55–59.5), page 2 alone (59.5–71), check together (71–73.5). Tutor asks “how do you know?” rather than giving answers. Questions marked ★ take-home are done at home. Lessons 5, 6 and 10 open with a short teach slide, and Lessons 4, 9 and 10 have slightly different times: each lesson script gives its own.	Printed workbook, tutor circulates.	Printed workbook. Child holds the page up to the camera. Tutor checks one or two answers each. Pairs stay in the main room.
73.5–76.5	**Flex (spare time)**	Three spare minutes kept on purpose (4.5 in Lesson 9, 4 in Lesson 10 and 5 in Lesson 4). Absorb a late start or a slow page 2 here. If you are on time, use it for the Investigation share or a second volunteer. Never add new content.	Same room.	Stay on camera. Keep the break short.
76.5–81.5	**Talk time**	Pairs talk for 90 seconds, then two volunteers share one answer and explain it. One question. Any child may pass. No cold-calling by name, no public tallies or rankings.	The tutor invites volunteers first.	The tutor invites volunteers; a child unmutes when invited. Quiet children can be invited, with a parent nearby, by a quick word in the open session; never by private chat or message.
81.5–90	**Exit ticket, Looking back and stamp**	Exit ticket (81.5–86.5, one question on a slip of paper, held up one child at a time). Looking back (86.5–88.5, in the box on the “Before you go” page after page 2, a pass is always allowed). Closing and stamps (88.5–90; Lesson 10: certificates and closing 87–90).	Paper slip, tutor stamps.	Child writes the answer and shows the tutor only. Tutor stamps the next time or posts it.

Teaching habits that matter

- Ask “How do you know?” after answers, right or wrong, but invite rather than single out: “Would anyone like to say how they started?” “I am not sure yet” is always a fine answer.
- Wait 10 seconds before helping. Let children struggle a little.
- Praise the thinking (“You checked it two ways”), not just the answer.
- Let children disagree with each other and settle it with objects or a drawing.
- If a child would like more, offer Think harder, Go further or the Investigation as a choice. Never name who is on which page, and never use extra drill.

Preparing the materials

- Send the workbook PDF to parents one week before. Ask them to print it single-sided and the cut-out pages on card if they can.
- Cut-outs used: place-value cards (Lessons 1, 2, 10), digit cards (Lessons 1 and 10), play coins and notes (Lesson 6), fraction strips (Lesson 7), cube nets and paper cube strips (Lesson 8), paper clock (Lesson 9).
- Keep the answer key with the tutor only. Check answers together on screen.
- Use the shared screen or a digital whiteboard for the mini-lesson table.

Setting up online

- Send parents the workbook and materials list one week before. Needed: the printed workbook and cut-outs, 20 counters or small objects, scissors, a split pin or short pencil for the clock, paper and a pencil.
- Use one approved platform with breakout rooms and screen share. Admit only registered children and never share the link publicly.
- Ask the parent to sit nearby but not give answers. The camera should show the child’s page and hands.
- Hold up the answer page to the camera at the end of each lesson, or take a photo and send it through the approved channel.
- Lessons are recorded on the approved platform so that children who miss one can catch up; follow “Recording the lesson” in this guide. Do not make your own recordings or photos. Add a second tutor or helper above 8 children.

Materials master list

Lesson	Title	Face to face	Online set-up
1	Digit Detectives	Digit cards 0 to 9 (one set per child), place-value mat (hundreds, tens, ones), base-ten blocks or paper strips and squares, mini whiteboards, the 1000-chart pieces on the workbook page (rows of 10).	Children cut out paper digit cards before the lesson and move them on camera; the tutor writes on the slide 1000-chart pieces.
2	Missing Digit Mysteries	Digit cards 0 to 9, base-ten blocks or place-value counters, mini whiteboards, a paper magic triangle with 6 circles and number tiles 110 to 160.	Children write column sums on whiteboards and hold them up; number tiles are cut from paper before the lesson. The tutor writes on the slide sums.
3	Legs, Wheels and Arrays	About 30 counters per pair, mini whiteboards, squared paper. Optional: picture cards of hens and goats, bicycles and tricycles.	Each child has 30 small objects (buttons, dried beans or Lego bricks) and a whiteboard or paper. The tutor shares the slides; children hold their arrays and tables up to the camera.
4	Pattern Machines	Linking cubes in 3 colours (about 20 per pair), mini whiteboards, matchsticks or lolly sticks (about 25 per pair), counters.	Children use coloured pencils and squared paper (or Lego bricks) for the cube train, and draw sticks for the growing pattern. The tutor shares the slides and the children hold up their tables.
5	Mystery Number Machines	Mini whiteboards, counters, paper cups, number machine cards (× 2, × 3, × 4, × 5, × 10 and + or − cards from 1 to 20), paper strips for bars.	Machine cards shown on the slide; children write on whiteboards and hold them up. For bars, children draw on paper or fold paper strips into equal parts.
6	Market Day	Play money: coins 5¢, 10¢, 20¢, 50¢, $1 and notes $2, $5, $10, $50 (about 6 of each per pair), price tags, mini whiteboards.	Cut out the play coins and notes from the workbook before the lesson (kit in hand at minute 0), or use real money at home. No play money? Draw each coin and note on paper. The shop and the toy stall are on the slides; children show their money to the camera.

7	Fair Shares	Paper strips of the same length (4 per child, cut out before the lesson), paper squares, scissors, crayons, mini whiteboards.	Parents cut the 4 strips for each child before the lesson (kit in hand at minute 0). Each child folds paper strips on camera; the tutor shows fraction bars on the slides and children colour their own bars.
8	Shape Builders	Paper circles (2 per child, cut out before the lesson), scissors, about 12 linking cubes per child (or sugar cubes), the cube-net cut-out page (nets A to F, near the back of the workbook), mini whiteboards.	Parents cut out the 2 paper circles and the nets A to D before the lesson (kit in hand at minute 0). Children fold and cut the circles and build stacks on camera; the six nets A to F are on a cut-out page near the back of the workbook (print and cut them before the lesson), or cut a net from an empty box. No linking cubes at home? Use sugar cubes, dice or small boxes, or draw each stack layer by layer.
9	Measure Masters	A coat hanger, two paper cups, string, a glue stick, a spoon and about 20 20¢ coins (a home-made balance; backup: a ruler balanced on a pencil); for the jug puzzles two cups or bottles of different size (about 3 and 5 units), a jug of dry rice or sand (or water), and tape marking 3 and 5 units; a cardboard clock with hands; mini whiteboards.	Children build the coat-hanger balance, hang it and test that it is level BEFORE the lesson (kit message, then a 30-second check at minute 0) and hold it up to the camera. No coat-hanger balance at home? The tutor balances a ruler on a pencil on camera and pairs share one view. The tutor shares the slide balances, jugs and clocks and annotates on screen. Each child needs a cardboard or paper-plate clock. For the pouring puzzles use dry rice or sand rather than water when online, and let children act out each pour with the two containers.
10	Challenge Day	Four station cards (all on one slide in the deck, and printed in the tutor guide), play coins, digit cards 0 to 9, paper strips for fractions, mini whiteboards, certificates, stamps.	Each breakout pair takes two station cards (the tutor pastes their text into the breakout room chat), about 2 minutes a card, and everyone reports back to the group. Each child needs digit cards (cut from paper), a pencil and a mini whiteboard.

How to use the scripts

Every lesson below is a minute-by-minute script. The slides match it: each part has its own few slides, and each slide shows the same minutes as the script. You do not have to say the words exactly, but keep the questions, because the questions are what makes children think.

SAY	Words to say aloud, in plain language for 8- and 9-year-olds. Said in your own voice, not read from the page.
DO	What you do: set up, hand out, show a slide, or move around the room.
ASK / EXPECT	Questions to put to the group, with the answer you are listening for. Let children answer before you explain.
IF STUCK	A ladder of help. Give hint 1, wait ten seconds, then give hint 2. Do not give the answer.

Five habits that run through every script

- Predict first. Before children test anything, ask for a guess and write it down. A wrong guess is useful because we can test it.
- Wait ten seconds after every question. Count silently. Most children answer in the last four.
- Ask “How do you know?” after right answers too (to volunteers), so that children do not read your face for the answer.
- Praise the thinking, not the answer: “You checked it two ways” beats “Well done”.
- When an answer is wrong, say “Let us test that,” and use objects, not “No”.

Answering a wrong answer

If...	Then...
A child gives a wrong answer	“Thank you. Let us test it with the counters.” Then build it together.
A child gives a right answer but cannot explain	“Show me with your counters.” Then: “Tell your partner how you know.”
A child says “I do not know”	“What do you notice? What could you try first?” Offer hint 1.
Two children disagree	“Both of you, convince the group using objects.” Let the objects decide.

Recording the lesson

Every live lesson is recorded so that a child who misses it can still catch up. A child watching alone cannot ask you a question, so the recording needs spoken cues, visible timers and quiet working time.

How to teach so the recording works

- Say the short REC line at the start of each part, once, to the camera, in a friendly “recording voice”. Keep it under ten seconds. The children in the live room will simply hear it as a routine.
- Say the workbook page and question number aloud every time, and read each question out loud as well as showing it.
- Put a countdown timer or a clock slide on screen while children work, and say “the timer is running” so the recording shows how long to pause.
- Repeat a child’s answer in your own words before you respond, so it is clear on the recording without relying on their voice.
- Do not talk through the working time. Silence is fine; the child watching later pauses the video anyway.
- Show the answer slides clearly and read each answer aloud, so a child can mark their own work.

Privacy and consent

- Only record on the approved platform, and only with parent consent given at registration. Do not make your own recordings or photos.
- Record the tutor and the shared screen only. Turn off or exclude the other children’s video in the recording (use active-speaker or screen-only recording), and keep children’s names off the screen.
- Do not read out a child’s name together with a right or wrong answer. Do not show a child’s page to the camera in the recording.
- Share the recording only through a private link to enrolled families, and delete it when the program ends, or after the period the parents agreed to.
- Tell the group at the start of each lesson that it is being recorded and what is recorded. A parent or child who does not wish to be in it can keep the camera off.

These are suggested working rules. Make sure the parent consent wording, the tutor agreement and the platform settings match them before the first recorded lesson.

After the lesson

- Add a chapter marker at each time in the chapter list at the end of the lesson, so a child can jump to the part they need.
- Turn on captions if the platform offers them.
- Put the workbook page numbers and the materials list in the description of the recording.

Lesson 1: Digit Detectives

Week 1 (first lesson) • Numbers to 1000 • Workbook: Lesson 1

Objective	Read and build 3-digit numbers by place value (also with more than 9 tens), make every number from a set of digit cards by listing in order, find the biggest and smallest, and use patterns on a 1000 chart to count on and back in 1s, 10s and 100s.
Thinking skills	Place value, making a list in order, looking for a pattern, reasoning about zero
Materials	Digit cards 0 to 9 (one set per child), place-value mat (hundreds, tens, ones), base-ten blocks or paper strips and squares, mini whiteboards, the 1000-chart pieces on the workbook page (rows of 10).
Online set-up	Children cut out paper digit cards before the lesson and move them on camera; the tutor writes on the slide 1000-chart pieces.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.
Watching the recording	A child who missed the live lesson watches the recording. They need: Digit cards 0 to 9 (one set per child), place-value mat (hundreds, tens, ones), base-ten blocks or paper strips and squares, mini whiteboards, the 1000-chart pieces on the workbook page (rows of 10). The REC lines tell them when to pause; the chapter list at the end of this lesson shows each part.

0–1.5 min Opening

REC *Hello! Before you press play, get a pencil, paper, your workbook and the materials on the slide. Pause or rewind any time.*

SAY *"Welcome, Digit Detectives! Every 3-digit number is made of just three digits, but WHERE a digit sits changes everything."*

"Today we find every number hiding in a set of digit cards, and we make sure we do not miss any. Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. The kit is on the desk from minute 0: ask everyone to show you the materials for today (a 30-second check; the materials list is above and on the title slide).

1.5–6 min Sneak-peek puzzles (4.5 minutes)

REC *Pause now and try the puzzles on your own paper for about 4 and a half minutes. Guessing is fine. Press play for the answers.*

SAY *"Two sneak-peek puzzles on ideas we have not learnt yet. Guessing is welcome, so is I do not know. Nobody is marked. Show your paper to me only."*

DO Show the Sneak-peek puzzles slide at minute 1.5 (4.5 minutes: about 45 seconds for question 1, 90 seconds for question 2, then the answers). Say the short line out loud (about 28 words): two

puzzles on ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked; show your paper to me only. Read each puzzle aloud; question 1 is the one to start with (objects, a drawing or counting).

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): Mei has 3 hundreds, 2 tens and 5 ones. Draw them if you like. What number is that?	325. Listen for: Hundreds first, then tens, then ones. A drawing of 3 flats, 2 rods and 5 units is welcome. This is a first win for everyone.
Question 2 (thinking): Mei has digit cards 4, 0 and 7. She uses each card once to make a 3-digit number. How many different numbers can she make? Write them.	4 numbers: 407, 470, 704, 740. Listen for: 0 cannot be the hundreds digit, so only 4 and 7 can start. Listing in order: 4 starts 2 numbers, 7 starts 2 numbers. A child who says 6 is using 3 x 2 x 1 without the zero rule: a good idea, not a mistake to correct yet.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

6–10.5 min Warm-up

REC *Join in! Say or write each answer before I do. Pause the video if you need longer.*

DO Place It Right. The tutor turns over digit cards one at a time. Write each one in the hundreds, tens or ones box straight away. No moving! Biggest number wins.

SAY *“Draw three boxes on your whiteboard: hundreds, tens, ones. I will turn over one digit card at a time. As soon as you see it, write it in a box. No moving it later!”*

“Try to make the BIGGEST number. Where will you put a 9? Where will you put a 1? Why?”

DO Keep the game to about two and a half minutes (6 to 8.5): stop it on the clock. Then the quick questions (8.5 to 10.5): ask questions 1 and 3 only (these are the two on the slide); questions 2, 4 and 5 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect

4 hundreds, 6 tens, 2 ones = ?	462
What is the 8 worth in 384?	80
10 more than 296?	306
100 less than 405?	305
The biggest 3-digit number?	999

10.5–23.5 min Explore (in pairs, with objects)

REC *Pause now. Try this with your objects, and a grown-up if you can, for about 13 minutes. Press play when you have an answer.*

DO Timing (13 minutes): set up and predict, 10.5 to 12. One round of breakout rooms with an auto-timer: 60 seconds to get in, 5.5 minutes of pair work (13 to 18.5), 60 seconds to come back. Then the group debrief, 19.5 to 23.5.

SAY *"You have the cards 3, 5 and 8. How many different 3-digit numbers do you think you can make with them? Write your guess in the corner of your whiteboard."*

DO 1. Take the digit cards 3, 5 and 8.

2. Make a 3-digit number using each card once. Write it down.

3. Make as many different numbers as you can.

4. Circle the biggest and the smallest. Have you got them all?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: put the 3 in the hundreds place first. What can the other two cards do?

Not sure if you have them all: "How many numbers start with 3? With 5? With 8?"

Finished early: add a 0 card and make numbers with 0, 3 and 5. Why are there fewer?

DO Go further (a second what-if, so nobody waits): it is the box at the foot of the Explore slide; post it in the chat as well when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: add a fourth card, 2. Use any THREE of the cards 2, 3, 5 and 8. How many different 3-digit numbers can you make? Answer: 24: 4 choices of which card to leave out, and each set of 3 cards makes 6 numbers (4 x 6 = 24).

DO At about minute 18.5, say "Pencils down" and bring the group back together (60 seconds). Take the three questions below, one at a time (about 4 minutes in all). Ask "How do you know?" on question 2 only; take about 15 seconds of talk on the others.

Ask	Expect
How many numbers did you make?	6: 358, 385, 538, 583, 835, 853.
How do you know you have them all?	List in order: each card starts 2 numbers, and 3 × 2 = 6.

Which is the biggest? The smallest?	853 and 358: put the biggest digit first for the biggest.

23.5–35.5 min Mini-lesson (you teach, they join in)

REC *Keep watching and copy the examples onto your paper. Missed a step? Rewind. You will try some yourself soon.*

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"5 hundreds, 12 tens and 4 ones. 12 tens is 120, so the number is 500 + 120 + 4 = 624. I can also see it as 6 hundreds, 2 tens, 4 ones, because 10 of the tens make another hundred."*

"Cards 1, 6 and 9. I keep the hundreds digit the same and swap the other two. 1 first: 169, 196. 6 first: 619, 691. 9 first: 916, 961. Each card starts 2 numbers, and 3 cards make 6 numbers. That is how I know I have them all."

"The biggest number has the biggest digit in the hundreds place: 961. The smallest has the smallest digit in the hundreds place: 169. But zero can never go first: 036 would be just 36 (thirty-six), which has only 2 digits, so it is not a 3-digit number."

Worked example on the slides	Answer
Place value: more than 9 tens. 5 hundreds, 12 tens and 4 ones. 12 tens = 120. 500 + 120 + 4 = ?	624. Ten of the tens make another hundred.
Every number from 1, 6, 9. Fix the hundreds digit, then swap the other two. 1 first: 169, 196. 6 first: 619, 691. 9 first: 916, 961. How many? Which is biggest?	6 numbers. Biggest 961, smallest 169.

SAY *"The key idea today is: list in order. Fix the hundreds digit, then swap the others. Each card starts the same number of numbers. Say it with me."*

DO Show the key idea slide at about minutes 34.5 to 35.5. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Writing 6 hundreds, 15 tens and 2 ones as 6152 instead of regrouping 15 tens as 1 hundred and 5 tens (752).
- Listing digit-card numbers at random, so one is missed or written twice, or counting 059 as a 3-digit number.
- Counting back in 10s across a hundred: 512, 502, 592 instead of 512, 502, 492.

35.5–38.5 min Brain break

REC *Movement break! Stand up and join in. The video keeps going.*

DO Swap the digits. Each child puts three digit cards in a row (hundreds, tens, ones). The tutor calls "swap the hundreds and the tens" or "swap the tens and the ones". Show your new number on camera. Is it bigger or smaller?

Water. Toilet break for anyone who needs it. Keep it to three minutes: two of movement, one for water.

38.5–73.5 min Worksheet (pairs, then alone, then stretch)

REC *Pause and do your workbook, starting at Start here on page 1, for about 35 minutes. Press play to check your work.*

DO Minute 38.5: give out the workbook. Read the title of Lesson 1 aloud. Say “Pairs first. Talk before you write.”

Minutes 38.5 to 52.5: pairs work together on the first page (the 2 minutes at the start are for finding the page and setting up; pairs stay in the main room). Minutes 52.5 to 55: check page 1 together (read only the answer line, children mark their own, one “How do you know?” for the page). Minutes 55 to 59.5: short teaching for page 2. Minutes 59.5 to 71: each child works alone on the second page. Minutes 71 to 73.5: check page 2 together (answer lines only, children mark their own). Think harder, Go further and the Investigation are open to any child at any time after question 3.

DO Take-home stars: Page 2: Q4(b) (two squares below 385) and the whole of Q5 (count on and back, four items). The teaching slide for counting past a hundred is not shown in class. Each is marked on the page with a ★ and the words take-home, stays Core (not Think harder), and keeps its answer on the check-together slide so a child can mark it at home. Do not spend class time on them. Late rule, printed here on purpose: at minute 59.5 (the start of page 2), if the clock is more than 2 minutes late, drop the last Core question of page 2 to take-home (children put a ★ beside it). Never shorten talk time, the exit ticket, looking back or the closing.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “How many of your numbers start with this card? Is that all of them?”
- Ask: “What is this digit worth? How many tens is that?”
- Ask: “Point to the number one square below this one on the chart. How much bigger is it?”

IF STUCK Hint 1: “Fix the hundreds digit first.”

Hint 2: “Build it with blocks: hundreds, tens and ones. Can 10 tens become a hundred?”

Hint 3: “Say the numbers out loud as you count on in 10s. What happens after 590?”

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer

Workbook Think harder 1: Use the digit cards 2, 6 and 9 once each. Make a list in order of every 3-digit number you can make that is bigger than 500. How do you know you have found them all?	4 numbers: 629, 692, 926, 962. The hundreds digit must be 6 or 9 (a number that starts with 2 is smaller than 500). With 6 in the hundreds, the other two cards make 629 and 692; with 9 in the hundreds, they make 926 and 962.
Workbook Think harder 2: Use the digit cards 4, 4 and 7. How many different 3-digit numbers can you make? Why is it not 6?	3 numbers: 447, 474, 744. The two 4s look the same, so swapping them does not make a new number.
Workbook Think harder 3: Ali has 46 ten-blocks. Siti has 3 hundred-blocks and 15 ten-blocks. Who has the bigger number? How much bigger?	Ali: 46 tens is 460. Siti: 3 hundreds and 15 tens is 300 + 150 = 450. Ali has the bigger number, by 10.
Workbook Think harder 4: Mei says: “If I swap the tens digit and the ones digit of a 3-digit number, the new number is always smaller.” Is she right? Show what happens with some examples. When does the number stay the same?	No. 452 becomes 425, which is smaller, but 425 becomes 452, which is bigger. It gets smaller when the tens digit is bigger than the ones digit, bigger when it is smaller, and stays the same when the two digits are the same (733).
Workbook Go further 5: Mei uses each of the digit cards 1, 2, 3, 4, 5 and 6 once to make two 3-digit numbers. She adds them. What is the smallest total she can make?	381, for example 135 + 246. The smallest digits 1 and 2 go in the hundreds places, 3 and 4 in the tens, 5 and 6 in the ones: 300 + 70 + 11 = 381.
Workbook Go further 6: Ali counts in 5s, starting at 3: 3, 8, 13, 18 … Will he say 500? Will he say 998? Explain.	He says only numbers ending in 3 or 8. So he will not say 500, but he will say 998.
Workbook Go further 7: Ben writes all the numbers from 300 to 399. How many times does he write the digit 3?	120. Hundreds place: 100 times. Tens place (330 to 339): 10 times. Ones place (303, 313, … 393): 10 times. 100 + 10 + 10 = 120.

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (on the “Before you go” page): Mirror numbers. A mirror number reads the same forwards and backwards, like 252 or 909. (a) Write 5 mirror numbers of your own. (b) Find ALL the mirror numbers from 200 to 299. How do you know you have found them all? (c) Ali says every mirror number is odd. Is he right? Easy first step: write 252, 909 and 131, then write three more mirror numbers.	(a) Any number whose first and last digits match, such as 101, 454, 777. (b) 10 numbers: 202, 212, 222, 232, 242, 252, 262, 272, 282, 292. The first and last digits must both be 2, and the middle digit can be 0 to 9: 10 choices. A child who lists them by the middle digit in order has not missed any. (c) No: 202 is even. A mirror number is even when its first (and last) digit is even. Extension: there are 90 mirror numbers from 100 to 999 (9 choices for the end digits x 10 for the middle). Look for: the child sees that the end digits match, lists in order, and can say why the list is complete.

73.5–76.5 min Flex (spare time)

REC *Pause or rewind whenever you need to.*

DO Spare time, kept on purpose (3 minutes). Use it to absorb a late start, a camera problem or a slow page 2. If you are on time, spend it on the Investigation share or a second talk-time volunteer, or an early start on talk time. Do not add new content. Never take time from the exit ticket, looking back or the stamp.

76.5–81.5 min Talk time

REC *Pause. Tell a grown-up your answer to the talk question, and why you think so. Then press play.*

DO Bring the group back to a circle (or camera on, mics on). Pairs talk for 90 seconds, then two volunteers share (about 5 minutes in all). One question only. You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *"Which is bigger, 401 or 399? Ali says 399 because it has two 9s. Who agrees?"*

Discussion question	Listen for
Which is bigger, 401 or 399? Ali says 399 because it has two 9s. Who is right, and why?	401 is bigger. Compare the hundreds first: 4 hundreds is more than 3 hundreds, so the tens and ones do not matter. Count on from 399: 400, 401.

81.5–86.5 min Exit ticket

REC *Pause and do the exit ticket on your own, without looking back. Then press play to check it.*

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. About 2 minutes of child work; then children hold their slips up one at a time (never all together) and the answer is on the next click.

Exit ticket	Answer
Use the digit cards 6, 0 and 2 once each. Write the biggest and the smallest 3-digit number.	Biggest 620, smallest 206.

86.5–88.5 min Looking back (2 minutes)

REC *Pause for about 2 minutes and fill in your looking-back page: what was hardest, and what will you try next time?*

SAY *"Looking back takes 2 minutes. Find the box called Looking back on the page "Before you go: Investigation and Looking back". Finish one line, two lines or all three, and tick what helped you. You may pass on any line. If you tick "I am still stuck", your tutor will check in during the session."*

DO Nothing is collected in front of the group and nobody has to read theirs out. The tutor may read a box while stamping journals (never by private message), and checks in on a child who ticked still

stuck during the open session, for example by a quick visit to the shared breakout room or a hand signal with a parent nearby, for example "I saw you drew it when you got stuck". A child may volunteer one sentence, never by name unless they choose to. If a child ticks "I am still stuck", check in during the next open session (a quick visit to the shared breakout room or a hand signal, with a parent nearby). Praise the strategy, not the speed.

88.5–90 min Closing and stamps (1.5 minutes)

REC *Well done! Show your journal and workbook to a grown-up and tell them one thing you found out.*

SAY *"You were real detectives today: you made a list in order, so no number could hide from you."*

"Hand me your exit ticket and I will stamp your inquiry journal."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Use any three of the cards 1, 3, 5 and 7. How many different 3-digit numbers can you make? (24: 6 for each set of three cards, and there are 4 sets.)

Answer key (workbook)

Q1 (a) 407 (b) 752 (c) 3 tens; 13 tens (d) 70

Q2 (a) Start with 2: 247, 274. Start with 4: 427, 472. Start with 7: 724, 742. (b) Biggest 742, smallest 247 (c) 472 (only 28 away from 500) (d) 2 odd numbers: 247 and 427

Q3 (a) 509, 590, 905, 950 (b) 0 cannot be the hundreds digit (059 is not a 3-digit number), so only 5 or 9 can start: 2 + 2 = 4 numbers, not 6.

Q4 Piece A: 348, 349 / 357, 359 / 367, 368. Piece B: 595, 596, 597 in the middle row and 606 at the bottom. (a) Down: 10 more. Right: 1 more. (b) 405 (395, then 405)

Q5 (a) 347, 447 (b) 502, 492 (c) 1000 (d) 700, 699

Q6 (a) 435 (b) 6 numbers: 102, 111, 120, 201, 210, 300

Investigation (Mirror numbers). (a) Any number whose first and last digits match, such as 101, 454, 777. (b) 10 numbers: 202, 212, 222, 232, 242, 252, 262, 272, 282, 292. The first and last digits must both be 2, and the middle digit can be 0 to 9: 10 choices. A child who lists them by the middle digit in order has not missed any. (c) No: 202 is even. A mirror number is even when its first (and last) digit is even. Extension: there are 90 mirror numbers from 100 to 999 (9 choices for the end digits x 10 for the middle). Look for: the child sees that the end digits match, lists in order, and can say why the list is complete.

Recording chapters and what a child watching later does

Time	Chapter marker	A child watching later

0–1.5 min	Opening	Gather your workbook, pencil, paper and the materials on the slide. Press play when ready.
1.5–6 min	Sneak-peek puzzles (4.5 minutes)	Pause. Try the puzzles on your own paper (about 4 and a half minutes). Guessing is fine. Press play for the answers.
6–10.5 min	Warm-up	Play along. Say or write each answer before the video does. Pause if you need longer.
10.5–23.5 min	Explore (in pairs, with objects)	Pause. Try the explore task with your objects, with someone at home if you can (about 13 minutes). Write down what you notice.
23.5–35.5 min	Mini-lesson (you teach, they join in)	Keep watching. Copy the worked examples. Rewind any step you missed.
35.5–38.5 min	Brain break	Stand up and join the brain break.
38.5–73.5 min	Worksheet (pairs, then alone, then stretch)	Pause. Do the workbook, starting with Start here on page 1 (about 35 minutes). Press play for each checking part and mark in a different colour.
73.5–76.5 min	Flex (spare time)	Pause or rewind whenever you need to.
76.5–81.5 min	Talk time	Pause. Explain your answer to someone at home, with a reason. Then play.
81.5–86.5 min	Exit ticket	Pause. Do the exit ticket on your own, then play to check it.
86.5–88.5 min	Looking back (2 minutes)	Pause. Fill in the looking-back page.
88.5–90 min	Closing and stamps (1.5 minutes)	Show your journal and workbook to a grown-up. Tell them one thing you found out.

Lesson 2: Missing Digit Mysteries

Week 1 (second lesson) • Addition and Subtraction within 1000 • Workbook: Lesson 2

Objective	Find missing digits in 3-digit addition and subtraction by starting at the ones and checking for a carry, use "=" as "is the same as" to balance number sentences, and complete magic squares and triangles with bigger numbers.
Thinking skills	Working backwards, guess and check, reasoning about regrouping, balancing, drawing a model
Materials	Digit cards 0 to 9, base-ten blocks or place-value counters, mini whiteboards, a paper magic triangle with 6 circles and number tiles 110 to 160.
Online set-up	Children write column sums on whiteboards and hold them up; number tiles are cut from paper before the lesson. The tutor writes on the slide sums.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.
Watching the recording	A child who missed the live lesson watches the recording. They need: Digit cards 0 to 9, base-ten blocks or place-value counters, mini whiteboards, a paper magic triangle with 6 circles and number tiles 110 to 160. The REC lines tell them when to pause; the chapter list at the end of this lesson shows each part.

0–1.5 min Opening

REC *Hello! Before you press play, get a pencil, paper, your workbook and the materials on the slide. Pause or rewind any time.*

SAY *"Welcome back, detectives! Someone has spilled ink on our sums and some digits have disappeared."*

"Today we find the missing digits, make both sides of "=" balance, and crack some magic squares. Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud. The kit is on the desk from minute 0: ask everyone to show you the materials for today (a 30-second check; the materials list is above and on the title slide).

1.5–6 min Sneak-peek puzzles (4.5 minutes)

REC *Pause now and try the puzzles on your own paper for about 4 and a half minutes. Guessing is fine. Press play for the answers.*

SAY *"Two sneak-peek puzzles on ideas we have not learnt yet. Guessing is welcome, so is I do not know. Nobody is marked. Show your paper to me only."*

DO Show the Sneak-peek puzzles slide at minute 1.5 (4.5 minutes: about 45 seconds for question 1, 90 seconds for question 2, then the answers). Say the short line out loud (about 28 words): two puzzles on ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is

marked; show your paper to me only. Read each puzzle aloud; question 1 is the one to start with (objects, a drawing or counting).

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): 3[ ]2 + 416 = 788. Which digit is hidden in the box? (Use counters or draw the columns if it helps.)	7 (372 + 416 = 788). Listen for: No carry in this one: look at the tens column, 1 and what makes 8. This is a first win for everyone.
Question 2 (thinking): 4[ ]6 + 237 = 683. Which digit is hidden in the box? How do you know?	4 (446 + 237 = 683). Listen for: Ones: 6 + 7 = 13, so 1 ten is carried. Tens: the box + 3 + 1 must end in 8, so the box is 4. Listen for: "carry", or a guess that was tested and checked.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

6–10.5 min Warm-up

REC *Join in! Say or write each answer before I do. Pause the video if you need longer.*

DO Partners to 100 and 1000. The tutor says a number; say its partner. 65 → 35, then 370 → 630, then 125 → 875.

SAY *“I say a number. You say its partner to make 100. Ready? 65!”*

“Now partners to 1000. 370! How did you find 630 so fast?”

DO Keep the game to about two and a half minutes (6 to 8.5): stop it on the clock. Then the quick questions (8.5 to 10.5): ask questions 1 and 3 only (these are the two on the slide); questions 2, 4 and 5 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
65 + __ = 100	35
300 + 450 = ?	750

700 – 250 = ?	450
238 + 10 = ?	248
500 – __ = 120	380

10.5–23.5 min Explore (in pairs, with objects)

REC *Pause now. Try this with your objects, and a grown-up if you can, for about 13 minutes. Press play when you have an answer.*

DO Timing (13 minutes): set up and predict, 10.5 to 12. One round of breakout rooms with an auto-timer: 60 seconds to get in, 5.5 minutes of pair work (13 to 18.5), 60 seconds to come back. Then the group debrief, 19.5 to 23.5.

SAY *"Pick any 3-digit number. I predict that its partner to make 999 will be easy to find, with no carrying at all. Do you think I am right? Let us test it."*

DO 1. Pick any 3-digit number, like 358.

2. Find its partner that makes 999. Check by adding.

3. Do it for three more numbers.

4. Look at each column. What do you notice?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: start with the hundreds. What goes with 3 to make 9?

Not sure: add your two numbers in a column sum to check the total is 999.

Finished early: try partners to 1000. What changes in the ones column?

DO Go further (a second what-if, so nobody waits): it is the box at the foot of the Explore slide; post it in the chat as well when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: now make 1000. What is the partner of 358 that makes 1000 (358 + ? = 1000)? What do you notice about the columns? Answer: 642. Ones: 8 + 2 = 10. Tens: 5 + 4 + 1 = 10. Hundreds: 3 + 6 + 1 = 10. Every column carries.

DO At about minute 18.5, say "Pencils down" and bring the group back together (60 seconds). Take the three questions below, one at a time (about 4 minutes in all). Ask "How do you know?" on question 2 only; take about 15 seconds of talk on the others.

Ask	Expect
What did you notice about each column?	The two digits in each column add up to 9.
Why is there never a carry?	No column adds up to more than 9.
Find the partner of 274 without adding.	725: 2 + 7, 7 + 2 and 4 + 5 all make 9.

23.5–35.5 min Mini-lesson (you teach, they join in)

REC *Keep watching and copy the examples onto your paper. Missed a step? Rewind. You will try some yourself soon.*

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"4[ ]7 + 28[ ] = 723. I start at the ones: 7 plus the ones box ends in 3. 7 + 6 = 13. So the ones box is 6, and I carry 1 ten."*

"Tens: the tens box + 8 + the carried 1 ends in 2. 3 + 8 + 1 = 12. So the tens box is 3, and I carry 1 hundred. Hundreds: 4 + 2 + 1 = 7. It matches! I check by adding: 437 + 286 = 723."

"6[ ]2 − 24[ ] = 385. Subtraction is harder, so I work backwards and turn it into an addition: 385 + 24[ ] = 6[ ]2. Ones: 5 + the ones box ends in 2, 5 + 7 = 12, so it is 7. Tens: 8 + 4 + 1 = 13, so the tens box is 3. Hundreds: 3 + 2 + 1 = 6 ✓."

"245 + 130 = 200 + [ ]. I do not need to add. 245 is 45 more than 200, so the box must be 45 more than 130. That is 175. Both sides are 375."

Worked example on the slides	Answer
Missing digits: start at the ones. 4[ ]7 + 28[ ] = 723 Ones: 7 + the ones box ends in 3, so it is 6 (13, carry 1). Tens: the tens box + 8 + 1 ends in 2, so it is 3 (12, carry 1). Hundreds: 4 + 2 + 1 = 7 ✓	437 + 286 = 723.
Balance: "=" means "is the same as". 245 + 130 = 200 + [ ] 245 is 45 more than 200, so the box is 45 more than 130.	175. Both sides are 375.

SAY *"The key idea today is: start at the ones. Start at the ones column and ask: was there a carry? Check by adding. Say it with me."*

DO Show the key idea slide at about minutes 34.5 to 35.5. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Forgetting the carried 1: in Q1c writing 6 in the tens box (8 + 6 = 14) instead of 5 (8 + 5 + 1 = 14).
- Reading "=" as "the answer comes next": 350 + 140 = 300 + [ ] answered 490.
- Magic triangle: putting the biggest numbers in the corners when the side total is small (390 needs the smallest corners).

35.5–38.5 min Brain break

REC *Movement break! Stand up and join in. The video keeps going.*

DO Balance bodies. Stand like a see-saw with arms out. The tutor says a number sentence: stay level if it is true, tip to one side if it is false.

Water. Toilet break for anyone who needs it. Keep it to three minutes: two of movement, one for water.

38.5–73.5 min Worksheet (pairs, then alone, then stretch)

REC *Pause and do your workbook, starting at Start here on page 1, for about 35 minutes. Press play to check your work.*

DO Minute 38.5: give out the workbook. Read the title of Lesson 2 aloud. Say “Pairs first. Talk before you write.”

Minutes 38.5 to 52.5: pairs work together on the first page (the 2 minutes at the start are for finding the page and setting up; pairs stay in the main room). Minutes 52.5 to 55: check page 1 together (read only the answer line, children mark their own, one “How do you know?” for the page). Minutes 55 to 59.5: short teaching for page 2. Minutes 59.5 to 71: each child works alone on the second page. Minutes 71 to 73.5: check page 2 together (answer lines only, children mark their own). Think harder, Go further and the Investigation are open to any child at any time after question 3.

DO Take-home stars: Page 1: the line under Q1 (check one answer by adding the full numbers), and Q3 c and d (480 + ___ = 500 + 260, and Ravi's 398 + 205). Page 2: the whole of Q5 (the magic triangle); its teaching slide is not shown in class. Each is marked on the page with a ★ and the words take-home, stays Core (not Think harder), and keeps its answer on the check-together slide so a child can mark it at home. Do not spend class time on them. Late rule, printed here on purpose: at minute 59.5 (the start of page 2), if the clock is more than 2 minutes late, drop the last Core question of page 2 to take-home (children put a ★ beside it). Never shorten talk time, the exit ticket, looking back or the closing.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “Which column did you start with? Why?”
- Ask: “Was there a carry into this column? How do you know?”
- Ask: “What is each side of the = sign worth? Are they the same?”

IF STUCK Hint 1: “Start at the ones column.”

Hint 2: “Try a digit in the box and add. Too big or too small?”

Hint 3: “Turn the subtraction into an addition: answer + bottom number = top number.”

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer

Workbook Think harder 1: A magic triangle has 3 corner circles and 3 circles in the middle of the sides. Put the numbers 110, 120, 130, 140, 150 and 160 in the circles, once each, so that every side adds up to 420. Draw the triangle. Which numbers are in the corners?	Corners 140, 150, 160. Between 140 and 150: 130. Between 150 and 160: 110. Between 160 and 140: 120 (each side: 140 + 130 + 150 = 420, 150 + 110 + 160 = 420, 160 + 120 + 140 = 420).
Workbook Think harder 2: Find the missing digits: 5[ ] + [ ]7 + 38 = 132. There are three numbers, so the carry from the ones may be more than 1.	57 + 37 + 38 = 132. Ones: 7 + 7 + 8 = 22, so write 2 and carry 2 tens. Tens: 5 + 3 + 3 + 2 = 13. The first box is 7 and the second box is 3.
Workbook Think harder 3: Mei did 523 − 267 and wrote 266. She checked by adding 266 + 267 = 533. Was her answer right? What mistake did she make? What is the right answer?	No. Her check gives 533, not 523, so 266 is wrong. The right answer is 256 (ones: 13 − 7 = 6; tens: the 2 tens lent 1 ten to the ones, so 1 ten is left and we borrow again: 11 − 6 = 5; hundreds: 4 − 2 = 2). Check: 256 + 267 = 523. Mei's tens digit is 1 too big: she forgot that the tens had already lent 1 ten to the ones.
Workbook Think harder 4: True or false? Say how you know without working out both sides. (a) 346 + 199 = 345 + 200 (b) 270 + 130 = 260 + 150 (c) 618 + 99 = 600 + 117	(a) True: 346 is 1 more than 345 and 199 is 1 less than 200. (b) False: 270 is 10 more than 260, but 130 is 20 less than 150, so the sides are 400 and 410. (c) True: 618 is 18 more than 600 and 99 is 18 less than 117 (both sides are 717).
Workbook Go further 5: Find ALL the ways to fill in the four boxes: [ ][ ]7 + 4[ ][ ] = 1000	10 ways: 507 + 493, 517 + 483, 527 + 473, 537 + 463, 547 + 453, 557 + 443, 567 + 433, 577 + 423, 587 + 413, 597 + 403. Ones: 7 + 3 = 10. Hundreds: 5 + 4 + 1 = 10. The two tens digits add up to 9 (with the carried 1 they make 10).
Workbook Go further 6: Box A and Box B hold 250 marbles together. Box B and Box C hold 350. Box A and Box C hold 400. How many marbles are in each box?	Box A 150, Box B 100, Box C 250. Box B + Box C is 100 more than Box A + Box B, so Box C has 100 more than Box A. Box A and Box C together are 400: take away the extra 100 to get 300, and half of 300 is 150. So Box A has 150 and Box C has 250. Then Box B has 250 − 150 = 100.
Workbook Go further 7: Use the digit cards 1, 2, 3, 4, 5 and 6 once each to make two 3-digit numbers. Make the difference as SMALL as possible.	47: 412 − 365. The hundreds digits are next to each other (4 and 3); then make the number starting with 4 as small as possible (412) and the one starting with 3 as big as possible (365).

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for

Workbook Investigation (on the “Before you go” page): Odd or even?. Ali says: "When I add two odd numbers, I always get an even number." (a) Test Ali's idea with 5 pairs of odd 3-digit numbers (for example 345 + 271). (b) Now test odd + even, and even + even. What do you notice? (c) Explain WHY, using pairs of counters or a drawing. Easy first step: use counters to try 3 + 5, 7 + 9 and 11 + 13. Are the answers odd or even?	(a) Ali is right: odd + odd is always even (checked for every pair of 3-digit numbers). (b) Odd + even is always odd; even + even is always even. (c) Why: an odd number is some pairs plus 1 left over, an even number is only pairs. Two odd numbers have two left-overs, and they make one more pair, so no one is left over. Look for: several tests before a claim, then a reason that uses pairs, counters or ones digits, not only "it worked every time". A ones-digit reason (odd ends in 1, 3, 5, 7, 9) is also fine. Extension (not printed): taking away behaves the same way: odd − odd is even, odd − even is odd.

73.5–76.5 min Flex (spare time)

REC *Pause or rewind whenever you need to.*

DO Spare time, kept on purpose (3 minutes). Use it to absorb a late start, a camera problem or a slow page 2. If you are on time, spend it on the Investigation share or a second talk-time volunteer, or an early start on talk time. Do not add new content. Never take time from the exit ticket, looking back or the stamp.

76.5–81.5 min Talk time

REC *Pause. Tell a grown-up your answer to the talk question, and why you think so. Then press play.*

DO Bring the group back to a circle (or camera on, mics on). Pairs talk for 90 seconds, then two volunteers share (about 5 minutes in all). One question only. You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *“Why do we start at the ones column when we look for missing digits?”*

Discussion question	Listen for
Why do we start at the ones column when we look for missing digits?	The ones column can make a carry that changes the tens column, and the tens can change the hundreds, so we must know the ones first.

81.5–86.5 min Exit ticket

REC *Pause and do the exit ticket on your own, without looking back. Then press play to check it.*

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. About 2 minutes of child work; then children hold their slips up one at a time (never all together) and the answer is on the next click.

Exit ticket	Answer

3[ ]4 + 257 = 601. Which digit goes in the box?	The box is 4 (344 + 257 = 601: ones 4 + 7 = 11; tens 4 + 5 + 1 = 10). Bonus for anyone who would like it: 4[ ]6 + 37[ ] = 801 gives 2 in the first box and 5 in the second box (426 + 375 = 801).

86.5–88.5 min Looking back (2 minutes)

REC *Pause for about 2 minutes and fill in your looking-back page: what was hardest, and what will you try next time?*

SAY *"Looking back takes 2 minutes. Find the box called Looking back on the page "Before you go: Investigation and Looking back". Finish one line, two lines or all three, and tick what helped you. You may pass on any line. If you tick "I am still stuck", your tutor will check in during the session."*

DO Nothing is collected in front of the group and nobody has to read theirs out. The tutor may read a box while stamping journals (never by private message), and checks in on a child who ticked still stuck during the open session, for example by a quick visit to the shared breakout room or a hand signal with a parent nearby, for example "I saw you drew it when you got stuck". A child may volunteer one sentence, never by name unless they choose to. If a child ticks "I am still stuck", check in during the next open session (a quick visit to the shared breakout room or a hand signal, with a parent nearby). Praise the strategy, not the speed.

88.5–90 min Closing and stamps (1.5 minutes)

REC *Well done! Show your journal and workbook to a grown-up and tell them one thing you found out.*

SAY *"You were digit detectives again: you started at the ones, checked every carry and proved each answer by adding."*

"Exit ticket, then your stamp."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Make your own missing-digit sum with two boxes that has exactly ONE answer. Swap with a partner. (Check: does any other digit also work?)

Answer key (workbook)

Q1 (a) 224 + 355 = 579 (b) 462 + 375 = 837 (c) 386 + 259 = 645

Q2 (a) 778 – 325 = 453 (b) 543 – 267 = 276

Q3 (a) 190 (b) 525 (c) 280 (d) Yes. 398 is 2 less than 400 and 205 is 2 more than 203, so both sides are the same (603).

Q4 (a) 150 (diagonal 80 + 50 + 20) (b) Top: 10. Middle: 30 and 70. Bottom: 90. (c) Yes, still magic. New total 450: every line has 3 numbers and each is 100 more, so 150 + 300.

Q5 (a) Corners 110, 120, 130. Between 110 and 120: 160. Between 120 and 130: 140. Between 130 and 110: 150. (Think harder 1 is the same triangle with sides of 420: corners 140, 150, 160; between 140 and 150: 130; between 150 and 160: 110; between 160 and 140: 120.)

Q6 250 and 450 (700 − 200 = 500, half of 500 is 250, 250 + 200 = 450).

Investigation (Odd or even?). (a) Ali is right: odd + odd is always even (checked for every pair of 3-digit numbers). (b) Odd + even is always odd; even + even is always even. (c) Why: an odd number is some pairs plus 1 left over, an even number is only pairs. Two odd numbers have two left-overs, and they make one more pair, so no one is left over. Look for: several tests before a claim, then a reason that uses pairs, counters or ones digits, not only "it worked every time". A ones-digit reason (odd ends in 1, 3, 5, 7, 9) is also fine. Extension (not printed): taking away behaves the same way: odd − odd is even, odd − even is odd.

Recording chapters and what a child watching later does

Time	Chapter marker	A child watching later
0–1.5 min	Opening	Gather your workbook, pencil, paper and the materials on the slide. Press play when ready.
1.5–6 min	Sneak-peek puzzles (4.5 minutes)	Pause. Try the puzzles on your own paper (about 4 and a half minutes). Guessing is fine. Press play for the answers.
6–10.5 min	Warm-up	Play along. Say or write each answer before the video does. Pause if you need longer.
10.5–23.5 min	Explore (in pairs, with objects)	Pause. Try the explore task with your objects, with someone at home if you can (about 13 minutes). Write down what you notice.
23.5–35.5 min	Mini-lesson (you teach, they join in)	Keep watching. Copy the worked examples. Rewind any step you missed.
35.5–38.5 min	Brain break	Stand up and join the brain break.
38.5–73.5 min	Worksheet (pairs, then alone, then stretch)	Pause. Do the workbook, starting with Start here on page 1 (about 35 minutes). Press play for each checking part and mark in a different colour.
73.5–76.5 min	Flex (spare time)	Pause or rewind whenever you need to.
76.5–81.5 min	Talk time	Pause. Explain your answer to someone at home, with a reason. Then play.
81.5–86.5 min	Exit ticket	Pause. Do the exit ticket on your own, then play to check it.

86.5–88.5 min	Looking back (2 minutes)	Pause. Fill in the looking-back page.
88.5–90 min	Closing and stamps (1.5 minutes)	Show your journal and workbook to a grown-up. Tell them one thing you found out.

Lesson 3: Legs, Wheels and Arrays

Week 2 (first lesson) • Multiplication and Division • Workbook: Lesson 3

Objective	Read an array as rows of equal groups and write two multiplication and two division sentences for it; solve "how many of each" legs and wheels puzzles with a table; share with leftovers and decide what the leftover means; find all the arrays for a number.
Thinking skills	Draw it (arrays), make a table, guess and check, working in order to find all the ways
Materials	About 30 counters per pair, mini whiteboards, squared paper. Optional: picture cards of hens and goats, bicycles and tricycles.
Online set-up	Each child has 30 small objects (buttons, dried beans or Lego bricks) and a whiteboard or paper. The tutor shares the slides; children hold their arrays and tables up to the camera.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.
Watching the recording	A child who missed the live lesson watches the recording. They need: About 30 counters per pair, mini whiteboards, squared paper. Optional: picture cards of hens and goats, bicycles and tricycles. The REC lines tell them when to pause; the chapter list at the end of this lesson shows each part.

0–1.5 min Opening

REC *Hello! Before you press play, get a pencil, paper, your workbook and the materials on the slide. Pause or rewind any time.*

SAY *"Welcome back, inquirers! Today we are on a farm and in a bicycle shop. We will count legs and wheels, line things up in neat rows called arrays, and share things out fairly."*

"Our big question today: when there are two kinds of things mixed up, how can we find out how many of each there are, without guessing wildly?"

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. The kit is on the desk from minute 0: ask everyone to show you the materials for today (a 30-second check; the materials list is above and on the title slide).

1.5–6 min Sneak-peek puzzles (4.5 minutes)

REC *Pause now and try the puzzles on your own paper for about 4 and a half minutes. Guessing is fine. Press play for the answers.*

SAY *"Two sneak-peek puzzles on ideas we have not learnt yet. Guessing is welcome, so is I do not know. Nobody is marked. Show your paper to me only."*

DO Show the Sneak-peek puzzles slide at minute 1.5 (4.5 minutes: about 45 seconds for question 1, 90 seconds for question 2, then the answers). Say the short line out loud (about 28 words): two puzzles on ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is

marked; show your paper to me only. Read each puzzle aloud; question 1 is the one to start with (objects, a drawing or counting).

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): A tray has 3 rows of counters with 5 counters in each row. Draw it if you like. How many counters are there?	15 counters (3 rows of 5: 5, 10, 15). Listen for: Counting in 5s, or drawing and counting. This is a first win for everyone.
Question 2 (thinking): A toy shop has cars (4 wheels) and scooters (2 wheels). 6 toys have 16 wheels altogether. How many cars and how many scooters? How do you know?	2 cars and 4 scooters (8 + 8 = 16). Listen for: Starting with all scooters (12 wheels) and swapping one at a time is the idea of today. A child who draws 6 vehicles and counts wheels is thinking well. Guess and check is welcome.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

6–10.5 min Warm-up

REC *Join in! Say or write each answer before I do. Pause the video if you need longer.*

DO Legs and wheels count, three picture cards: 4 cars, 6 hens, 5 bicycles. Count the wheels or legs in 2s and 4s. (Spare cards: 3 tricycles, 5 dogs, 2 cars and 1 bicycle.)

SAY *“Here are some cards. Each card shows some cars, bicycles or animals. Count the wheels or the legs. Count in 2s, 3s, 4s or 5s, not in 1s!”*

“Show me the number with your fingers or on your whiteboard. Then we will click to check.”

DO Keep the game to about two and a half minutes (6 to 8.5): stop it on the clock. Then the quick questions (8.5 to 10.5): ask questions 1 and 3 only (these are the two on the slide); questions 2, 4 and 5 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect

3 × 5 = ?	15
4 groups of 4. How many?	16
20 ÷ 2 = ?	10
30 shared equally by 10 children. How many each?	3
How many 5s make 35?	7

10.5–23.5 min Explore (in pairs, with objects)

REC *Pause now. Try this with your objects, and a grown-up if you can, for about 13 minutes. Press play when you have an answer.*

DO Timing (13 minutes): set up and predict, 10.5 to 12. One round of breakout rooms with an auto-timer: 60 seconds to get in, 5.5 minutes of pair work (13 to 18.5), 60 seconds to come back. Then the group debrief, 19.5 to 23.5.

SAY *"The farmer has only hens and goats in a pen. He counts 16 legs. How many animals do you think there are? Is there only one answer? Write a guess."*

DO 1. A pen has only hens (2 legs) and goats (4 legs).

2. The farmer counts 16 legs altogether.

3. Use counters or drawings to find a mix of hens and goats.

4. Find every mix. Record each one.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: try all hens first. How many hens make 16 legs?

Then swap one hen for a goat. What happens to the number of legs? Fix it.

Finished early: which mix has the most animals? Which has the fewest? How do you know there are no more mixes?

DO Go further (a second what-if, so nobody waits): it is the box at the foot of the Explore slide; post it in the chat as well when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: what if the farmer counts 20 legs? How many different mixes of hens and goats are there now? Answer: 6 mixes: 10 hens; 8 hens and 1 goat; 6 hens and 2 goats; 4 hens and 3 goats; 2 hens and 4 goats; 5 goats.

DO At about minute 18.5, say "Pencils down" and bring the group back together (60 seconds). Take the three questions below, one at a time (about 4 minutes in all). Ask "How do you know?" on question 2 only; take about 15 seconds of talk on the others.

Ask	Expect
How many different mixes did you find?	5 mixes: 8 hens; 6 hens and 1 goat; 4 hens and 2 goats; 2 hens and 3 goats; 4 goats.

How do you know you have them all?	The goats went 0, 1, 2, 3, 4. Five goats would need 20 legs, too many.
What happens each time you swap 2 hens for 1 goat?	The legs stay at 16: 2 hens have 4 legs, the same as 1 goat.

23.5–35.5 min Mini-lesson (you teach, they join in)

REC *Keep watching and copy the examples onto your paper. Missed a step? Rewind. You will try some yourself soon.*

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"3 rows of 4 dots. I count the rows: 3. I count how many in each row: 4. 4, 8, 12. So 3 × 4 = 12. If I turn my head, it is 4 rows of 3, so 4 × 3 = 12 too. Sharing 12 into 3 rows gives 4 in each row: 12 ÷ 3 = 4. Putting 12 in groups of 4 gives 3 groups: 12 ÷ 4 = 3."*

"Part of this tray is under a cloth. I do not need to see every cookie. The top row shows 5 in each row. The side shows 4 rows. 4 rows of 5 is 20. I can see 8, so 20 – 8 = 12 are hidden."

"6 vehicles, bicycles and cars, 16 wheels. I start with all bicycles: 6 bicycles is 12 wheels. Too few. I swap one bicycle for a car: that adds 2 wheels, 14. Swap again: 16. So 4 bicycles and 2 cars. Check: 8 + 8 = 16 wheels, 4 + 2 = 6 vehicles."

Worked example on the slides	Answer
Arrays: rows of equal groups. 3 rows of 4 dots. 3 × 4 = 12 and 4 × 3 = 12. 12 ÷ 3 = 4 and 12 ÷ 4 = 3. A tray of 4 rows of 5 cookies has some under a cloth. We can see 8. How many are hidden?	12 hidden (4 rows of 5 is 20, and 20 – 8 = 12).
How many of each? Make a table. 6 vehicles: bicycles (2 wheels) and cars (4 wheels). 16 wheels altogether. Start with all bicycles: 6 bicycles, 12 wheels. Swap one bicycle for a car each time: 14, then 16 wheels. How many of each?	4 bicycles and 2 cars (8 + 8 = 16 wheels).

SAY *"The key idea today is: draw it, then make a table. An array is rows of equal groups: rows × number in each row. For "how many of each", start with all of one kind and swap one at a time in a table. Say it with me."*

DO Show the key idea slide at about minutes 34.5 to 35.5. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Counting the hidden part of an array one by one (or guessing) instead of using rows × number in each row and taking away the ones you can see.
- Giving the number of whole groups as the answer when the leftover still needs a place (28 children, vans of 5: writing 5 vans instead of 6).

- In "how many of each" puzzles, matching only one clue (the legs) and forgetting to check the other (the number of animals).

35.5–38.5 min Brain break

REC *Movement break! Stand up and join in. The video keeps going.*

DO Array moves. The tutor calls "3 rows of 4!": do 3 rows of 4 claps, counting 4, 8, 12. Then "5 rows of 2" jumps and "2 rows of 10" stomps.

Water. Toilet break for anyone who needs it. Keep it to three minutes: two of movement, one for water.

38.5–73.5 min Worksheet (pairs, then alone, then stretch)

REC *Pause and do your workbook, starting at Start here on page 1, for about 35 minutes. Press play to check your work.*

DO Minute 38.5: give out the workbook. Read the title of Lesson 3 aloud. Say “Pairs first. Talk before you write.”

Minutes 38.5 to 52.5: pairs work together on the first page (the 2 minutes at the start are for finding the page and setting up; pairs stay in the main room). Minutes 52.5 to 55: check page 1 together (read only the answer line, children mark their own, one “How do you know?” for the page). Minutes 55 to 59.5: short teaching for page 2. Minutes 59.5 to 71: each child works alone on the second page. Minutes 71 to 73.5: check page 2 together (answer lines only, children mark their own). Think harder, Go further and the Investigation are open to any child at any time after question 3.

DO Take-home stars: None: all six Core questions fit in class, so no Core item moves to take-home in this lesson. Each is marked on the page with a ★ and the words take-home, stays Core (not Think harder), and keeps its answer on the check-together slide so a child can mark it at home. Do not spend class time on them. Late rule, printed here on purpose: at minute 59.5 (the start of page 2), if the clock is more than 2 minutes late, drop the last Core question of page 2 to take-home (children put a ★ beside it). Never shorten talk time, the exit ticket, looking back or the closing.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: "How many rows? How many in each row? Which multiplication is that?"
- Ask: "What changes each time you swap one animal for the other kind?"
- Ask: "Did you check BOTH clues: the number of animals and the number of legs?"

IF STUCK Hint 1: "Draw it. Dots in rows, or circles for heads and sticks for legs."

Hint 2: "Start with all of one kind. Then swap one at a time."

Hint 3: "Make a table so you can see the pattern and do not miss a row."

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have

finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: A shop has some bicycles (2 wheels) and some tricycles (3 wheels), at least one of each. There are 17 wheels altogether. Find ALL the ways. Start with 1 tricycle, then 2 tricycles, and so on. Make a table or use counters. How do you know you have found them all?	3 ways: 7 bicycles and 1 tricycle (14 + 3); 4 bicycles and 3 tricycles (8 + 9); 1 bicycle and 5 tricycles (2 + 15). 0, 2 or 4 tricycles leave an odd number of wheels, which bicycles cannot make, and 6 or more tricycles need 18 wheels or more.
Workbook Think harder 2: A farm has 9 animals, each a hen (2 legs) or a goat (4 legs). Mei counts 25 legs. Siti counts 26 legs. Only one of them can be right. Who? How many hens and goats?	Siti is right: 5 hens and 4 goats (10 + 16 = 26). Mei cannot be right: every hen has 2 legs and every goat has 4 legs, so the total is always an even number, and 25 is odd. (Start with 9 hens, 18 legs; each swap to a goat adds 2: 18, 20, 22, 24, 26.)
Workbook Think harder 3: Ali shares 34 marbles equally among 4 friends, giving as many as he can. How many does each friend get? How many are left over?	8 each (4 × 8 = 32) and 2 left over.
Workbook Think harder 4: An array of stickers has 4 rows. Ben counts 12 stickers in the first 3 rows. How many stickers are in the whole array?	16 stickers. 3 rows have 12, so each row has 4 (12 ÷ 3 = 4); 4 rows of 4 is 16.
Workbook Go further 5: A shop has 12 vehicles. Each one is a bicycle (2 wheels) or a tricycle (3 wheels). Mei says: “Ravi could count 24 wheels, or 25, or 26, or any number of wheels up to 36.” Is she right? Could he count 37? Then think about 12 hens (2 legs) and goats (4 legs): why can Ravi count 25 wheels but never 25 legs?	Mei is right. All 12 bicycles make 24 wheels and all 12 tricycles make 36. Each swap of a bicycle for a tricycle adds exactly 1 wheel, so the totals go up 1 at a time and none is skipped: every number from 24 to 36, and 37 is too many. For hens and goats each swap adds 2 legs, so the total can only be 24, 26, 28 … 48: always even, so never 25.
Workbook Go further 6: Mr Lim has between 20 and 30 sweets. When he puts them in groups of 4, 3 are left over. When he puts them in groups of 5, 2 are left over. How many sweets does he have?	27. Groups of 4 with 3 left: 23 or 27. Groups of 5 with 2 left: 22 or 27. Only 27 is in both lists.
Workbook Go further 7: A pen has hens and goats. There are 2 more hens than goats. There are 22 legs altogether. How many hens and how many goats?	5 hens and 3 goats. Guess and check: 1 goat and 3 hens make 10 legs; 2 and 4 make 16; 3 and 5 make 22. Check: 12 + 10 = 22.

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (on the “Before you go” page): Where tables meet. (a) List the numbers up to 50 that are in BOTH the 2 times table and the 5 times table. What do you notice about them? (b) Now find the numbers up to 50 that are in both the 3 and 5 tables. (c) Is every number in both the 2 and 5 tables also in the 10 table? Explain. Easy first step: write the 5 times table up to 50, then circle the even numbers.	(a) 10, 20, 30, 40, 50: they jump by 10 and are the 10 times table. (b) 3 and 5: 15, 30, 45 (jump by 15). (Extension: 4 and 5 give 20, 40, jump by 20.) (c) Yes to both, brute-forced up to 100: a number in the 10 table is 10 x something = 2 x (5 x something) = 5 x (2 x something). Child-level reason: a number in the 2 table is even and a number in the 5 table ends in 0 or 5, so a number in both ends in 0, which is the 10 table; and every number in the 10 table ends in 0, so it is even and in the 5 table. Look for: organised lists (writing both tables out and circling), the noticing that the jumps are 10, 15, 20, and a test of “always”.

73.5–76.5 min Flex (spare time)

REC *Pause or rewind whenever you need to.*

DO Spare time, kept on purpose (3 minutes). Use it to absorb a late start, a camera problem or a slow page 2. If you are on time, spend it on the Investigation share or a second talk-time volunteer, or an early start on talk time. Do not add new content. Never take time from the exit ticket, looking back or the stamp.

76.5–81.5 min Talk time

REC *Pause. Tell a grown-up your answer to the talk question, and why you think so. Then press play.*

DO Bring the group back to a circle (or camera on, mics on). Pairs talk for 90 seconds, then two volunteers share (about 5 minutes in all). One question only. You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *“Ravi shares stickers equally among 4 friends. Can the number left over ever be 4 or more?”*

Discussion question	Listen for
Ravi shares stickers equally among 4 friends. Can the number left over ever be 4 or more? Why?	No. If 4 or more were left, Ravi could give each friend one more, so the leftover is always 0, 1, 2 or 3.

81.5–86.5 min Exit ticket

REC *Pause and do the exit ticket on your own, without looking back. Then press play to check it.*

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. About 2 minutes of child work; then children hold their slips up

one at a time (never all together) and the answer is on the next click.

Exit ticket	Answer
A pen has hens and goats. There are 5 animals and 14 legs. How many hens and how many goats?	3 hens and 2 goats (6 + 8 = 14 legs).

86.5–88.5 min Looking back (2 minutes)

REC *Pause for about 2 minutes and fill in your looking-back page: what was hardest, and what will you try next time?*

SAY *"Looking back takes 2 minutes. Find the box called Looking back on the page "Before you go: Investigation and Looking back". Finish one line, two lines or all three, and tick what helped you. You may pass on any line. If you tick "I am still stuck", your tutor will check in during the session."*

DO Nothing is collected in front of the group and nobody has to read theirs out. The tutor may read a box while stamping journals (never by private message), and checks in on a child who ticked still stuck during the open session, for example by a quick visit to the shared breakout room or a hand signal with a parent nearby, for example "I saw you drew it when you got stuck". A child may volunteer one sentence, never by name unless they choose to. If a child ticks "I am still stuck", check in during the next open session (a quick visit to the shared breakout room or a hand signal, with a parent nearby). Praise the strategy, not the speed.

88.5–90 min Closing and stamps (1.5 minutes)

REC *Well done! Show your journal and workbook to a grown-up and tell them one thing you found out.*

SAY *"Today you drew arrays, made tables and checked both clues. Next time a puzzle has two kinds of things mixed up, start with all of one kind and swap one at a time."*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Find all the arrays (at least 2 rows, at least 2 in each row) for 16 counters. (3 ways: 2 rows of 8, 4 rows of 4, 8 rows of 2.)

Answer key (workbook)

Q1 (a) 4 × 5 = 20 and 5 × 4 = 20. (b) 20 ÷ 4 = 5 and 20 ÷ 5 = 4.

Q2 (a) 30 cups (3 rows of 10). (b) 18 cups. We can see 12 (10 in the top row and 1 more in each of the other 2 rows), and 30 − 12 = 18. (c) e.g. "3 rows of 10 is 30, then take away the 12 I can see."

Q3 4 hens and 3 goats. Table: 7 hens 0 goats 14 legs; 6, 1, 16; 5, 2, 18; 4, 3, 20. Each swap adds 2 legs. Check: 8 + 12 = 20 legs and 4 + 3 = 7 animals.

Q4 (a) 5 stickers each (4 × 5 = 20). (b) 3 left over. (c) 1 more sticker: 24 stickers give 6 each with none left over.

Q5 (a) 6 vans. 5 vans hold 25 children and the other 3 children need a van too. (b) 2 empty seats (6 vans hold 30, and 30 − 28 = 2).

Q6 4 ways: 2 rows of 10, 4 rows of 5, 5 rows of 4, 10 rows of 2. 3 rows does not work (3 rows of 6 is 18, 3 rows of 7 is 21). More than 10 rows would leave fewer than 2 chairs in a row.

Investigation (Where tables meet). (a) 10, 20, 30, 40, 50: they jump by 10 and are the 10 times table. (b) 3 and 5: 15, 30, 45 (jump by 15). (Extension: 4 and 5 give 20, 40, jump by 20.) (c) Yes to both, brute-forced up to 100: a number in the 10 table is 10 x something = 2 x (5 x something) = 5 x (2 x something). Child-level reason: a number in the 2 table is even and a number in the 5 table ends in 0 or 5, so a number in both ends in 0, which is the 10 table; and every number in the 10 table ends in 0, so it is even and in the 5 table. Look for: organised lists (writing both tables out and circling), the noticing that the jumps are 10, 15, 20, and a test of “always”.

Recording chapters and what a child watching later does

Time	Chapter marker	A child watching later
0–1.5 min	Opening	Gather your workbook, pencil, paper and the materials on the slide. Press play when ready.
1.5–6 min	Sneak-peek puzzles (4.5 minutes)	Pause. Try the puzzles on your own paper (about 4 and a half minutes). Guessing is fine. Press play for the answers.
6–10.5 min	Warm-up	Play along. Say or write each answer before the video does. Pause if you need longer.
10.5–23.5 min	Explore (in pairs, with objects)	Pause. Try the explore task with your objects, with someone at home if you can (about 13 minutes). Write down what you notice.
23.5–35.5 min	Mini-lesson (you teach, they join in)	Keep watching. Copy the worked examples. Rewind any step you missed.
35.5–38.5 min	Brain break	Stand up and join the brain break.
38.5–73.5 min	Worksheet (pairs, then alone, then stretch)	Pause. Do the workbook, starting with Start here on page 1 (about 35 minutes). Press play for each checking part and mark in a different colour.
73.5–76.5 min	Flex (spare time)	Pause or rewind whenever you need to.
76.5–81.5 min	Talk time	Pause. Explain your answer to someone at home, with a reason. Then play.
81.5–86.5 min	Exit ticket	Pause. Do the exit ticket on your own, then play to check it.

86.5–88.5 min	Looking back (2 minutes)	Pause. Fill in the looking-back page.
88.5–90 min	Closing and stamps (1.5 minutes)	Show your journal and workbook to a grown-up. Tell them one thing you found out.

Lesson 4: Pattern Machines

Week 2 (second lesson) • Patterns with Multiplication • Workbook: Lesson 4

Objective	Find far places in a repeating pattern by counting in groups (the ends of the groups are in a times table); count how many of one shape there are; describe a growing pattern with a table and a rule and use it to go forwards and backwards; find the numbers that two counting patterns share.
Thinking skills	Look for a pattern, make a table, make a list, explain a rule
Materials	Linking cubes in 3 colours (about 20 per pair), mini whiteboards, matchsticks or lolly sticks (about 25 per pair), counters.
Online set-up	Children use coloured pencils and squared paper (or Lego bricks) for the cube train, and draw sticks for the growing pattern. The tutor shares the slides and the children hold up their tables.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.
Watching the recording	A child who missed the live lesson watches the recording. They need: Linking cubes in 3 colours (about 20 per pair), mini whiteboards, matchsticks or lolly sticks (about 25 per pair), counters. The REC lines tell them when to pause; the chapter list at the end of this lesson shows each part.

0–1.5 min Opening

REC *Hello! Before you press play, get a pencil, paper, your workbook and the materials on the slide. Pause or rewind any time.*

SAY *"Hello, pattern machines! Today we find out what is far, far along a pattern, like the 41st shape, without drawing all 41."*

"The secret is groups. If we know the group, the times tables tell us where every group ends."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. The kit is on the desk from minute 0: ask everyone to show you the materials for today (a 30-second check; the materials list is above and on the title slide).

1.5–6 min Sneak-peek puzzles (4.5 minutes)

REC *Pause now and try the puzzles on your own paper for about 4 and a half minutes. Guessing is fine. Press play for the answers.*

SAY *"Two sneak-peek puzzles on ideas we have not learnt yet. Guessing is welcome, so is I do not know. Nobody is marked. Show your paper to me only."*

DO Show the Sneak-peek puzzles slide at minute 1.5 (4.5 minutes: about 45 seconds for question 1, 90 seconds for question 2, then the answers). Say the short line out loud (about 28 words): two puzzles on ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is

marked; show your paper to me only. Read each puzzle aloud; question 1 is the one to start with (objects, a drawing or counting).

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): Draw this pattern up to the 9th shape: circle, circle, square, circle, circle, square ... What is the 9th shape?	A square (the squares are the 3rd, 6th and 9th shapes). Listen for: Drawing it out is a fine method. Listen for "every 3rd one is a square".
Question 2 (thinking): A pattern repeats: star, moon, moon, sun, star, moon, moon, sun ... What is the 22nd shape? How do you know?	A moon. Listen for: Groups of 4: 20 is the end of a group (5 groups of 4), so the 21st is a star and the 22nd is a moon. Drawing 22 shapes is also fine.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

6–10.5 min Warm-up

REC *Join in! Say or write each answer before I do. Pause the video if you need longer.*

DO Pattern rhythm: clap, clap, stomp, clap, clap, stomp ... Keep it going. What do we do on beat 9? Beat 10? Beat 20?

SAY *“Clap, clap, stomp. Clap, clap, stomp. Keep it going with me!”*

“Freeze! What will we do on beat 9? On beat 10? How do you know?”

DO Keep the game to about two and a half minutes (6 to 8.5): stop it on the clock. Then the quick questions (8.5 to 10.5): ask questions 1 and 3 only (these are the two on the slide); questions 2, 4 and 5 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
4, 8, 12, __	16
50, 45, 40, __	35

3, 6, 9, 12, __	15
Count in 4s from 4. What is the 10th number?	40
2, 5, 8, 11, __	14

10.5–23.5 min Explore (in pairs, with objects)

REC *Pause now. Try this with your objects, and a grown-up if you can, for about 13 minutes. Press play when you have an answer.*

DO Timing (13 minutes): set up and predict, 10.5 to 12. One round of breakout rooms with an auto-timer: 60 seconds to get in, 5.5 minutes of pair work (13 to 18.5), 60 seconds to come back. Then the group debrief, 19.5 to 23.5.

SAY *"Our cube train goes red, red, blue, red, red, blue. What colour do you think cube 20 will be? Write your guess before you build."*

DO 1. Build a cube train: red, red, blue, red, red, blue ...

2. Stop after 9 cubes.

3. Predict: what colour is cube 20? Write it down.

4. Build to check. Which cubes are blue?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: put a finger on each blue cube. Say their numbers. What do you notice?

The blue cubes are at 3, 6, 9 ... Which table is that? Keep counting in 3s.

Finished early: what colour is cube 30? Cube 31? Explain without building.

DO Go further (a second what-if, so nobody waits): it is the box at the foot of the Explore slide; post it in the chat as well when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: what if the cube train is red, red, blue, green, red, red, blue, green ...? What colour is cube 20? What colour is cube 30? Answer: Cube 20 is green (the green cubes are 4, 8, 12, 16, 20). Cube 30 is red (28 is green, 29 and 30 are red).

DO At about minute 18.5, say "Pencils down" and bring the group back together (60 seconds). Take the three questions below, one at a time (about 4 minutes in all). Ask "How do you know?" on question 2 only; take about 15 seconds of talk on the others.

Ask	Expect
What colour is cube 20?	Red.
How did you know without building?	The blue cubes are at 3, 6, 9 ... 18, so 19 and 20 are red.
Which cubes are blue?	The 3 times table: 3, 6, 9, 12, 15, 18 ...

23.5–35.5 min Mini-lesson (you teach, they join in)

REC *Keep watching and copy the examples onto your paper. Missed a step? Rewind. You will try some yourself soon.*

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Sun, moon, star, sun, moon, star. The group is 3 shapes long. Every group ends with a star, at 3, 6, 9, 12: the 3 times table. Place 14? 12 is the end of a group, so 13 starts again with a sun, and 14 is a moon."*

"Where is the 5th star? There is one star at the end of each group, so the 5th star ends the 5th group: 5 × 3 = 15."

"Squares made of sticks: 4, 7, 10. Each new square needs only 3 more sticks because it shares a side. I make a table and keep adding 3."

Worked example on the slides	Answer
Far places: count in groups. Sun, moon, star, sun, moon, star … The group is 3 shapes. Groups end at 3, 6, 9, 12. What is in place 14?	A moon (12 ends a group; 13 sun, 14 moon).
Where is the 5th star?. One star in each group, at the end. Stars are at 3, 6, 9, 12, 15. Where is the 5th star? How many moons are in the first 21 shapes?	Place 15. 7 moons (21 shapes is 7 groups).

SAY *"The key idea today is: count in groups. Find the group size. The groups end at the times-table numbers. Jump to the end of a group, then count on. Say it with me."*

DO Show the key idea slide at about minutes 34.5 to 35.5. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Saying that a place in the times table (like 20 for a group of 5) is the FIRST shape of a group, when it is the LAST shape.
- Counting the whole pattern one by one to the far place instead of jumping to the end of a group.
- Growing patterns: multiplying by the step (10 squares × 3 = 30) and forgetting the start (the extra stick), instead of using the table or the rule.

35.5–38.5 min Brain break

REC *Movement break! Stand up and join in. The video keeps going.*

DO Pattern moves: jump, jump, squat, jump, jump, squat … 15 moves. What is move 15?

Water. Toilet break for anyone who needs it. Keep it to three minutes: two of movement, one for water.

38.5–71.5 min Worksheet (pairs, then alone, then stretch)

REC *Pause and do your workbook, starting at Start here on page 1, for about 33 minutes. Press play to check your work.*

DO Minute 38.5: give out the workbook. Read the title of Lesson 4 aloud. Say “Pairs first. Talk before you write.”

Minutes 38.5 to 52.5: pairs work together on the first page (the 2 minutes at the start are for finding the page and setting up; pairs stay in the main room). Minutes 52.5 to 55: check page 1 together (read only the answer line, children mark their own, one “How do you know?” for the page). Minutes 55 to 57.5: short teaching for page 2. Minutes 57.5 to 69: each child works alone on the second page. Minutes 69 to 71.5: check page 2 together (answer lines only, children mark their own). Think harder, Go further and the Investigation are open to any child at any time after question 3.

DO Take-home stars: Page 1: Q2(c) (30 letters, how many Ts) and Q3(b) (beads altogether). Page 2: Q4(d) (23 petals) and Q5(d) (why does it go up by 2). The two machine and growing-pattern teaching slides moved into the mini-lesson. Each is marked on the page with a ★ and the words take-home, stays Core (not Think harder), and keeps its answer on the check-together slide so a child can mark it at home. Do not spend class time on them. Late rule, printed here on purpose: at minute 57.5 (the start of page 2), if the clock is more than 2 minutes late, drop the last Core question of page 2 to take-home (children put a ★ beside it). Never shorten talk time, the exit ticket, looking back or the closing.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: "How long is one group? Where does each group end?"
- Ask: "Is that place the first shape of a group or the last?"
- Ask: "What is the rule? Does it work for every pair in your table?"

IF STUCK Hint 1: "Draw a line after each group. Count the groups."

Hint 2: "Find the nearest times-table number before your place. That shape ends a group."

Hint 3: "Make a table and look at how much it goes up each time."

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Look at the shape pattern in the Start here page again (places 1 to 10). How many circles are there in the first 30 shapes? How do you know?	12 circles (30 shapes is 6 groups of 5, with 2 circles in each group).

Workbook Think harder 2: Beads go red, red, blue, green, red, red, blue, green … Mei says the 30th bead is blue. Is she right? Which places up to 30 have a green bead?	Mei is not right. The pattern repeats every 4 beads and 28 ends a group (4 × 7 = 28), so the 29th bead is red and the 30th bead is red. Green beads are the 4th bead of each group: places 4, 8, 12, 16, 20, 24 and 28.
Workbook Think harder 3: Beads go red, yellow, yellow, red, yellow, yellow … How many yellow beads are in the first 18 beads?	12 yellow beads (18 beads is 6 groups of 3, with 2 yellow in each).
Workbook Think harder 4: Houses are made of sticks in a row. 1 house uses 5 sticks, 2 houses use 9 sticks, 3 houses use 13 sticks. How many sticks are needed for 6 houses?	25 sticks. The rule is add 4: 5, 9, 13, 17, 21, 25.
Workbook Go further 5: Ali counts in 3s from 3. Mei counts in 4s from 4. They both say 12. Predict the next number they will BOTH say. Then list every number up to 100 that they both say. What do you notice? Why?	24 is next. They both say 12, 24, 36, 48, 60, 72, 84 and 96 (8 numbers): the numbers go up by 12 each time, because 12 is the first number in both the 3 and 4 times tables, and each next one is 12 more.
Workbook Go further 6: Siti builds towers of cubes. Tower 1 has 2 cubes, tower 2 has 5 cubes, tower 3 has 8 cubes. The towers keep growing in the same way. Which tower has 29 cubes? Is there a tower with exactly 50 cubes? With exactly 100 cubes? Explain how you know.	Tower 10 has 29 cubes (the rule is add 3: 2, 5, 8, 11, … 29). Tower 17 has 50 cubes. No tower has 100: every tower has 1 less than a number in the 3 times table (tower 10: 30 − 1 = 29), and 101 is not in the 3 times table (99 and 102 are).
Workbook Go further 7: Frog A starts at 0 and jumps 4 each time: 4, 8, 12 … Frog B starts at 2 and jumps 3 each time: 5, 8, 11 … Which numbers up to 30 do both frogs land on?	8 and 20. Frog A: 4, 8, 12, 16, 20, 24, 28. Frog B: 5, 8, 11, 14, 17, 20, 23, 26, 29.

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (on the “Before you go” page): Three counting patterns. Pattern A: 1, 4, 7, 10 … (add 3 each time). Pattern B: 2, 5, 8, 11 … Pattern C: 3, 6, 9, 12 … (a) Write each pattern up to 30. (b) Pick any number from 1 to 30. Which pattern is it in? Can a number be in two patterns? Can a number be in none? (c) Why does this happen? Easy first step: write pattern A, 1, 4, 7, 10 …, up to 30.	(a) A: 1, 4, 7, 10, 13, 16, 19, 22, 25, 28. B: 2, 5, 8, 11, 14, 17, 20, 23, 26, 29. C: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. (b) Every number is in exactly ONE pattern (brute-forced for 1 to 99). (c) Counting in 3s, a number is either on the 3s (pattern C), 1 more than a number on the 3s (A), or 2 more (B); there is no other choice. Extension (not printed): if each pattern added 4 you would need 4 patterns, starting at 1, 2, 3 and 4. Look for: a complete listing, the claim “exactly one” tested on more than one number, and a reason that uses the 3 times table.

71.5–76.5 min Flex (spare time)

REC *Pause or rewind whenever you need to.*

DO Spare time, kept on purpose (5 minutes). Use it to absorb a late start, a camera problem or a slow page 2. If you are on time, spend it on the Investigation share or a second talk-time volunteer, or an early start on talk time. Do not add new content. Never take time from the exit ticket, looking back or the stamp.

76.5–81.5 min Talk time

REC *Pause. Tell a grown-up your answer to the talk question, and why you think so. Then press play.*

DO Bring the group back to a circle (or camera on, mics on). Pairs talk for 90 seconds, then two volunteers share (about 5 minutes in all). One question only. You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *"A pattern repeats every 5 shapes. Is the 10th shape always the same as the 20th?"*

Discussion question	Listen for
A pattern repeats every 5 shapes. Is the 10th shape always the same as the 20th? What if the pattern repeats every 3 shapes?	Yes: 10 and 20 are both in the 5 times table, so both are the last shape of a group. (If it repeated every 3 shapes the answer would be no: 9 ends a group, so 10 is the 1st shape; 18 ends a group, so 20 is the 2nd shape.)

81.5–86.5 min Exit ticket

REC *Pause and do the exit ticket on your own, without looking back. Then press play to check it.*

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. About 2 minutes of child work; then children hold their slips up one at a time (never all together) and the answer is on the next click.

Exit ticket	Answer
The pattern goes triangle, circle, square, triangle, circle, square … What is the 17th shape? How do you know?	A circle. 15 ends a group (5 groups of 3), so 16 is a triangle and 17 is a circle.

86.5–88.5 min Looking back (2 minutes)

REC *Pause for about 2 minutes and fill in your looking-back page: what was hardest, and what will you try next time?*

SAY *"Looking back takes 2 minutes. Find the box called Looking back on the page "Before you go: Investigation and Looking back". Finish one line, two lines or all three, and tick what helped you. You may pass on any line. If you tick "I am still stuck", your tutor will check in during the session."*

DO Nothing is collected in front of the group and nobody has to read theirs out. The tutor may read a box while stamping journals (never by private message), and checks in on a child who ticked still stuck during the open session, for example by a quick visit to the shared breakout room or a hand signal with a parent nearby, for example "I saw you drew it when you got stuck". A child may volunteer one sentence, never by name unless they choose to. If a child ticks "I am still stuck", check in during the next open session (a quick visit to the shared breakout room or a hand signal, with a parent nearby). Praise the strategy, not the speed.

88.5–90 min Closing and stamps (1.5 minutes)

REC *Well done! Show your journal and workbook to a grown-up and tell them one thing you found out.*

SAY *"You did not draw 41 shapes today. You found the group, used a times table and counted on. That is how pattern machines think."*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Make your own repeating pattern of 4 shapes. Write 3 "far place" questions about it for a partner, with the answers.

Answer key (workbook)

Q1 (a) Square (20 ends a group of 5). (b) Circle (21 triangle, 22 circle, 23 circle). (c) Triangle: 40 ends a group ($5 \times 8 = 40$), so 41 starts again.

Q2 (a) A (16 ends a group of 4; 17 is M, 18 is A). (b) Place 40 (every H ends a group: 4, 8, 12 ... the 10th is 4×10). (c) 7 Ts. They are in places 3, 7, 11, 15, 19, 23, 27; 28 ends the 7th group, and places 29 and 30 are M and A.

Q3 (a) 18 yellow beads (12 red is 6 groups of 2 red; each group has 3 yellow). (b) 30 beads (6 groups of 5).

Q4 (a) 5, 10, 15, 20, 25, 30. (b) 50 petals. (c) 7 flowers. (d) 4 flowers, with 3 petals left over.

Q5 (a) 4, 6, 8, 10, 12, 14. (b) 22 children. (c) 9 tables. (d) Each new table adds 1 seat on the top and 1 on the bottom; the 2 seats at the ends stay the same.

Q6 Ali (12 numbers, ending at 36): 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36. Mei (9 numbers, ending at 36): 4, 8, 12, 16, 20, 24, 28, 32, 36. Both: 12, 24, 36. They go up by 12 each time.

Investigation (Three counting patterns). (a) A: 1, 4, 7, 10, 13, 16, 19, 22, 25, 28. B: 2, 5, 8, 11, 14, 17, 20, 23, 26, 29. C: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. (b) Every number is in exactly ONE pattern (brute-forced for 1 to 99). (c) Counting in 3s, a number is either on the 3s (pattern C), 1 more than a number on the 3s (A), or 2 more (B); there is no other choice. Extension (not printed): if each pattern added 4 you would need 4 patterns, starting at 1, 2, 3 and 4. Look for: a complete listing, the claim "exactly one" tested on more than one number, and a reason that uses the 3 times table.

Recording chapters and what a child watching later does

Time	Chapter marker	A child watching later
0–1.5 min	Opening	Gather your workbook, pencil, paper and the materials on the slide. Press play when ready.
1.5–6 min	Sneak-peek puzzles (4.5 minutes)	Pause. Try the puzzles on your own paper (about 4 and a half minutes). Guessing is fine. Press play for the answers.
6–10.5 min	Warm-up	Play along. Say or write each answer before the video does. Pause if you need longer.
10.5–23.5 min	Explore (in pairs, with objects)	Pause. Try the explore task with your objects, with someone at home if you can (about 13 minutes). Write down what you notice.
23.5–35.5 min	Mini-lesson (you teach, they join in)	Keep watching. Copy the worked examples. Rewind any step you missed.
35.5–38.5 min	Brain break	Stand up and join the brain break.
38.5–71.5 min	Worksheet (pairs, then alone, then stretch)	Pause. Do the workbook, starting with Start here on page 1 (about 33 minutes). Press play for each checking part and mark in a different colour.
71.5–76.5 min	Flex (spare time)	Pause or rewind whenever you need to.
76.5–81.5 min	Talk time	Pause. Explain your answer to someone at home, with a reason. Then play.
81.5–86.5 min	Exit ticket	Pause. Do the exit ticket on your own, then play to check it.
86.5–88.5 min	Looking back (2 minutes)	Pause. Fill in the looking-back page.
88.5–90 min	Closing and stamps (1.5 minutes)	Show your journal and workbook to a grown-up. Tell them one thing you found out.

Lesson 5: Mystery Number Machines

Week 3 (first lesson) • Working Backwards • Workbook: Lesson 5

Objective	Find a mystery number in two-step puzzles with +, −, × and ÷ by working backwards (undo the last step first with the opposite operation), check by going forwards, and use before-and-after bar models for stories where an amount changes.
Thinking skills	Inverse operations, working backwards, guess and check, before-and-after bar models, checking forwards
Materials	Mini whiteboards, counters, paper cups, number machine cards (× 2, × 3, × 4, × 5, × 10 and + or − cards from 1 to 20), paper strips for bars.
Online set-up	Machine cards shown on the slide; children write on whiteboards and hold them up. For bars, children draw on paper or fold paper strips into equal parts.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.
Watching the recording	A child who missed the live lesson watches the recording. They need: Mini whiteboards, counters, paper cups, number machine cards (× 2, × 3, × 4, × 5, × 10 and + or − cards from 1 to 20), paper strips for bars. The REC lines tell them when to pause; the chapter list at the end of this lesson shows each part.

0–1.5 min Opening

REC *Hello! Before you press play, get a pencil, paper, your workbook and the materials on the slide. Pause or rewind any time.*

SAY *"Good morning, inquirers! Today you are code-breakers. A number went into a secret machine and something came out. Your job is to run the machine backwards and find the number that went in."*

"Remember our rule: notice it, test it, explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud. The kit is on the desk from minute 0: ask everyone to show you the materials for today (a 30-second check; the materials list is above and on the title slide).

1.5–6 min Sneak-peek puzzles (4.5 minutes)

REC *Pause now and try the puzzles on your own paper for about 4 and a half minutes. Guessing is fine. Press play for the answers.*

SAY *"Two sneak-peek puzzles on ideas we have not learnt yet. Guessing is welcome, so is I do not know. Nobody is marked. Show your paper to me only."*

DO Show the Sneak-peek puzzles slide at minute 1.5 (4.5 minutes: about 45 seconds for question 1, 90 seconds for question 2, then the answers). Say the short line out loud (about 28 words): two puzzles on ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is

marked; show your paper to me only. Read each puzzle aloud; question 1 is the one to start with (objects, a drawing or counting).

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): A machine adds 7. The OUT is 12. What was the IN? (Count back with counters if you like.)	5 (5 + 7 = 12). Listen for: IN is what goes in and OUT is what comes out. Undo = do the opposite: 12 - 7. This is a first win for everyone.
Question 2 (thinking): A machine doubles, then adds 7. The OUT is 17. What was the IN? How do you know?	5 (5 doubled is 10, and 10 + 7 = 17). Listen for: Undo the last step first: 17 - 7 = 10, then half of 10 is 5. A guess that was tested (5 x 2 + 7 = 17) is just as good.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

6–10.5 min Warm-up

REC *Join in! Say or write each answer before I do. Pause the video if you need longer.*

DO Guess my machine, three rounds. The tutor is a secret one-step machine. Two rounds: call out an IN number, hear the OUT, and guess the rule. One round: the tutor gives only an OUT: what went IN?

SAY *“I am a machine. Give me a number and I will tell you what comes out. Ready? You say 4, I say 12. You say 10, I say 30. What am I doing?”*

“Now I only tell you what comes out: 18. What went in? How did you work it out?”

DO Keep the game to about two and a half minutes (6 to 8.5): stop it on the clock. Then the quick questions (8.5 to 10.5): ask questions 1 and 3 only (these are the two on the slide); questions 2, 4 and 5 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect

I think of a number. I multiply it by 2. I get 18. What is it?	9
I think of a number. I divide it by 5. I get 4. What is it?	20
I think of a number. I take away 25. I get 40. What is it?	65
What is the opposite of × 4?	÷ 4
I think of a number. I multiply it by 3, then add 4. I get 19. What is it?	5

10.5–23.5 min Explore (in pairs, with objects)

REC *Pause now. Try this with your objects, and a grown-up if you can, for about 13 minutes. Press play when you have an answer.*

DO Timing (13 minutes): set up and predict, 10.5 to 12. One round of breakout rooms with an auto-timer: 60 seconds to get in, 5.5 minutes of pair work (13 to 18.5), 60 seconds to come back. Then the group debrief, 19.5 to 23.5.

SAY *"Here are two machine cards: × 2 and + 6. If I put 4 in, what comes out? Now I swap the cards: + 6 first, then × 2. Will the same number come out? Thumbs up or down before we test it."*

DO 1. Pick one × card and one + or − card. Put them in a row: that is your machine.

2. Secretly choose a start number. Run it through both cards. Say only the OUT.

3. Your partner works backwards to find your start number. Check it together.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: write the machine as boxes and arrows, then put your partner's OUT number in the last box.

Mixed up: "What was the LAST thing the machine did? Undo that first."

Finished early: find an OUT number your partner can NOT undo with whole numbers. Why does it not work?

DO Go further (a second what-if, so nobody waits): it is the box at the foot of the Explore slide; post it in the chat as well when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: what if the machine adds 7 FIRST and then doubles? Put IN 5 into it. Is the OUT the same as before (17)? Answer: No: 5 + 7 = 12, doubled is 24. The order of the steps matters.

DO At about minute 18.5, say "Pencils down" and bring the group back together (60 seconds). Take the three questions below, one at a time (about 4 minutes in all). Ask "How do you know?" on question 2 only; take about 15 seconds of talk on the others.

Ask	Expect

How did you find your partner's start number?	Undo the last card first with its opposite, then undo the first card.
What is the opposite of × 3? Of − 8?	÷ 3. + 8.

23.5–35.5 min Mini-lesson (you teach, they join in)

REC *Keep watching and copy the examples onto your paper. Missed a step? Rewind. You will try some yourself soon.*

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Mystery number × 3, then + 4, gives 25. I go backwards. The last step was + 4, so I undo it first with − 4: 25 − 4 = 21. The step before was × 3, so I undo it with ÷ 3: 21 ÷ 3 = 7. Check forwards: 7 × 3 = 21, 21 + 4 = 25. ✓"*

"Ben gave away 18 stamps, then bought 25, and now has 60. I draw the bars and start at the end. Before he bought 25 he had 60 − 25 = 35. Before he gave away 18 he had 35 + 18 = 53. Check: 53 − 18 = 35, 35 + 25 = 60. ✓"

"Mei gave away half her beads, then used 5. She has 9 left. The half she kept is 5 + 9 = 14. The other half is also 14. So she had 14 + 14 = 28."

Worked example on the slides	Answer
Two steps: undo the last step first. ? → × 3 → + 4 → 25 Undo + 4: 25 − 4 = 21. Undo × 3: 21 ÷ 3 = ? Check forwards.	7 (7 × 3 = 21, 21 + 4 = 25).
Before and after: draw the bars. Ben had some stamps. He gave 18 to Mei. Then he bought 25 more. Now he has 60. Before he bought 25: 60 − 25 = 35. Before he gave 18: 35 + 18 = ?	53 stamps (check 53 − 18 = 35, 35 + 25 = 60).

SAY *"The key idea today is: undo it backwards. Start at the end. Undo the LAST step first with its opposite, then the step before. Check by going forwards. Say it with me."*

DO Show the key idea slide at about minutes 34.5 to 35.5. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Doing the same step again instead of the opposite (27 × 2 instead of 27 ÷ 2).
- Undoing the steps in the wrong order: the first step first instead of the last step first.
- Bar stories: stopping at one part (one bag or one half) instead of finding the whole, or forgetting to undo one of the changes.

35.5–38.5 min Brain break

REC *Movement break! Stand up and join in. The video keeps going.*

DO Machine moves. The tutor calls a machine (“× 2 jumps!”, “+ 3 claps!”) on a number on the board, and everyone does the OUT number of moves.

Water. Toilet break for anyone who needs it. Keep it to three minutes: two of movement, one for water.

38.5–73.5 min Worksheet (pairs, then alone, then stretch)

REC *Pause and do your workbook, starting at Start here on page 1, for about 35 minutes. Press play to check your work.*

DO Minute 38.5: give out the workbook. Read the title of Lesson 5 aloud. Say “Pairs first. Talk before you write.”

Minutes 38.5 to 41: a short teach slide first (this lesson). Minutes 41 to 52.5: pairs work together on the first page (the 2 minutes at the start are for finding the page and setting up; pairs stay in the main room). Minutes 52.5 to 55: check page 1 together (read only the answer line, children mark their own, one “How do you know?” for the page). Minutes 55 to 59.5: short teaching for page 2. Minutes 59.5 to 71: each child works alone on the second page. Minutes 71 to 73.5: check page 2 together (answer lines only, children mark their own). Think harder, Go further and the Investigation are open to any child at any time after question 3.

DO Take-home stars: Page 1: Q1(d) (× 4, then take away 15). No other Core question moves in this lesson. Each is marked on the page with a ★ and the words take-home, stays Core (not Think harder), and keeps its answer on the check-together slide so a child can mark it at home. Do not spend class time on them. Late rule, printed here on purpose: at minute 59.5 (the start of page 2), if the clock is more than 2 minutes late, drop the last Core question of page 2 to take-home (children put a ★ beside it). Never shorten talk time, the exit ticket, looking back or the closing.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “What was the last step? What is its opposite?”
- Ask: “Show me your check going forwards. Do you get the OUT number?”
- Ask: “Point to your bar. Which part do you know? Which part is the mystery?”

IF STUCK Hint 1: “Draw the machine boxes. Write the OUT in the last box.”

Hint 2: “Cover the first step with your finger. Undo only the last step.”

Hint 3: “For bars: start from what is left now and put back each change, from the end to the start.”

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: A machine multiplies by a number, then adds a number. IN 2 gives OUT 11 and IN 5 gives OUT 20. Find what the machine does (try × 2, × 3, × 4 … with IN 2, then test IN 5). Then: what OUT does IN 10 give? Which IN gives OUT 26?	× 3, then + 5. IN 10 gives OUT 35. OUT 26 comes from IN 7 (26 − 5 = 21, 21 ÷ 3 = 7).
Workbook Think harder 2: Ben had some money. He spent $15 on a book. Then his dad gave him as much as he had left. Now he has $42. How much did Ben have at first? Draw bars.	$36. Undo the last step: dad gave him as much as he had left, so $42 is two equal bars of $21. So Ben had $21 after the book, and $21 + $15 = $36 at first. Check: 36 − 15 = 21, 21 + 21 = 42.
Workbook Think harder 3: Siti had some beads. She gave half of them to Mei. Then she bought 12 more. Now she has 30 beads. How many beads did Siti have at first?	36 (30 − 12 = 18 is half, 18 + 18 = 36)
Workbook Think harder 4: Ali, Ben and Siti have 40 marbles altogether. Ben has 5 more marbles than Ali. Siti has 8 more marbles than Ali. How many marbles does each child have? Draw a bar for each child.	Ali 9, Ben 14, Siti 17 (40 − 5 − 8 = 27, 27 ÷ 3 = 9; check 9 + 14 + 17 = 40)
Workbook Go further 5: Siti had some sweets. She gave Ben 1 more than half of them. Now she has 7 sweets left. How many sweets did she have at first?	16 (7 + 1 = 8 is half, 8 + 8 = 16; she gave Ben 9)
Workbook Go further 6: A machine does × 2, then − 1. Mei puts a number through the machine 3 times in a row (each OUT goes back in). The final OUT is 17. What number did she start with?	3 (17 + 1 = 18, 18 ÷ 2 = 9; 9 + 1 = 10, 10 ÷ 2 = 5; 5 + 1 = 6, 6 ÷ 2 = 3; check 3 → 5 → 9 → 17)
Workbook Go further 7: Three boxes have 30 pens altogether. 5 pens are moved from box A to box B. Then 3 pens are moved from box B to box C. Now all three boxes have the same number of pens. How many pens were in each box at first?	A 15, B 8, C 7 (each ends with 10; undo the moves from the end)

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for

Workbook Investigation (on the “Before you go” page): A magic trick. Think of a number. Add 5. Double it. Take away 10. Halve it. (a) Try 5 different starting numbers. What do you notice about your final answer? (b) Is it ALWAYS true? Try a big number and a small number, then say why. (c) Make your own trick that always gives you back the number you started with. Test it on 3 numbers. Easy first step: do the trick with 3, then with 10, and write down each step.	(a) The final answer is always the number you started with (brute-forced for 1 to 999). (b) Adding 5 then doubling adds 10 and doubles the start; taking away 10 and halving undoes both: (n + 5) x 2 − 10 = n x 2, and half of that is n. A child reason: “double and halve undo each other, and add 5 doubled is 10, which I take away”. (c) Any valid trick, for example: add 3, double, take away 6, halve; or times 4, add 8, divide by 4, take away 2. Check by testing three numbers. Look for: testing small and big numbers, a reason in words, and a trick that really works.

73.5–76.5 min Flex (spare time)

REC *Pause or rewind whenever you need to.*

DO Spare time, kept on purpose (3 minutes). Use it to absorb a late start, a camera problem or a slow page 2. If you are on time, spend it on the Investigation share or a second talk-time volunteer, or an early start on talk time. Do not add new content. Never take time from the exit ticket, looking back or the stamp.

76.5–81.5 min Talk time

REC *Pause. Tell a grown-up your answer to the talk question, and why you think so. Then press play.*

DO Bring the group back to a circle (or camera on, mics on). Pairs talk for 90 seconds, then two volunteers share (about 5 minutes in all). One question only. You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *“To undo × 2, then + 6, with OUT 20, Mei takes away 6 first. Ali divides by 2 first. Who is right? Test both.”*

Discussion question	Listen for
To undo “× 2, then + 6” with OUT 20, Mei takes away 6 first. Ali divides by 2 first. Who is right?	Mei is right: 20 − 6 = 14, 14 ÷ 2 = 7, and 7 × 2 + 6 = 20. Ali gets 20 ÷ 2 = 10, 10 − 6 = 4, but 4 × 2 + 6 = 14, not 20. The last step happened last, so it must be undone first, like taking off shoes before socks.

81.5–86.5 min Exit ticket

REC *Pause and do the exit ticket on your own, without looking back. Then press play to check it.*

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. About 2 minutes of child work; then children hold their slips up one at a time (never all together) and the answer is on the next click.

Exit ticket	Answer
I think of a number. I multiply it by 4, then take away 6. I get 14. What is my number?	5 (14 + 6 = 20, 20 ÷ 4 = 5; check 5 × 4 = 20, 20 − 6 = 14)

86.5–88.5 min Looking back (2 minutes)

REC *Pause for about 2 minutes and fill in your looking-back page: what was hardest, and what will you try next time?*

SAY *"Looking back takes 2 minutes. Find the box called Looking back on the page "Before you go: Investigation and Looking back". Finish one line, two lines or all three, and tick what helped you. You may pass on any line. If you tick "I am still stuck", your tutor will check in during the session."*

DO Nothing is collected in front of the group and nobody has to read theirs out. The tutor may read a box while stamping journals (never by private message), and checks in on a child who ticked still stuck during the open session, for example by a quick visit to the shared breakout room or a hand signal with a parent nearby, for example "I saw you drew it when you got stuck". A child may volunteer one sentence, never by name unless they choose to. If a child ticks "I am still stuck", check in during the next open session (a quick visit to the shared breakout room or a hand signal, with a parent nearby). Praise the strategy, not the speed.

88.5–90 min Closing and stamps (1.5 minutes)

REC *Well done! Show your journal and workbook to a grown-up and tell them one thing you found out.*

SAY *"Today you ran machines backwards and drew before-and-after bars. Code-breakers always check forwards, and you did. Well done, inquirers!"*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Make a three-step machine with a × card, a ÷ card and a + card. Give a partner only the OUT. Can they find the IN? Which OUT numbers make your machine impossible to undo?

Answer key (workbook)

Q1 (a) 10 (b) 16 (c) 24 (d) 9 (undo the last step first: e.g. (a) 27 − 7 = 20, 20 ÷ 2 = 10)

Q2 (a) IN 4 → OUT 14; IN 10 → OUT 32; OUT 17 ← IN 5; OUT 29 ← IN 9. (b) IN 3, 5, 10, 20 give OUT 7, 11, 21, 41.

Q3 59 stickers (50 − 15 = 35, 35 + 24 = 59; check 59 − 24 = 35, 35 + 15 = 50)

Q4 30 sweets (one bag: 6 + 4 = 10; 3 bags: 10 × 3 = 30)

Q5 (a) Mei 18 (b) Ali 14 (c) No: the order of the steps matters (d) 7 (18 − 4 = 14, 14 ÷ 2 = 7)

Investigation (A magic trick). (a) The final answer is always the number you started with (brute-forced for 1 to 999). (b) Adding 5 then doubling adds 10 and doubles the start; taking away 10 and halving

undoes both: (n + 5) x 2 − 10 = n x 2, and half of that is n. A child reason: “double and halve undo each other, and add 5 doubled is 10, which I take away”. (c) Any valid trick, for example: add 3, double, take away 6, halve; or times 4, add 8, divide by 4, take away 2. Check by testing three numbers. Look for: testing small and big numbers, a reason in words, and a trick that really works.

Recording chapters and what a child watching later does

Time	Chapter marker	A child watching later
0–1.5 min	Opening	Gather your workbook, pencil, paper and the materials on the slide. Press play when ready.
1.5–6 min	Sneak-peek puzzles (4.5 minutes)	Pause. Try the puzzles on your own paper (about 4 and a half minutes). Guessing is fine. Press play for the answers.
6–10.5 min	Warm-up	Play along. Say or write each answer before the video does. Pause if you need longer.
10.5–23.5 min	Explore (in pairs, with objects)	Pause. Try the explore task with your objects, with someone at home if you can (about 13 minutes). Write down what you notice.
23.5–35.5 min	Mini-lesson (you teach, they join in)	Keep watching. Copy the worked examples. Rewind any step you missed.
35.5–38.5 min	Brain break	Stand up and join the brain break.
38.5–73.5 min	Worksheet (pairs, then alone, then stretch)	Pause. Do the workbook, starting with Start here on page 1 (about 35 minutes). Press play for each checking part and mark in a different colour.
73.5–76.5 min	Flex (spare time)	Pause or rewind whenever you need to.
76.5–81.5 min	Talk time	Pause. Explain your answer to someone at home, with a reason. Then play.
81.5–86.5 min	Exit ticket	Pause. Do the exit ticket on your own, then play to check it.
86.5–88.5 min	Looking back (2 minutes)	Pause. Fill in the looking-back page.
88.5–90 min	Closing and stamps (1.5 minutes)	Show your journal and workbook to a grown-up. Tell them one thing you found out.

Lesson 6: Market Day

Week 3 (second lesson) • Money • Workbook: Lesson 6

Objective	Read and write money in dollars and cents, pay exactly with the fewest notes and coins, make a list in order to find every way to pay, give change by counting on, decide whether there is enough money, and work backwards to find how much someone had at first.
Thinking skills	Systematic listing (make a list in order), working backwards, counting on, comparing, justifying "there are no more ways"
Materials	Play money: coins 5¢, 10¢, 20¢, 50¢, $1 and notes $2, $5, $10, $50 (about 6 of each per pair), price tags, mini whiteboards.
Online set-up	Cut out the play coins and notes from the workbook before the lesson (kit in hand at minute 0), or use real money at home. No play money? Draw each coin and note on paper. The shop and the toy stall are on the slides; children show their money to the camera.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.
Watching the recording	A child who missed the live lesson watches the recording. They need: Play money: coins 5¢, 10¢, 20¢, 50¢, $1 and notes $2, $5, $10, $50 (about 6 of each per pair), price tags, mini whiteboards. The REC lines tell them when to pause; the chapter list at the end of this lesson shows each part.

0–1.5 min Opening

REC *Hello! Before you press play, get a pencil, paper, your workbook and the materials on the slide. Pause or rewind any time.*

SAY *"Welcome to Market Day, inquirers! Today you are shoppers AND shopkeepers. You will pay exactly, give change, and find every way to pay without missing one."*

"Notice it, test it with play money, and explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud. The kit is on the desk from minute 0: ask everyone to show you the materials for today (a 30-second check; the materials list is above and on the title slide).

1.5–6 min Sneak-peek puzzles (4.5 minutes)

REC *Pause now and try the puzzles on your own paper for about 4 and a half minutes. Guessing is fine. Press play for the answers.*

SAY *"Two sneak-peek puzzles on ideas we have not learnt yet. Guessing is welcome, so is I do not know. Nobody is marked. Show your paper to me only."*

DO Show the Sneak-peek puzzles slide at minute 1.5 (4.5 minutes: about 45 seconds for question 1, 90 seconds for question 2, then the answers). Say the short line out loud (about 28 words): two puzzles on ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is

marked; show your paper to me only. Read each puzzle aloud; question 1 is the one to start with (objects, a drawing or counting).

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): Mei has one 50¢ coin, one 20¢ coin and one 10¢ coin. How much money is that? Use play coins or draw them.	80¢ (50 + 20 + 10). Listen for: Counting on from the biggest coin. This is a first win for everyone.
Question 2 (thinking): Mei has $5. Three drinks cost $1.60 each. Does she have enough? How much money is left?	Yes. 3 drinks cost $4.80, so 20¢ is left. Listen for: Listen for adding $1.60 three times, or $1 + $1 + $1 and then 60¢ three times (180¢ = $1.80). Counting up from $4.80 to $5 is a good method.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

6–10.5 min Warm-up

REC *Join in! Say or write each answer before I do. Pause the video if you need longer.*

DO Market Day shop, two price tags: the eraser (85¢) and the scooter ($57.20). Pay exactly with play notes and coins. Who can pay with the fewest pieces? (Spare tags: pencil $1.35, sticker book $3.45, water bottle $6.50, toy car $12.70.)

SAY *“Here is our market. The sticker book costs $3.45. That means 3 dollars and 45 cents. Pay exactly with your play money.”*

“Now count your pieces. Who used the fewest? How did you choose?”

DO Keep the game to about two and a half minutes (6 to 8.5): stop it on the clock. Then the quick questions (8.5 to 10.5): ask questions 1 and 3 only (these are the two on the slide); questions 2, 4 and 5 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect

Write 345¢ in dollars and cents.	$3.45
How many cents is $2.05?	205¢
Which is more: $1.50 or 105¢?	$1.50 (150¢)
Pay 75¢ with the fewest coins.	50¢ + 20¢ + 5¢
You have 60¢. A sticker costs 85¢. How much more do you need?	25¢

10.5–23.5 min Explore (in pairs, with objects)

REC *Pause now. Try this with your objects, and a grown-up if you can, for about 13 minutes. Press play when you have an answer.*

DO Timing (13 minutes): set up and predict, 10.5 to 12. One round of breakout rooms with an auto-timer: 60 seconds to get in, 5.5 minutes of pair work (13 to 18.5), 60 seconds to come back. Then the group debrief, 19.5 to 23.5.

SAY *"The toy stall has a kite, a ball, a yo-yo, bubbles and a sticker. You have $10. How many different baskets do you think cost EXACTLY $10? Write a guess."*

DO 1. You have $10. Look at the toy stall: kite $6, ball $4, yo-yo $3, bubbles $2, sticker $1.

2. Find every basket that costs EXACTLY $10 (at most one of each toy).

3. Make a list in order so you do not miss one or count one twice.

4. Say how you know there are no more.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: start with the most expensive toy. Put it in the basket. How much is left to spend?

Repeating baskets: "Kite and ball is the same basket as ball and kite."

Finished early: which basket gives you the most toys for $10? Which gives exactly $1 change?

DO Go further (a second what-if, so nobody waits): it is the box at the foot of the Explore slide; post it in the chat as well when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: what if the toy stall also sells a robot for $5? Now how many different baskets cost exactly $10? (At most one of each toy.) Answer: 5 baskets: kite + ball; kite + yo-yo + sticker; ball + sticker + robot; yo-yo + bubbles + robot; ball + yo-yo + bubbles + sticker.

DO At about minute 18.5, say "Pencils down" and bring the group back together (60 seconds). Take the three questions below, one at a time (about 4 minutes in all). Ask "How do you know?" on question 2 only; take about 15 seconds of talk on the others.

Ask	Expect
How many baskets cost exactly $10?	3: kite + ball; kite + yo-yo + sticker; ball + yo-yo + bubbles + sticker.

How do you know you have them all?	Start with the kite in the basket, then without the kite, and list in order.
You buy the kite and the bubbles. How much change?	$10 – $8 = $2.

23.5–35.5 min Mini-lesson (you teach, they join in)

REC *Keep watching and copy the examples onto your paper. Missed a step? Rewind. You will try some yourself soon.*

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Ways to pay $6 with $5 notes, $2 notes and $1 coins. I start with the biggest note: one $5. $1 is left, so one $1 coin. Now no $5: three $2, then two $2 and two $1, one $2 and four $1, no $2 and six $1. The $2 column went 3, 2, 1, 0, so I know I am finished: 5 ways."*

"Mei spent 80¢ on a drink, then 65¢ on a bun, and has 55¢ left. I work backwards from the end. Before the bun she had 55 + 65 = 120¢. Before the drink she had 120 + 80 = 200¢. That is $2."

"Change from $1 for 45¢: I count on like a shopkeeper. 45¢ and 5¢ is 50¢. 50¢ and 50¢ is $1. So the change is 5¢ + 50¢ = 55¢."

Worked example on the slides	Answer
Make a list in order: pay $6 with $5, $2 and $1. Start with the biggest note: one $5, then $1. No $5: three $2, two $2, one $2, no $2. Fill up with $1 coins.	5 ways.
Work backwards with money. Mei bought a drink for 80¢. Then she bought a bun for 65¢. She has 55¢ left. Before the bun: 55¢ + 65¢ = 120¢. Before the drink: 120¢ + 80¢ = ?	200¢ = $2.

SAY *"The key idea today is: in order, and backwards. To find every way, make a list in order: the most of the biggest note first, then one fewer each time. To find what someone had at first, work backwards and put back what was spent. Say it with me."*

DO Show the key idea slide at about minutes 34.5 to 35.5. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Mixing up dollars and cents: writing 345¢ as $34.5, or reading $3.05 as $3.50.
- Using a $20 note (there is no $20 note), or not starting with the biggest note that fits when finding the fewest.
- Listing ways with no order, so some ways are missed or the same way is written twice.

35.5–38.5 min Brain break

REC *Movement break! Stand up and join in. The video keeps going.*

DO Shopkeeper stretch. The tutor calls an amount: reach up high for dollars, touch your toes for cents. “$3.50!” “75¢!” “205¢!” (that is $2.05: reach up!).

Water. Toilet break for anyone who needs it. Keep it to three minutes: two of movement, one for water.

38.5–73.5 min Worksheet (pairs, then alone, then stretch)

REC *Pause and do your workbook, starting at Start here on page 1, for about 35 minutes. Press play to check your work.*

DO Minute 38.5: give out the workbook. Read the title of Lesson 6 aloud. Say “Pairs first. Talk before you write.”

Minutes 38.5 to 41: a short teach slide first (this lesson). Minutes 41 to 52.5: pairs work together on the first page (the 2 minutes at the start are for finding the page and setting up; pairs stay in the main room). Minutes 52.5 to 55: check page 1 together (read only the answer line, children mark their own, one “How do you know?” for the page). Minutes 55 to 59.5: short teaching for page 2. Minutes 59.5 to 71: each child works alone on the second page. Minutes 71 to 73.5: check page 2 together (answer lines only, children mark their own). Think harder, Go further and the Investigation are open to any child at any time after question 3.

DO Take-home stars: Page 1: the last row of Q1 (the bicycle, $76) and Q2(b) (the two ways that use exactly 4 pieces). Page 2: Q3(b) (change from $50, fewest notes) and Q4(b) (pairs with exactly $3 change). Each is marked on the page with a ★ and the words take-home, stays Core (not Think harder), and keeps its answer on the check-together slide so a child can mark it at home. Do not spend class time on them. Late rule, printed here on purpose: at minute 59.5 (the start of page 2), if the clock is more than 2 minutes late, drop the last Core question of page 2 to take-home (children put a ★ beside it). Never shorten talk time, the exit ticket, looking back or the closing.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “Is this in cents or in dollars? How do you know?”
- Ask: “What is the biggest note that is not too much?”
- Ask: “How do you know you have found every way?”

IF STUCK Hint 1: “Lay out real play money and count it.”

Hint 2: “Make a table with a column for each note or coin. Start with as many of the biggest as you can.”

Hint 3: “Start from the money that is left and put back what was spent.”

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both

children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Ben spent half of his money on a kite. Then he spent $4 on a ball. He had $6 left. How much money did Ben have at first? Tip: draw his money as a bar cut in half.	$20 ($6 + $4 = $10 is the half he kept after the kite, so the other half is also $10: $10 + $10 = $20).
Workbook Think harder 2: How many ways can you pay exactly $10 using $5 notes, $2 notes and $1 coins?	10 ways (1 + 3 + 6)
Workbook Think harder 3: Ali has 4 coins. They are worth $1.25 altogether. Find all the sets of 4 coins he could have.	2 sets: $1 + 10¢ + 10¢ + 5¢, and 50¢ + 50¢ + 20¢ + 5¢
Workbook Think harder 4: How many different ways can you pay exactly 40¢ using 5¢, 10¢ and 20¢ coins? Make a list in order, starting with the most 20¢ coins.	9 ways: two 20¢ (1 way); one 20¢ and 20¢ more (3 ways); no 20¢ (5 ways).
Workbook Go further 5: Siti had some money. She spent half of it on a book. Then she spent half of what was left on a pen. She had $5 left. How much money did she have at first?	$20 ($5 + $5 = $10 before the pen; $10 + $10 = $20)
Workbook Go further 6: Sweets cost 5¢ and lollipops cost 20¢. Ben buys 8 things, at least one of each kind. Mei says he spent $1.00. Siti says he spent $1.50. Could either of them be right? Find ALL the amounts Ben could have spent. What do you notice about the amounts, and why?	Mei could be right (4 sweets and 4 lollipops: 20¢ + 80¢ = $1.00). Siti cannot. All the amounts: 55¢, 70¢, 85¢, $1.00, $1.15, $1.30, $1.45 (1 to 7 lollipops). They go up by 15¢ each time, because swapping one sweet for one lollipop adds 15¢. $1.50 is not on the list.
Workbook Go further 7: Mei has one $10 note, one $5 note, one $2 note and one $1 coin. How many different amounts can she pay exactly, using one or more of them?	15 amounts: $1, $2, $3, $5, $6, $7, $8, $10, $11, $12, $13, $15, $16, $17, $18

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for

Workbook Investigation (on the “Before you go” page): Most coins. Use only 5¢, 10¢, 20¢ and 50¢ coins. (a) Pay 35¢ using the fewest coins. How many coins is that? (b) Try the other amounts from 5¢ to 95¢ (in steps of 5¢) and make a table of how many coins each needs. (c) Which amount needs the MOST coins, even when you use the fewest possible? Easy first step: pay 35¢ with the fewest coins from the 5¢, 10¢, 20¢ and 50¢ coins. Needs: play coins (5¢, 10¢, 20¢, 50¢), or paper and a pencil.	Fewest coins (5¢ to 95¢): 5:1, 10:1, 15:2, 20:1, 25:2, 30:2, 35:3, 40:2, 45:3, 50:1, 55:2, 60:2, 65:3, 70:2, 75:3, 80:3, 85:4, 90:3, 95:4 (brute-forced). (a) 35¢ = 20¢ + 10¢ + 5¢, 3 coins. (c) 85¢ and 95¢ need 4 coins each: 85¢ = 50¢ + 20¢ + 10¢ + 5¢ and 95¢ = 50¢ + 20¢ + 20¢ + 5¢. An amount ending in 5 needs a 5¢ coin, and the rest then needs 3 more coins. Look for: a systematic table, trying biggest coin first, and checking that no smaller number of coins works.

73.5–76.5 min Flex (spare time)

REC *Pause or rewind whenever you need to.*

DO Spare time, kept on purpose (3 minutes). Use it to absorb a late start, a camera problem or a slow page 2. If you are on time, spend it on the Investigation share or a second talk-time volunteer, or an early start on talk time. Do not add new content. Never take time from the exit ticket, looking back or the stamp.

76.5–81.5 min Talk time

REC *Pause. Tell a grown-up your answer to the talk question, and why you think so. Then press play.*

DO Bring the group back to a circle (or camera on, mics on). Pairs talk for 90 seconds, then two volunteers share (about 5 minutes in all). One question only. You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *“Mei says: To pay $40 with the fewest notes, I use two $20 notes. What would you tell her?”*

Discussion question	Listen for
Mei says: “To pay $40 with the fewest notes, I use two $20 notes.” What would you tell her?	There is no $20 note in Singapore. $50 is too much, so we use $10 notes: $10 + $10 + $10 + $10, which is 4 notes.

81.5–86.5 min Exit ticket

REC *Pause and do the exit ticket on your own, without looking back. Then press play to check it.*

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. About 2 minutes of child work; then children hold their slips up one at a time (never all together) and the answer is on the next click.

Exit ticket	Answer

A pen costs $3.65. Pay exactly with the fewest notes and coins.	$2 + $1 + 50¢ + 10¢ + 5¢ (5 pieces)

86.5–88.5 min Looking back (2 minutes)

REC *Pause for about 2 minutes and fill in your looking-back page: what was hardest, and what will you try next time?*

SAY *"Looking back takes 2 minutes. Find the box called Looking back on the page "Before you go: Investigation and Looking back". Finish one line, two lines or all three, and tick what helped you. You may pass on any line. If you tick "I am still stuck", your tutor will check in during the session."*

DO Nothing is collected in front of the group and nobody has to read theirs out. The tutor may read a box while stamping journals (never by private message), and checks in on a child who ticked still stuck during the open session, for example by a quick visit to the shared breakout room or a hand signal with a parent nearby, for example "I saw you drew it when you got stuck". A child may volunteer one sentence, never by name unless they choose to. If a child ticks "I am still stuck", check in during the next open session (a quick visit to the shared breakout room or a hand signal, with a parent nearby). Praise the strategy, not the speed.

88.5–90 min Closing and stamps (1.5 minutes)

REC *Well done! Show your journal and workbook to a grown-up and tell them one thing you found out.*

SAY *"Great shopping, inquirers! You paid exactly, gave change and made lists in order. A good shopkeeper always checks, and so did you."*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Find every way to pay exactly $10 using $5 notes, $2 notes and $1 coins. (There are 10.)

Answer key (workbook)

Q1 Curry puff $1.65: $1 + 50¢ + 10¢ + 5¢ (4). Notebook $4.30: $2 + $2 + 20¢ + 10¢ (4). Umbrella $18.75: $10 + $5 + $2 + $1 + 50¢ + 20¢ + 5¢ (7). Bicycle $76: $50 + $10 + $10 + $5 + $1 (5).

Q2 7 ways ($5, $2, $1): 1,1,1 / 1,0,3 / 0,4,0 / 0,3,2 / 0,2,4 / 0,1,6 / 0,0,8. Row H is spare: tell children "you may not need every row". (a) 7 ways (b) $5 + $1 + $1 + $1, and $2 + $2 + $2 + $2

Q3 (a) 35¢ (b) $24 change: $10 + $10 + $2 + $2 (there is no $20 note)

Q4 (a) No. The bag and the cap cost $14 + $8 = $22. She has $20, so she is $2 short. (b) She spends $17: book and cap ($9 + $8), or bag and pen ($14 + $3). Those are the only two pairs.

Q5 (a) 25¢ + 30¢ + 45¢ = 100¢ = $1.00 (Think harder 1 is the half-money puzzle with the kite and the ball: $20.)

Q6 3 ways: 4 erasers; 2 erasers and 5 stickers; 10 stickers. (1 or 3 erasers leave 150¢ or 50¢, which 20¢ stickers cannot make exactly.)

Investigation (Most coins). Fewest coins (5¢ to 95¢): 5:1, 10:1, 15:2, 20:1, 25:2, 30:2, 35:3, 40:2, 45:3, 50:1, 55:2, 60:2, 65:3, 70:2, 75:3, 80:3, 85:4, 90:3, 95:4 (brute-forced). (a) 35¢ = 20¢ + 10¢ + 5¢, 3 coins. (c) 85¢ and 95¢ need 4 coins each: 85¢ = 50¢ + 20¢ + 10¢ + 5¢ and 95¢ = 50¢ + 20¢ + 20¢ + 5¢. An amount ending in 5 needs a 5¢ coin, and the rest then needs 3 more coins. Look for: a systematic table, trying biggest coin first, and checking that no smaller number of coins works.

Recording chapters and what a child watching later does

Time	Chapter marker	A child watching later
0–1.5 min	Opening	Gather your workbook, pencil, paper and the materials on the slide. Press play when ready.
1.5–6 min	Sneak-peek puzzles (4.5 minutes)	Pause. Try the puzzles on your own paper (about 4 and a half minutes). Guessing is fine. Press play for the answers.
6–10.5 min	Warm-up	Play along. Say or write each answer before the video does. Pause if you need longer.
10.5–23.5 min	Explore (in pairs, with objects)	Pause. Try the explore task with your objects, with someone at home if you can (about 13 minutes). Write down what you notice.
23.5–35.5 min	Mini-lesson (you teach, they join in)	Keep watching. Copy the worked examples. Rewind any step you missed.
35.5–38.5 min	Brain break	Stand up and join the brain break.
38.5–73.5 min	Worksheet (pairs, then alone, then stretch)	Pause. Do the workbook, starting with Start here on page 1 (about 35 minutes). Press play for each checking part and mark in a different colour.
73.5–76.5 min	Flex (spare time)	Pause or rewind whenever you need to.
76.5–81.5 min	Talk time	Pause. Explain your answer to someone at home, with a reason. Then play.
81.5–86.5 min	Exit ticket	Pause. Do the exit ticket on your own, then play to check it.
86.5–88.5 min	Looking back (2 minutes)	Pause. Fill in the looking-back page.
88.5–90 min	Closing and stamps (1.5 minutes)	Show your journal and workbook to a grown-up. Tell them one thing you found out.

Lesson 7: Fair Shares

Week 4 (first lesson) • Fractions • Workbook: Lesson 7

Objective	Recognise equal parts and name fractions with bottom numbers up to 12; compare unit fractions and like fractions; add and take away like fractions within one whole; solve fair-sharing stories with a bar.
Thinking skills	Checking for equal parts, comparing with the same whole, drawing bars, explaining
Materials	Paper strips of the same length (4 per child, cut out before the lesson), paper squares, scissors, crayons, mini whiteboards.
Online set-up	Parents cut the 4 strips for each child before the lesson (kit in hand at minute 0). Each child folds paper strips on camera; the tutor shows fraction bars on the slides and children colour their own bars.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.
Watching the recording	A child who missed the live lesson watches the recording. They need: Paper strips of the same length (4 per child, cut out before the lesson), paper squares, scissors, crayons, mini whiteboards. The REC lines tell them when to pause; the chapter list at the end of this lesson shows each part.

0–1.5 min Opening

REC *Hello! Before you press play, get a pencil, paper, your workbook and the materials on the slide. Pause or rewind any time.*

SAY *"Good morning, inquirers! Today we are cake cutters and pizza sharers. There is one rule: every share must be FAIR."*

"By the end you will name fair shares with fractions, tell which share is bigger, and explain how you know."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud. The kit is on the desk from minute 0: ask everyone to show you the materials for today (a 30-second check; the materials list is above and on the title slide).

1.5–6 min Sneak-peek puzzles (4.5 minutes)

REC *Pause now and try the puzzles on your own paper for about 4 and a half minutes. Guessing is fine. Press play for the answers.*

SAY *"Two sneak-peek puzzles on ideas we have not learnt yet. Guessing is welcome, so is I do not know. Nobody is marked. Show your paper to me only."*

DO Show the Sneak-peek puzzles slide at minute 1.5 (4.5 minutes: about 45 seconds for question 1, 90 seconds for question 2, then the answers). Say the short line out loud (about 28 words): two puzzles on ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is

marked; show your paper to me only. Read each puzzle aloud; question 1 is the one to start with (objects, a drawing or counting).

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): Fold a paper strip (or draw a bar) into 2 equal parts. Colour 1 part. What fraction is coloured?	1/2 (one half). Listen for: Equal parts matter. Folding on camera is welcome. This is a first win for everyone.
Question 2 (thinking): Ali has 1/5 of a roti prata. Mei has 1/8 of the same prata. Who has more? How do you know?	Ali (1/5). The same prata cut into fewer parts gives bigger parts. Listen for: Many children say 1/8 because 8 is bigger. Draw two strips, one cut in 5 and one cut in 8, and compare. A drawing is a good answer.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

6–10.5 min Warm-up

REC *Join in! Say or write each answer before I do. Pause the video if you need longer.*

DO Fair or not fair? Four of the six snacks (pizza, prata, cake, pancake; spare: kaya toast, chocolate bar) are cut into parts: thumbs up if all the parts are the same size, thumbs down if not.

SAY *“Look at each snack. Ali and Mei want to share it. Is it fair? Thumbs up for fair, thumbs down for not fair.”*

“Be ready to tell me why.”

DO Keep the game to about two and a half minutes (6 to 8.5): stop it on the clock. Then the quick questions (8.5 to 10.5): ask questions 1 and 3 only (these are the two on the slide); questions 2, 4 and 5 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect

A pizza is cut into 4 equal slices. What is 1 slice called?	One quarter, 1/4
How many halves make 1 whole?	2
How many thirds make 1 whole?	3
A cake is cut into 5 equal pieces. What is 1 piece called?	One fifth, 1/5
2 of 7 equal parts are shaded. What fraction is shaded?	2/7

10.5–23.5 min Explore (in pairs, with objects)

REC *Pause now. Try this with your objects, and a grown-up if you can, for about 13 minutes. Press play when you have an answer.*

DO Timing (13 minutes): set up and predict, 10.5 to 12. One round of breakout rooms with an auto-timer: 60 seconds to get in, 5.5 minutes of pair work (13 to 18.5), 60 seconds to come back. Then the group debrief, 19.5 to 23.5.

SAY *"Before we fold: which will be bigger, one quarter of a strip or one eighth of the same strip? Show me 4 fingers for a quarter or 8 fingers for an eighth."*

DO 1. Take 4 paper strips that are the same length.

2. Keep one whole. Fold one in half, one into quarters and one into eighths.

3. Open them out. Write the fraction on ONE part of each strip.

4. Line them up under the whole.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck folding: match the two ends exactly, then press the fold flat.

Not sure what to write: "How many equal parts are there? That number goes at the bottom."

Finished early: can you fold a strip into 3 equal parts? How do you check they are equal?

DO Go further (a second what-if, so nobody waits): it is the box at the foot of the Explore slide; post it in the chat as well when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: fold a strip into 3 equal parts. How do you know the parts are equal? Then fold one into 6 equal parts: how many sixths make a half? A third? Answer: Thirds: fold and adjust until the 3 parts match, then check one part on top of another. 3 sixths make a half and 2 sixths make a third.

DO At about minute 18.5, say "Pencils down" and bring the group back together (60 seconds). Take the three questions below, one at a time (about 4 minutes in all). Ask "How do you know?" on question 2 only; take about 15 seconds of talk on the others.

Ask	Expect

Which part is bigger, 1/4 or 1/8? How do you know?	1/4: the same strip cut into fewer parts, so each part is longer.
How many eighths fit on one quarter? How many quarters make 1 whole?	2 eighths; 4 quarters.

23.5–35.5 min Mini-lesson (you teach, they join in)

REC *Keep watching and copy the examples onto your paper. Missed a step? Rewind. You will try some yourself soon.*

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Before I name a fraction I check the parts are equal. This bar has 5 parts, all the same size, so the parts are fifths: 5 goes at the bottom. 3 parts are shaded: 3 goes on top. Three fifths."*

"One half or one sixth: which is bigger? Same whole. One half is 1 of 2 parts, one sixth is 1 of 6 parts. More parts means smaller parts, so one half is bigger."

"Two ninths and four ninths: ninths are the name of the parts, like apples. 2 ninths and 4 ninths are 6 ninths. The parts do not change size, so the bottom number stays 9."

Worked example on the slides	Answer
Name the fraction. A bar has 5 equal parts and 3 are shaded. Check the parts are equal, count all the parts (bottom number), then count the shaded parts (top number). What fraction is shaded? What fraction is not?	3/5 shaded, 2/5 not shaded; together 5/5, 1 whole.
Compare with bars. Bars of the same length cut into halves, thirds and sixths, one part shaded in each. Which is bigger: 1/2, 1/3 or 1/6? Then: which is bigger, 2/7 or 5/7?	1/6 < 1/3 < 1/2 (more equal parts, smaller parts). 5/7 > 2/7 (same-size parts, more of them).

SAY *"The key idea today is: equal parts. A fraction names equal parts of a whole: the bottom number is how many equal parts, the top number is how many we take. Say it with me."*

DO Show the key idea slide at about minutes 34.5 to 35.5. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Naming a fraction when the parts are not equal: circling B or D in Q1.
- Thinking 1/6 is bigger than 1/4 because 6 is bigger than 4 (Q3).
- Adding the bottom numbers too: 2/5 + 1/5 written as 3/10 instead of 3/5 (Q4).

35.5–38.5 min Brain break

REC *Movement break! Stand up and join in. The video keeps going.*

DO Fraction turns. Stand up and turn a whole turn, a half turn, a quarter turn, three quarter turns, then freeze.

Water. Toilet break for anyone who needs it. Keep it to three minutes: two of movement, one for water.

38.5–73.5 min Worksheet (pairs, then alone, then stretch)

REC *Pause and do your workbook, starting at Start here on page 1, for about 35 minutes. Press play to check your work.*

DO Minute 38.5: give out the workbook. Read the title of Lesson 7 aloud. Say “Pairs first. Talk before you write.”

Minutes 38.5 to 52.5: pairs work together on the first page (the 2 minutes at the start are for finding the page and setting up; pairs stay in the main room). Minutes 52.5 to 55: check page 1 together (read only the answer line, children mark their own, one “How do you know?” for the page). Minutes 55 to 59.5: short teaching for page 2. Minutes 59.5 to 71: each child works alone on the second page. Minutes 71 to 73.5: check page 2 together (answer lines only, children mark their own). Think harder, Go further and the Investigation are open to any child at any time after question 3.

DO Take-home stars: Page 1: the explanation in Q1(a) (why is one shape not cut into quarters). Page 2: Q4(d) and Q4(e). Also at home: fold a strip into 3 equal parts (the Go further card). Each is marked on the page with a ★ and the words take-home, stays Core (not Think harder), and keeps its answer on the check-together slide so a child can mark it at home. Do not spend class time on them. Late rule, printed here on purpose: at minute 59.5 (the start of page 2), if the clock is more than 2 minutes late, drop the last Core question of page 2 to take-home (children put a ★ beside it). Never shorten talk time, the exit ticket, looking back or the closing.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “Are all the parts the same size? How do you know?”
- Ask: “What does the bottom number tell us? What does the top number tell us?”
- Ask: “Did you draw both wholes the same length?”

IF STUCK Hint 1: “Count all the equal parts first. That number goes at the bottom.”

Hint 2: “Fold or draw a strip to check which part is bigger.”

Hint 3: “Colour the parts on a bar, then count them.”

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer

Workbook Think harder 1: Write these in order, smallest first: 1/3, 1/12, 1/2, 1/5.	1/12, 1/5, 1/3, 1/2
Workbook Think harder 2: Mei says: “3/8 of a pizza is more than 3/5 of the same pizza, because 8 is bigger than 5.” Is she right? Draw two bars of the same length to show.	No. Eighths are smaller than fifths (the more equal parts, the smaller each part). So 3 eighths is less than 3 fifths: 3/5 is bigger than 3/8.
Workbook Think harder 3: Siti says: “3/8 + 2/8 = 5/16, because 8 + 8 = 16.” Is she right? Draw a bar to show. What is the right answer, and why does the bottom number not change?	No. 3 eighths and 2 eighths make 5 eighths: 5/8. The parts are still eighths, like 3 apples and 2 apples make 5 apples. Adding the bottom numbers would change the size of the parts.
Workbook Think harder 4: Ali says: “Half of my pizza is bigger than half of your pizza.” Can he be right? Explain.	Yes, if his pizza is bigger. Halves of different-size wholes are different sizes.
Workbook Go further 5: A cake is cut into 12 equal pieces. Mei gets 4/12. Ali and Ben share the rest, but Ali gets 2/12 more than Ben. What fraction does Ben get?	3/12 (and Ali gets 5/12): 8 pieces are shared; take away Ali's extra 2, share 6 as 3 and 3.
Workbook Go further 6: Three children share a cake cut into 10 equal pieces. Each child gets a DIFFERENT number of pieces (at least 1), and the whole cake is shared out. Find ALL the ways to share it. Write each child’s share as a fraction.	4 ways (tenths): 1, 2, 7; 1, 3, 6; 1, 4, 5; 2, 3, 5.
Workbook Go further 7: I am a fraction with 1 on top. I am smaller than 1/4 but bigger than 1/7. What could I be? Find all the answers.	1/5 and 1/6

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (on the “Before you go” page): Fraction strips. (a) Cover a HALF strip exactly with pieces of ONE kind only. Which kinds work? How many pieces of each? (b) Do the same for a THIRD strip and a QUARTER strip. (c) Can thirds ever cover a half exactly? Why not? Easy first step: lay the two quarter strips on top of the half strip. Do they fit exactly? Needs: the fraction strips from the cut-out page (halves, thirds, quarters, sixths, eighths).	(a) Half: 2 quarters, 3 sixths or 4 eighths (not thirds). (b) Third: 2 sixths only. Quarter: 2 eighths only (checked with exact fractions). (c) No: one third is shorter than a half but two thirds is longer, so no number of thirds fits exactly. Look for: children lay the strips side by side and say “fits exactly”. Keep to the strips and drawings: do not teach “equivalent fractions” as a rule (that is P3).

73.5–76.5 min Flex (spare time)

REC *Pause or rewind whenever you need to.*

DO Spare time, kept on purpose (3 minutes). Use it to absorb a late start, a camera problem or a slow page 2. If you are on time, spend it on the Investigation share or a second talk-time volunteer, or an early start on talk time. Do not add new content. Never take time from the exit ticket, looking back or the stamp.

76.5–81.5 min Talk time

REC *Pause. Tell a grown-up your answer to the talk question, and why you think so. Then press play.*

DO Bring the group back to a circle (or camera on, mics on). Pairs talk for 90 seconds, then two volunteers share (about 5 minutes in all). One question only. You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *"Is 1/2 always bigger than 1/4? Can you think of a time when it is not?"*

Discussion question	Listen for
Is 1/2 always bigger than 1/4? When could it not be?	Only when the wholes are the same size: half a small cupcake is less than a quarter of a big cake.

81.5–86.5 min Exit ticket

REC *Pause and do the exit ticket on your own, without looking back. Then press play to check it.*

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. About 2 minutes of child work; then children hold their slips up one at a time (never all together) and the answer is on the next click.

Exit ticket	Answer
Which is bigger, 1/7 or 1/9? Then work out 2/9 + 5/9.	1/7 (the same whole cut into fewer parts gives bigger parts). 2/9 + 5/9 = 7/9.

86.5–88.5 min Looking back (2 minutes)

REC *Pause for about 2 minutes and fill in your looking-back page: what was hardest, and what will you try next time?*

SAY *"Looking back takes 2 minutes. Find the box called Looking back on the page "Before you go: Investigation and Looking back". Finish one line, two lines or all three, and tick what helped you. You may pass on any line. If you tick "I am still stuck", your tutor will check in during the session."*

DO Nothing is collected in front of the group and nobody has to read theirs out. The tutor may read a box while stamping journals (never by private message), and checks in on a child who ticked still stuck during the open session, for example by a quick visit to the shared breakout room or a hand

signal with a parent nearby, for example "I saw you drew it when you got stuck". A child may volunteer one sentence, never by name unless they choose to. If a child ticks "I am still stuck", check in during the next open session (a quick visit to the shared breakout room or a hand signal, with a parent nearby). Praise the strategy, not the speed.

88.5–90 min Closing and stamps (1.5 minutes)

REC *Well done! Show your journal and workbook to a grown-up and tell them one thing you found out.*

SAY *"You checked for equal parts, compared fractions with bars and explained why. That is fair and careful maths."*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Find all the ways to add two fractions with bottom number 8 to make 1 whole (1/8 + 7/8, 2/8 + 6/8, 3/8 + 5/8, 4/8 + 4/8: 4 ways).

Answer key (workbook)

Q1 A, C, E and F are cut into quarters. (a) B or D: their 4 parts are not the same size, so a part is not a quarter. F is the hardest: all four small triangles are the same size, even though the middle one is upside down.

Q2 (a) shaded 3/8, not shaded 5/8 (b) 5/6 and 1/6 (c) 7/10 and 3/10

Q3 (a) 1/3; 7/9 (b) 1/10, 1/5, 1/4, 1/2 (c) No. Cutting the same whole into 6 equal parts makes smaller parts than cutting it into 4, so 1/6 is smaller than 1/4.

Q4 (a) 3/5 (b) 7/8 (c) 9/12 (d) 5/6 (e) 4/10

Q5 (a) 5/8 (b) 3/8 (c) Mei, by 1/8

Q6 Siti. The pizzas are the same size and one quarter covers 2 eighths on the bars; 3 eighths is more than 2 eighths. (Equivalent fractions are a P3 topic: show this with bars only and do not write 1/4 = 2/8 as a rule.)

Investigation (Fraction strips). (a) Half: 2 quarters, 3 sixths or 4 eighths (not thirds). (b) Third: 2 sixths only. Quarter: 2 eighths only (checked with exact fractions). (c) No: one third is shorter than a half but two thirds is longer, so no number of thirds fits exactly. Look for: children lay the strips side by side and say "fits exactly". Keep to the strips and drawings: do not teach "equivalent fractions" as a rule (that is P3).

Recording chapters and what a child watching later does

Time	Chapter marker	A child watching later
0–1.5 min	Opening	Gather your workbook, pencil, paper and the materials on the slide. Press play when ready.

1.5–6 min	Sneak-peek puzzles (4.5 minutes)	Pause. Try the puzzles on your own paper (about 4 and a half minutes). Guessing is fine. Press play for the answers.
6–10.5 min	Warm-up	Play along. Say or write each answer before the video does. Pause if you need longer.
10.5–23.5 min	Explore (in pairs, with objects)	Pause. Try the explore task with your objects, with someone at home if you can (about 13 minutes). Write down what you notice.
23.5–35.5 min	Mini-lesson (you teach, they join in)	Keep watching. Copy the worked examples. Rewind any step you missed.
35.5–38.5 min	Brain break	Stand up and join the brain break.
38.5–73.5 min	Worksheet (pairs, then alone, then stretch)	Pause. Do the workbook, starting with Start here on page 1 (about 35 minutes). Press play for each checking part and mark in a different colour.
73.5–76.5 min	Flex (spare time)	Pause or rewind whenever you need to.
76.5–81.5 min	Talk time	Pause. Explain your answer to someone at home, with a reason. Then play.
81.5–86.5 min	Exit ticket	Pause. Do the exit ticket on your own, then play to check it.
86.5–88.5 min	Looking back (2 minutes)	Pause. Fill in the looking-back page.
88.5–90 min	Closing and stamps (1.5 minutes)	Show your journal and workbook to a grown-up. Tell them one thing you found out.

Lesson 8: Shape Builders

Week 4 (second lesson) • 2D and 3D Shapes • Workbook: Lesson 8

Objective	Name half circles and quarter circles and use them to make and break up shapes; count every square and triangle in a figure by working in order; count the cubes in a stack, including hidden ones, layer by layer; decide which nets fold into a cube.
Thinking skills	Spatial reasoning, systematic counting, folding in your head, explaining
Materials	Paper circles (2 per child, cut out before the lesson), scissors, about 12 linking cubes per child (or sugar cubes), the cube-net cut-out page (nets A to F, near the back of the workbook), mini whiteboards.
Online set-up	Parents cut out the 2 paper circles and the nets A to D before the lesson (kit in hand at minute 0). Children fold and cut the circles and build stacks on camera; the six nets A to F are on a cut-out page near the back of the workbook (print and cut them before the lesson), or cut a net from an empty box. No linking cubes at home? Use sugar cubes, dice or small boxes, or draw each stack layer by layer.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.
Watching the recording	A child who missed the live lesson watches the recording. They need: Paper circles (2 per child, cut out before the lesson), scissors, about 12 linking cubes per child (or sugar cubes), the cube-net cut-out page (nets A to F, near the back of the workbook), mini whiteboards. The REC lines tell them when to pause; the chapter list at the end of this lesson shows each part.

0–1.5 min Opening

REC *Hello! Before you press play, get a pencil, paper, your workbook and the materials on the slide. Pause or rewind any time.*

SAY *"Good morning, inquirers! Today we are shape builders. We will cut circles into pieces, hunt for hidden squares and triangles, count cubes we cannot even see, and fold flat paper into a cube."*

"The secret for every puzzle today: count in order, so nothing is missed and nothing is counted twice."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. The kit is on the desk from minute 0: ask everyone to show you the materials for today (a 30-second check; the materials list is above and on the title slide).

1.5–6 min Sneak-peek puzzles (4.5 minutes)

REC *Pause now and try the puzzles on your own paper for about 4 and a half minutes. Guessing is fine. Press play for the answers.*

SAY *"Two sneak-peek puzzles on ideas we have not learnt yet. Guessing is welcome, so is I do not know. Nobody is marked. Show your paper to me only."*

DO Show the Sneak-peek puzzles slide at minute 1.5 (4.5 minutes: about 45 seconds for question 1, 90 seconds for question 2, then the answers). Say the short line out loud (about 28 words): two puzzles on ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked; show your paper to me only. Read each puzzle aloud; question 1 is the one to start with (objects, a drawing or counting).

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): A stack has 2 layers. Each layer has 2 rows of 3 cubes. How many cubes are in the stack? (Build it with cubes or draw it.)	12 cubes (6 in each layer, 2 layers). Listen for: Counting one layer first, then doubling, is a good method. This is a first win for everyone.
Question 2 (thinking): A shape is made of 2 half circles and 1 quarter circle. If you cut every piece into quarter circles, how many quarter circles do you get?	5 (each half circle is 2 quarters: 2 + 2 + 1). Listen for: Listen for "a half circle is half of a circle, which is 2 quarters". Drawing the pieces and cutting them is fine.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

6–10.5 min Warm-up

REC *Join in! Say or write each answer before I do. Pause the video if you need longer.*

DO Name the solid: a dice, a tissue box, a tin can, a party hat (spare: a drum, a traffic cone). Make the shape with your hands, then name it.

SAY *“I will show you something from home. Make its shape with your hands: a cube, a cuboid, a cylinder or a cone. Then we say its name together.”*

DO Keep the game to about two and a half minutes (6 to 8.5): stop it on the clock. Then the quick questions (8.5 to 10.5): ask questions 1 and 3 only (these are the two on the slide); questions 2, 4 and 5 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
How many faces does a cube have?	6
How many flat faces does a cylinder have?	2
Cut a circle into 2 equal parts. What is each part called?	A half circle
How many quarter circles make 1 whole circle?	4
Which solid has a tip and can roll?	A cone

10.5–23.5 min Explore (in pairs, with objects)

REC *Pause now. Try this with your objects, and a grown-up if you can, for about 13 minutes. Press play when you have an answer.*

DO Timing (13 minutes): set up and predict, 10.5 to 12. One round of breakout rooms with an auto-timer: 60 seconds to get in, 5.5 minutes of pair work (13 to 18.5), 60 seconds to come back. Then the group debrief, 19.5 to 23.5.

SAY *"Before we cut: if I cut a circle in half, and then cut one half in half again, how many pieces will I have? What will they look like?"*

DO 1. Take your 2 paper circles (cut out before the lesson, the same size).

2. Fold one in half and cut along the fold.

3. Fold the other in half, then in half again. Open it and cut along both folds.

4. Use your pieces to make new shapes. Name them.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck folding: fold the circle so the edges match exactly, then press the fold flat.

Not sure of a name: "Count the straight sides. One straight side: half circle. Two straight sides: quarter circle."

Finished early: find 3 different ways to make a whole circle from your pieces.

DO Go further (a second what-if, so nobody waits): it is the box at the foot of the Explore slide; post it in the chat as well when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: you have 2 half circles and 3 quarter circles. How many whole circles can you make? What is left over? Answer: 7 quarter circles in all: 1 whole circle (4 quarters) and 3 quarters left over.

DO At about minute 18.5, say "Pencils down" and bring the group back together (60 seconds). Take the three questions below, one at a time (about 4 minutes in all). Ask "How do you know?" on question 2 only; take about 15 seconds of talk on the others.

Ask	Expect

How many quarter circles cover a half circle exactly?	2
Find 3 different ways to make a whole circle from your pieces.	2 half circles; 4 quarter circles; 1 half circle and 2 quarter circles.
How many straight sides has a quarter circle? Are they the same length?	2, and yes: they meet at a square corner.

23.5–35.5 min Mini-lesson (you teach, they join in)

REC *Keep watching and copy the examples onto your paper. Missed a step? Rewind. You will try some yourself soon.*

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"To name a piece I count its straight sides. One straight side and one curved side: a half circle. Two straight sides that meet at a square corner, and one curved side: a quarter circle. Turning it round does not change its name."*

"Squares hide inside bigger squares. I count by size: 1 by 1 first, then I slide a 2 by 2 window and count the places it fits, then 3 by 3. Then I add them up."

"For cubes I count layer by layer. Cubes cannot float, so if I see a cube on top, there must be a cube under it, even if I cannot see it."

Worked example on the slides	Answer
Find the pieces. A mushroom is made of a half circle cap, a square stem and 2 quarter circles of grass. Count one kind at a time: 1 straight and 1 curved side, then 2 straight sides and 1 curved side, then 4 straight sides. How many pieces of each kind?	Half circle 1, quarter circles 2, square 1: 4 pieces.
Count every square. A grid 3 squares long and 3 squares high. Count by size: 1 by 1, then slide a 2 by 2 window, then the whole grid. How many squares altogether?	9 + 4 + 1 = 14 squares.

SAY *"The key idea today is: count in order. Sort what you count, by size, by the line it sits on or by layer, so you never miss one or count one twice. Say it with me."*

DO Show the key idea slide at about minutes 34.5 to 35.5. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Counting only the 1 by 1 squares in Q2a (16 instead of 30).
- Counting only the cubes you can see in Q3 and forgetting the hidden cubes underneath and behind.
- Thinking any 6 joined squares fold into a cube (Q4): B, D and F do not (cutting out and folding shows why).

35.5–38.5 min Brain break

REC *Movement break! Stand up and join in. The video keeps going.*

DO Be a shape! Make a half circle, a cone, a cube and a cylinder with your body when the tutor calls it.

Water. Toilet break for anyone who needs it. Keep it to three minutes: two of movement, one for water.

38.5–73.5 min Worksheet (pairs, then alone, then stretch)

REC *Pause and do your workbook, starting at Start here on page 1, for about 35 minutes. Press play to check your work.*

DO Minute 38.5: give out the workbook. Read the title of Lesson 8 aloud. Say “Pairs first. Talk before you write.”

Minutes 38.5 to 52.5: pairs work together on the first page (the 2 minutes at the start are for finding the page and setting up; pairs stay in the main room). Minutes 52.5 to 55: check page 1 together (read only the answer line, children mark their own, one “How do you know?” for the page). Minutes 55 to 59.5: short teaching for page 2. Minutes 59.5 to 71: each child works alone on the second page. Minutes 71 to 73.5: check page 2 together (answer lines only, children mark their own). Think harder, Go further and the Investigation are open to any child at any time after question 3.

DO Take-home stars: Page 1: Q1(c) (Mei's half and quarter circles) and Q2(b) (count the triangles). Page 2: Q4(b) (add one square to make a cube net). The triangle-counting teaching slide goes home with Q2(b); it is not shown in class. Each is marked on the page with a ★ and the words take-home, stays Core (not Think harder), and keeps its answer on the check-together slide so a child can mark it at home. Do not spend class time on them. Late rule, printed here on purpose: at minute 59.5 (the start of page 2), if the clock is more than 2 minutes late, drop the last Core question of page 2 to take-home (children put a ★ beside it). Never shorten talk time, the exit ticket, looking back or the closing.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “How many straight sides does it have? So what is it called?”
- Ask: “Where else can your window fit? Have you tried every place?”
- Ask: “What is holding up that top cube?”

IF STUCK Hint 1: “Count one kind or one size at a time, and tick each one.”

Hint 2: “Build it with cubes, or cut out the net and fold it.”

Hint 3: “For a net, pick a bottom square and fold the others up one at a time.”

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce

who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: A stack of cubes has a bottom layer 4 cubes long and 3 cubes deep, a middle layer 3 cubes long and 2 cubes deep, and 1 cube on top. How many cubes are in the stack? Draw the layers.	19 (12 + 6 + 1)
Workbook Think harder 2: How many rectangles can you find in a grid 2 squares long and 2 squares high? Remember that a square is a rectangle too. Count them by size.	9: four small squares (1 by 1), two long ones (1 high and 2 long), two tall ones (2 high and 1 long), and the whole grid (2 by 2). 4 + 2 + 2 + 1 = 9.
Workbook Think harder 3: Cut out nets E and F from the cut-out page and try folding them. Which one folds up into a cube? How do you know?	E folds into a cube (it is a staircase of squares, one of the 11 cube nets). F does not: the row of 5 squares wraps round and the 1st and 5th squares land on the same face.
Workbook Think harder 4: Six squares are joined in a straight line. Can they fold into a cube? Explain.	No. When you fold the line round, the 5th square lands on the same face as the 1st and the 6th on the 2nd, and the two ends are left open.
Workbook Go further 5: A pyramid of cubes has a top layer of 1 cube, then a layer 2 by 2, then a layer 3 by 3, then a bottom layer 4 by 4. How many cubes are there?	30 (1 + 4 + 9 + 16)
Workbook Go further 6: How many squares can you find in a grid 5 squares long and 5 squares high?	55 (25 + 16 + 9 + 4 + 1)
Workbook Go further 7: A 3 by 3 by 3 cube is built from 27 small cubes and the whole outside is painted. Sort the small cubes: how many have paint on 3 faces, on 2 faces, on 1 face and on no faces? Check that your four numbers make 27. What if the big cube is 4 by 4 by 4: how many small cubes have no paint?	Paint on 3 faces: 8 (the corners). On 2 faces: 12 (one in the middle of each edge). On 1 face: 6 (the middle of each face). No paint: 1 (the centre). 8 + 12 + 6 + 1 = 27. For a 4 by 4 by 4 cube the unpainted cubes make a 2 by 2 by 2 block inside: 8 cubes.

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for

Workbook Investigation (on the “Before you go” page): Squares joined edge to edge. Squares must touch along a whole side. A shape that is turned round or flipped over counts as the SAME shape. (a) Find ALL the different shapes you can make with 3 squares. (b) Now find ALL the different shapes you can make with 4 squares. Draw each one on the grid. (c) How do you know you have found them all? Easy first step: join 2 squares edge to edge. How many different shapes can you make? Then add a third square. Needs: square tiles or small squares of paper, and the grid on this page.	(a) 2 shapes: a straight line and an L. (b) 5 shapes: a straight line of 4, a 2 x 2 square, an L, a T and a zigzag (S shape). Brute-forced: 1, 1, 2, 5 shapes for 1, 2, 3, 4 squares (turns and flips count as the same). If a child counts flips as different, they find 7 (L and its mirror, S and its mirror): praise this and explain the rule. (c) Look for a system: start with the straight line, then move one square, and check each new shape against the others by turning the tile.

73.5–76.5 min Flex (spare time)

REC *Pause or rewind whenever you need to.*

DO Spare time, kept on purpose (3 minutes). Use it to absorb a late start, a camera problem or a slow page 2. If you are on time, spend it on the Investigation share or a second talk-time volunteer, or an early start on talk time. Do not add new content. Never take time from the exit ticket, looking back or the stamp.

76.5–81.5 min Talk time

REC *Pause. Tell a grown-up your answer to the talk question, and why you think so. Then press play.*

DO Bring the group back to a circle (or camera on, mics on). Pairs talk for 90 seconds, then two volunteers share (about 5 minutes in all). One question only. You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *“How can you be sure you counted every cube in a stack, even the ones you cannot see?”*

Discussion question	Listen for
How can you be sure you counted every cube in a stack, even the ones you cannot see?	Count layer by layer; every cube on top sits on a cube below, so the hidden ones must be there.

81.5–86.5 min Exit ticket

REC *Pause and do the exit ticket on your own, without looking back. Then press play to check it.*

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. About 2 minutes of child work; then children hold their slips up one at a time (never all together) and the answer is on the next click.

Exit ticket	Answer

How many cubes are in a stack with a bottom layer 3 cubes long and 2 cubes deep, and 1 cube on top?	7 (6 + 1)

86.5–88.5 min Looking back (2 minutes)

REC *Pause for about 2 minutes and fill in your looking-back page: what was hardest, and what will you try next time?*

SAY *"Looking back takes 2 minutes. Find the box called Looking back on the page "Before you go: Investigation and Looking back". Finish one line, two lines or all three, and tick what helped you. You may pass on any line. If you tick "I am still stuck", your tutor will check in during the session."*

DO Nothing is collected in front of the group and nobody has to read theirs out. The tutor may read a box while stamping journals (never by private message), and checks in on a child who ticked still stuck during the open session, for example by a quick visit to the shared breakout room or a hand signal with a parent nearby, for example "I saw you drew it when you got stuck". A child may volunteer one sentence, never by name unless they choose to. If a child ticks "I am still stuck", check in during the next open session (a quick visit to the shared breakout room or a hand signal, with a parent nearby). Praise the strategy, not the speed.

88.5–90 min Closing and stamps (1.5 minutes)

REC *Well done! Show your journal and workbook to a grown-up and tell them one thing you found out.*

SAY *"You named pieces by their sides, counted in order and folded nets in your head. That is real shape building."*

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

How many squares are in a grid 5 squares long and 5 squares high? (25 + 16 + 9 + 4 + 1 = 55)

Answer key (workbook)

Q1 Square 2, triangle 3, half circle 4, quarter circle 2. (a) half circles (b) 11 pieces (c) 3 circles: 3 half circles are 6 quarter circles, 6 + 6 = 12 quarter circles, and 4 quarter circles make a circle.

Q2 (a) 1 by 1: 16, 2 by 2: 9, 3 by 3: 4, 4 by 4: 1; 30 squares (b) 6 on the bottom line + 6 on the line across = 12 triangles

Q3 (a) A 5, B 8, C 14 (9 + 4 + 1) (b) 13 more cubes (27 − 14)

Q4 (a) A and C fold into a cube; B and D do not (children cut out and fold to check). Nets E and F are Think harder 3: E folds, F does not (b) the new square goes under any one of the 4 squares in the row (4 possible places)

Investigation (Squares joined edge to edge). (a) 2 shapes: a straight line and an L. (b) 5 shapes: a straight line of 4, a 2 x 2 square, an L, a T and a zigzag (S shape). Brute-forced: 1, 1, 2, 5 shapes for 1, 2, 3, 4 squares (turns and flips count as the same). If a child counts flips as different, they find 7 (L and its

mirror, S and its mirror): praise this and explain the rule. (c) Look for a system: start with the straight line, then move one square, and check each new shape against the others by turning the tile.

Recording chapters and what a child watching later does

Time	Chapter marker	A child watching later
0–1.5 min	Opening	Gather your workbook, pencil, paper and the materials on the slide. Press play when ready.
1.5–6 min	Sneak-peek puzzles (4.5 minutes)	Pause. Try the puzzles on your own paper (about 4 and a half minutes). Guessing is fine. Press play for the answers.
6–10.5 min	Warm-up	Play along. Say or write each answer before the video does. Pause if you need longer.
10.5–23.5 min	Explore (in pairs, with objects)	Pause. Try the explore task with your objects, with someone at home if you can (about 13 minutes). Write down what you notice.
23.5–35.5 min	Mini-lesson (you teach, they join in)	Keep watching. Copy the worked examples. Rewind any step you missed.
35.5–38.5 min	Brain break	Stand up and join the brain break.
38.5–73.5 min	Worksheet (pairs, then alone, then stretch)	Pause. Do the workbook, starting with Start here on page 1 (about 35 minutes). Press play for each checking part and mark in a different colour.
73.5–76.5 min	Flex (spare time)	Pause or rewind whenever you need to.
76.5–81.5 min	Talk time	Pause. Explain your answer to someone at home, with a reason. Then play.
81.5–86.5 min	Exit ticket	Pause. Do the exit ticket on your own, then play to check it.
86.5–88.5 min	Looking back (2 minutes)	Pause. Fill in the looking-back page.
88.5–90 min	Closing and stamps (1.5 minutes)	Show your journal and workbook to a grown-up. Tell them one thing you found out.

Lesson 9: Measure Masters

Week 5 (first lesson) • Length, Mass, Volume and Time • Workbook: Lesson 9

Objective	Solve balance-scale puzzles in g and kg by swapping things that balance; solve pouring puzzles with litre jugs by keeping a table of every pour; join and cut lengths in m and cm (overlaps, pieces and cuts); tell the time to 5 minutes with am and pm and find start times, end times and how long, also across the hour.
Thinking skills	Reasoning with equal amounts (swapping), working step by step, keeping a table, counting on in time, checking
Materials	A coat hanger, two paper cups, string, a glue stick, a spoon and about 20 20¢ coins (a home-made balance; backup: a ruler balanced on a pencil); for the jug puzzles two cups or bottles of different size (about 3 and 5 units), a jug of dry rice or sand (or water), and tape marking 3 and 5 units; a cardboard clock with hands; mini whiteboards.
Online set-up	Children build the coat-hanger balance, hang it and test that it is level BEFORE the lesson (kit message, then a 30-second check at minute 0) and hold it up to the camera. No coat-hanger balance at home? The tutor balances a ruler on a pencil on camera and pairs share one view. The tutor shares the slide balances, jugs and clocks and annotates on screen. Each child needs a cardboard or paper-plate clock. For the pouring puzzles use dry rice or sand rather than water when online, and let children act out each pour with the two containers.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.
Watching the recording	A child who missed the live lesson watches the recording. They need: A coat hanger, two paper cups, string, a glue stick, a spoon and about 20 20¢ coins (a home-made balance; backup: a ruler balanced on a pencil); for the jug puzzles two cups or bottles of different size (about 3 and 5 units), a jug of dry rice or sand (or water), and tape marking 3 and 5 units; a cardboard clock with hands; mini whiteboards. The REC lines tell them when to pause; the chapter list at the end of this lesson shows each part.

0–1.5 min Opening

REC *Hello! Before you press play, get a pencil, paper, your workbook and the materials on the slide. Pause or rewind any time.*

SAY *"Good morning, Measure Masters! Today we measure how long, how heavy, how much water and how much time."*

"Some puzzles today have no ruler, no scale with numbers and no marks on the jugs. We will use our heads: notice it, test it, explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. The kit is on the desk from minute 0: ask everyone to show you the materials for today (a 30-second check; the materials list is above and on the title slide).

1.5–6 min Sneak-peek puzzles (4.5 minutes)

REC *Pause now and try the puzzles on your own paper for about 4 and a half minutes. Guessing is fine. Press play for the answers.*

SAY *"Two sneak-peek puzzles on ideas we have not learnt yet. Guessing is welcome, so is I do not know. Nobody is marked. Show your paper to me only."*

DO Show the Sneak-peek puzzles slide at minute 1.5 (4.5 minutes: about 45 seconds for question 1, 90 seconds for question 2, then the answers). Say the short line out loud (about 28 words): two puzzles on ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked; show your paper to me only. Read each puzzle aloud; question 1 is the one to start with (objects, a drawing or counting).

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): You have a 3 ℓ jug and a 5 ℓ jug. You fill the 5 ℓ jug and pour from it into the 3 ℓ jug until the 3 ℓ jug is full. How much water is left in the 5 ℓ jug? (Try it with two cups and rice, or draw it.)	2 ℓ (5 - 3 = 2). Listen for: One pour is enough. This is a first win for everyone and previews the pour table.
Question 2 (thinking): A melon balances 2 apples. One apple balances 3 plums. How many plums balance the melon? How do you know?	6 plums (3 + 3). Listen for: Listen for swapping: the melon can be swapped for 2 apples, and each apple for 3 plums. Drawing the scale is a good method.

DO The neutral line "After question 3 you may choose: carry on, or try Think harder." is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

6–10.5 min Warm-up

REC *Join in! Say or write each answer before I do. Pause the video if you need longer.*

DO Which unit? Show three things one at a time (a door, a bag of rice, brushing teeth; spare: a pencil, a strawberry, a bucket of water). Children write m, cm, kg, g, ℓ or minutes on whiteboards and explain.

SAY *"I will show you something. Write the unit you would use to measure it on your whiteboard: m, cm, kg, g, ℓ or minutes. Show me on three."*

"Be ready to say why. Why not grams for a bag of rice?"

DO Keep the game to about two and a half minutes (6 to 8.5): stop it on the clock. Then the quick questions (8.5 to 10.5): ask questions 1 and 3 only (these are the two on the slide); questions 2, 4 and 5 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect
35 cm + 40 cm = ?	75 cm
The minute hand points to 6. How many minutes past the hour?	30 minutes
A 5 ℓ bucket has 2 ℓ of water. How much more fills it?	3 ℓ
Lunch at 12:30. am or pm?	pm
One bag of rice is 5 kg. How heavy are 2 bags?	10 kg

10.5–23.5 min Explore (in pairs, with objects)

REC *Pause now. Try this with your objects, and a grown-up if you can, for about 13 minutes. Press play when you have an answer.*

DO Timing (13 minutes): set up and predict, 10.5 to 12. One round of breakout rooms with an auto-timer: 60 seconds to get in, 5.5 minutes of pair work (13 to 18.5), 60 seconds to come back. Then the group debrief, 19.5 to 23.5.

SAY *"Look at your balance. A glue stick goes in one cup. How many 20¢ coins will balance it? Write your guess before you test."*

DO 1. Your two cups are already hanging from the coat hanger (done before the lesson). Check it is level when the cups are empty.

2. Put a glue stick in one cup. Guess: how many 20¢ coins will balance it?

3. Add coins one at a time until the hanger is level.

4. Write your number. How do you know it is level?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: check the hanger is level with both cups empty before you start.

Coins keep sliding: put them in the cup one at a time and stop when the hanger is level.

Finished early: find something that balances exactly the same number of coins as your glue stick.

DO Go further (a second what-if, so nobody waits): it is the box at the foot of the Explore slide; post it in the chat as well when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: one glue stick balances 8 coins. A pencil case balances 3 glue sticks.

How many coins balance the pencil case? If an eraser balances 4 coins, how many erasers balance the pencil case? Answer: 24 coins (3 x 8). 6 erasers (24 / 4).

DO At about minute 18.5, say “Pencils down” and bring the group back together (60 seconds). Take the three questions below, one at a time (about 4 minutes in all). Ask “How do you know?” on question 2 only; take about 15 seconds of talk on the others.

Ask	Expect
How many coins balanced your glue stick?	Open answers. The count depends on the coins and the objects.
The glue stick balances 8 coins and the spoon balances 5 coins. Which is heavier?	The glue stick: it needs more of the same coin.
Your partner used erasers and got a different number. Why?	Erasers and coins are different units. To compare, use the same unit.

23.5–34 min Mini-lesson (you teach, they join in)

REC *Keep watching and copy the examples onto your paper. Missed a step? Rewind. You will try some yourself soon.*

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *“A watermelon balances 3 pineapples. One pineapple balances a 2 kg weight. The scale is level, so both sides are equally heavy. I can swap each pineapple for 2 kg: 2, 4, 6. The watermelon is 6 kg.”*

“Now the watermelon balances one pineapple and 4 bags of sugar. The watermelon is 6 kg and the pineapple is 2 kg, so the bags must be the rest: 6 − 2 = 4 kg. 4 bags make 4 kg, so each bag is 1 kg. I check: 2 + 1 + 1 + 1 + 1 = 6 ✓.”

“A jug puzzle: a 4 ℓ jug and a 9 ℓ jug, and I want exactly 1 ℓ. I fill the 9 ℓ jug. I pour into the 4 ℓ jug until it is full: 9 − 4 = 5 left. I empty the 4 ℓ jug and pour again: 5 − 4 = 1. I write every pour in a table so I can check it.”

Worked example on the slides	Answer
Balance puzzle: swap it. A watermelon balances 3 pineapples. One pineapple balances a 2 kg weight. The watermelon also balances 1 pineapple and 4 bags of sugar. How heavy is the watermelon? One bag of sugar?	Watermelon 6 kg (2 + 2 + 2). The 4 bags are 6 − 2 = 4 kg, so each bag is 1 kg. Check: 2 + 1 + 1 + 1 + 1 = 6.
Jug puzzle: keep a table. You have a 4 ℓ jug and a 9 ℓ jug with no marks. You can fill a jug, empty it, or pour from one into the other. How can you get exactly 1 ℓ?	Fill the 9 ℓ jug (0, 9). Pour into the 4 ℓ jug (4, 5). Empty the 4 ℓ jug (0, 5). Pour again (4, 1). 1 ℓ is left in the 9 ℓ jug.

SAY *"The key idea today is: same on both sides. A level balance means both sides are equally heavy, so you can swap. For jugs, write every pour in a table. Say it with me."*

DO Show the key idea slide at about minutes 33 to 34. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Balance puzzles: adding the masses on the two sides instead of using "both sides are the same".
- Jug puzzles: writing "pour" without saying how much is in each jug after the pour, so the steps cannot be checked.
- Time: counting on past the hour as if there were 100 minutes in an hour (4:50 + 25 minutes written as 4:75 or 5:75).

34–37 min Brain break

REC *Movement break! Stand up and join in. The video keeps going.*

DO Be a balance! Stand on one leg with your arms out. The tutor calls "heavy left!" or "heavy right!" and you tip, then come back level.

Water. Toilet break for anyone who needs it. Keep it to three minutes: two of movement, one for water.

37–72 min Worksheet (pairs, then alone, then stretch)

REC *Pause and do your workbook, starting at Start here on page 1, for about 35 minutes. Press play to check your work.*

DO Minute 37: give out the workbook. Read the title of Lesson 9 aloud. Say "Pairs first. Talk before you write."

Minutes 37 to 51: pairs work together on the first page (the 2 minutes at the start are for finding the page and setting up; pairs stay in the main room). Minutes 51 to 53.5: check page 1 together (read only the answer line, children mark their own, one "How do you know?" for the page). Minutes 53.5 to 58: short teaching for page 2. Minutes 58 to 69.5: each child works alone on the second page. Minutes 69.5 to 72: check page 2 together (answer lines only, children mark their own). Think harder, Go further and the Investigation are open to any child at any time after question 3.

DO Take-home stars: Page 1: Q2(b) (get exactly 1 ℓ). Page 2: Q4(b) (the ribbon with a 5 cm overlap) and Q5(c) (how long is the trip). Also at home: try the spoon on the balance. The ribbon teaching slide and the time try-it slide are not shown in class. Each is marked on the page with a ★ and the words take-home, stays Core (not Think harder), and keeps its answer on the check-together slide so a child can mark it at home. Do not spend class time on them. Late rule, printed here on purpose: at minute 58 (the start of page 2), if the clock is more than 2 minutes late, drop the last Core question of page 2 to take-home (children put a ★ beside it). Never shorten talk time, the exit ticket, looking back or the closing.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: “The scale is level. What does that tell you?”
- Ask: “What can you swap the melon for?”
- Ask: “How much is in each jug now? Show me in your table.”

IF STUCK Hint 1: “Find the thing you know first. Which scale has a number on it?”

Hint 2: “Swap: put what balances it on the scale instead.”

Hint 3: “For jugs, write the two amounts after every pour.”

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: 2 apples and 1 lemon balance 1 pomelo. 1 pomelo balances 5 lemons. How many lemons balance 1 apple? How many lemons balance 1 pomelo and 1 apple together?	1 apple balances 2 lemons. Swap the pomelo for 5 lemons: 2 apples and 1 lemon balance 5 lemons. Take 1 lemon off both sides: 2 apples balance 4 lemons, so 1 apple balances 2 lemons. A pomelo and an apple balance 5 + 2 = 7 lemons.
Workbook Think harder 2: Mei’s family leaves home at 10:45 am. They reach Grandma’s flat at 12:10 pm. How long is the trip?	1 hour 25 minutes (10:45 to 11:00 is 15 minutes, 11:00 to 12:00 is 1 hour, 12:00 to 12:10 is 10 minutes).
Workbook Think harder 3: 3 apples balance 1 pomelo. The pomelo and 2 apples together balance a 500 g weight. How heavy is one apple? How heavy is the pomelo?	Apple 100 g, pomelo 300 g (5 apples balance 500 g).
Workbook Think harder 4: You have a 2 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 3 ℓ? List each pour.	Fill 7, pour into 2, empty 2, pour again: 3 ℓ is left in the 7 ℓ jug.
Workbook Go further 5: You have a 3 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 5 ℓ? List each pour.	(3 ℓ, 7 ℓ): fill 7 (0, 7); pour (3, 4); empty 3 (0, 4); pour (3, 1); empty 3 (0, 1); pour (1, 0); fill 7 (1, 7); pour until the 3 ℓ jug is full (3, 5). 5 ℓ in the 7 ℓ jug.
Workbook Go further 6: Scale 1: 2 cubes and 1 ball balance 2 balls. Scale 2: 1 ball and 1 cube balance a 300 g weight. How heavy is one cube? How heavy is one ball?	Take one ball off both sides of scale 1: 2 cubes balance 1 ball. On scale 2, swap the ball for 2 cubes: 3 cubes balance 300 g, so a cube is 100 g and a ball is 200 g. Check scale 1: 100 + 100 + 200 = 400 = 200 + 200.

Workbook Go further 7: Mei has 3 lessons in a row. Each lesson is 35 minutes long, with a 10-minute break between lessons. The first lesson starts at 8:15 am. What time does the last lesson end?	10:20 am (8:15 to 8:50, break to 9:00, 9:00 to 9:35, break to 9:45, 9:45 to 10:20)

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (on the “Before you go” page): Weights puzzle. You have three different weights. Each weight is 1 g, 2 g, 3 g, 4 g or 5 g. You put some of them on one pan to make a mass. (a) With weights of 1 g, 2 g and 4 g, which masses from 1 g to 7 g can you make? (b) Find ALL the sets of three different weights from 1 g to 5 g that can make every mass from 1 g to 6 g. (c) What do all the good sets have in common? Easy first step: with weights of 1 g, 2 g and 4 g, can you make 3 g? Can you make 5 g? Needs: paper and a pencil (cubes or coins can stand for the weights).	(a) All of them: 1, 2, 1+2 = 3, 4, 1+4 = 5, 2+4 = 6, 1+2+4 = 7. (b) Exactly 2 sets work: 1, 2, 3 (makes 1 to 6) and 1, 2, 4 (makes 1 to 7); brute-forced over all 10 sets. (c) Both contain 1 g and 2 g: only a 1 g weight can make 1 g, and with different weights only a 2 g weight can make 2 g. Look for: a systematic list of the possible sets, checking each mass in order, and the noticing that 1 g is needed.

72–76.5 min Flex (spare time)

REC *Pause or rewind whenever you need to.*

DO Spare time, kept on purpose (4.5 minutes). Use it to absorb a late start, a camera problem or a slow page 2. If you are on time, spend it on the Investigation share or a second talk-time volunteer, or an early start on talk time. Do not add new content. Never take time from the exit ticket, looking back or the stamp.

76.5–81.5 min Talk time

REC *Pause. Tell a grown-up your answer to the talk question, and why you think so. Then press play.*

DO Bring the group back to a circle (or camera on, mics on). Pairs talk for 90 seconds, then two volunteers share (about 5 minutes in all). One question only. You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *“How did swapping help you with the balance puzzles?”*

Discussion question	Listen for

A melon balances 2 apples. Can you swap the melon on a scale for 2 apples and keep it level? Why?	A level balance means the two sides weigh the same, so the melon can be swapped for 2 apples (or 400 g) and the scale stays level. (If a pair finishes early: a table shows how much is in each jug after every pour, so we can check every step and find our way back.)

81.5–86.5 min Exit ticket

REC *Pause and do the exit ticket on your own, without looking back. Then press play to check it.*

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. About 2 minutes of child work; then children hold their slips up one at a time (never all together) and the answer is on the next click.

Exit ticket	Answer
A lesson starts at 4:50 pm and lasts 25 minutes. What time does it end?	5:15 pm (4:50 to 5:00 is 10 minutes, 15 more minutes is 5:15)

86.5–88.5 min Looking back (2 minutes)

REC *Pause for about 2 minutes and fill in your looking-back page: what was hardest, and what will you try next time?*

SAY *"Looking back takes 2 minutes. Find the box called Looking back on the page "Before you go: Investigation and Looking back". Finish one line, two lines or all three, and tick what helped you. You may pass on any line. If you tick "I am still stuck", your tutor will check in during the session."*

DO Nothing is collected in front of the group and nobody has to read theirs out. The tutor may read a box while stamping journals (never by private message), and checks in on a child who ticked still stuck during the open session, for example by a quick visit to the shared breakout room or a hand signal with a parent nearby, for example "I saw you drew it when you got stuck". A child may volunteer one sentence, never by name unless they choose to. If a child ticks "I am still stuck", check in during the next open session (a quick visit to the shared breakout room or a hand signal, with a parent nearby). Praise the strategy, not the speed.

88.5–90 min Closing and stamps (1.5 minutes)

REC *Well done! Show your journal and workbook to a grown-up and tell them one thing you found out.*

SAY *"You were real Measure Masters today. You swapped things on a balance, you found exact amounts with no marks on the jugs, and you counted time across the hour."*

"Next lesson is Challenge Day: bring your best thinking!"

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Make your own jug puzzle with two jug sizes and a target amount. Check that it can be done, and list every pour.

Answer key (workbook)

Q1 (a) 200 g (b) 400 g (c) 50 g (the melon is 400 g, the apple is 200 g, so 4 lemons are 200 g) (d) 8 lemons

Q2 (a) Fill the 5 ℓ jug. Pour it into the 3 ℓ jug until the 3 ℓ jug is full. 2 ℓ is left in the 5 ℓ jug. (b) Fill the 3 ℓ jug and pour it into the 5 ℓ jug. Fill the 3 ℓ jug again and pour until the 5 ℓ jug is full (2 ℓ goes in). 1 ℓ is left in the 3 ℓ jug.

Q3 (3 ℓ jug, 5 ℓ jug) Start 0, 0. 1: fill the 5 ℓ jug: 0, 5. 2: pour into the 3 ℓ jug until it is full: 3, 2. 3: empty the 3 ℓ jug: 0, 2. 4: pour the 2 ℓ into the 3 ℓ jug: 2, 0. 5: fill the 5 ℓ jug: 2, 5. 6: pour into the 3 ℓ jug until it is full (1 ℓ fits): 3, 4. 4 ℓ ✓ (6 steps is the shortest way; checked by search.)

Q4 (a) 46 cm (b) 105 cm (64 + 46 = 110, take away the 5 cm overlap)

Q5 (a) A 7:25 am B 4:50 pm (b) 5:25 pm (c) 45 minutes (10:45 to 11:00 is 15 minutes, 30 more)

Investigation (Weights puzzle). (a) All of them: 1, 2, 1+2 = 3, 4, 1+4 = 5, 2+4 = 6, 1+2+4 = 7. (b) Exactly 2 sets work: 1, 2, 3 (makes 1 to 6) and 1, 2, 4 (makes 1 to 7); brute-forced over all 10 sets. (c) Both contain 1 g and 2 g: only a 1 g weight can make 1 g, and with different weights only a 2 g weight can make 2 g. Look for: a systematic list of the possible sets, checking each mass in order, and the noticing that 1 g is needed.

Recording chapters and what a child watching later does

Time	Chapter marker	A child watching later
0–1.5 min	Opening	Gather your workbook, pencil, paper and the materials on the slide. Press play when ready.
1.5–6 min	Sneak-peek puzzles (4.5 minutes)	Pause. Try the puzzles on your own paper (about 4 and a half minutes). Guessing is fine. Press play for the answers.
6–10.5 min	Warm-up	Play along. Say or write each answer before the video does. Pause if you need longer.
10.5–23.5 min	Explore (in pairs, with objects)	Pause. Try the explore task with your objects, with someone at home if you can (about 13 minutes). Write down what you notice.
23.5–34 min	Mini-lesson (you teach, they join in)	Keep watching. Copy the worked examples. Rewind any step you missed.
34–37 min	Brain break	Stand up and join the brain break.
37–72 min	Worksheet (pairs, then alone, then stretch)	Pause. Do the workbook, starting with Start here on page 1 (about 35 minutes). Press play for each checking part and mark in a different colour.

72–76.5 min	Flex (spare time)	Pause or rewind whenever you need to.
76.5–81.5 min	Talk time	Pause. Explain your answer to someone at home, with a reason. Then play.
81.5–86.5 min	Exit ticket	Pause. Do the exit ticket on your own, then play to check it.
86.5–88.5 min	Looking back (2 minutes)	Pause. Fill in the looking-back page.
88.5–90 min	Closing and stamps (1.5 minutes)	Show your journal and workbook to a grown-up. Tell them one thing you found out.

Lesson 10: Challenge Day

Week 5 (second lesson) • Mixed Puzzles • Workbook: Lesson 10

Objective	Choose a way to start a puzzle (draw it, list it in order, make a table) and use it on mixed puzzles: picture graphs with a key of 5 or 10, a logic grid with exactly one answer, numbers to 1000, money, fractions, and × and ÷ with the 2, 3, 4, 5 and 10 tables. Check the answer and explain it.
Thinking skills	Choosing a strategy, drawing models, listing in order, making tables, logical elimination, checking, explaining
Materials	Four station cards (all on one slide in the deck, and printed in the tutor guide), play coins, digit cards 0 to 9, paper strips for fractions, mini whiteboards, certificates, stamps.
Online set-up	Each breakout pair takes two station cards (the tutor pastes their text into the breakout room chat), about 2 minutes a card, and everyone reports back to the group. Each child needs digit cards (cut from paper), a pencil and a mini whiteboard.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.
Watching the recording	A child who missed the live lesson watches the recording. They need: Four station cards (all on one slide in the deck, and printed in the tutor guide), play coins, digit cards 0 to 9, paper strips for fractions, mini whiteboards, certificates, stamps. The REC lines tell them when to pause; the chapter list at the end of this lesson shows each part.

0–1.5 min Opening

REC *Hello! Before you press play, get a pencil, paper, your workbook and the materials on the slide. Pause or rewind any time.*

SAY *"Welcome to Challenge Day, inquirers! Today's puzzles mix everything from our ten lessons."*

"Clever inquirers do not know every answer straight away. They know how to START: draw it, list it, or make a table."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud. The kit is on the desk from minute 0: ask everyone to show you the materials for today (a 30-second check; the materials list is above and on the title slide).

1.5–6 min Sneak-peek puzzles (4.5 minutes)

REC *Pause now and try the puzzles on your own paper for about 4 and a half minutes. Guessing is fine. Press play for the answers.*

SAY *"Two sneak-peek puzzles on ideas we have not learnt yet. Guessing is welcome, so is I do not know. Nobody is marked. Show your paper to me only."*

DO Show the Sneak-peek puzzles slide at minute 1.5 (4.5 minutes: about 45 seconds for question 1, 90 seconds for question 2, then the answers). Say the short line out loud (about 28 words): two

puzzles on ideas we have not learnt yet; guessing and "I do not know" are welcome; nobody is marked; show your paper to me only. Read each puzzle aloud; question 1 is the one to start with (objects, a drawing or counting).

Answers stay private. Each child holds their own paper to the camera for you only (look, do not read out), or writes it on their own paper and shows only you on camera. Never raise whiteboards together and never show a board in front of the group. Never ask a named child "how do you know?" Ask for volunteers: "Would anyone like to tell us how they started?"

Question 1 is an access question: every child can try it with objects, a drawing or counting, so everyone has a first win. Question 2 is the thinking question. Do not mark either one. Click to show the answers at the end as "one way to think about it", without saying who had what.

Use what you see only to plan. Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page and never name a child in front of the group. Children choose, and may move either way at any time. Pairs are not split unless both children agree. Everybody joins the explore, the mini-lesson, the check-together slides and talk time.

Sneak-peek puzzle	Answer and what to listen for
Question 1 (access: objects, drawing or counting): A box has 3 triangles and 2 squares. A triangle has 3 sides and a square has 4 sides. Draw the shapes. How many sides are there altogether?	17 sides (9 + 8). Listen for: Drawing the shapes and counting the sides is a good start. This is a first win for everyone.
Question 2 (thinking): A box has 9 shapes: triangles (3 sides) and squares (4 sides). There are 31 sides altogether. How many squares? Say how you started: draw it, list it or make a table.	4 squares and 5 triangles (16 + 15 = 31 sides). Listen for: Any start is welcome: drawing 9 shapes, listing "1 square, 2 squares ..." or a table. A guess that is checked counts.

DO The neutral line “After question 3 you may choose: carry on, or try Think harder.” is printed on page 1 of the workbook, next to the lesson title. Nobody is labelled and nothing is announced.

6–10.5 min Warm-up

REC *Join in! Say or write each answer before I do. Pause the video if you need longer.*

DO Relay review. Three cards: digit cards, change and fractions (spare: missing digits, legs, patterns). A volunteer answers, then chooses a friend who has not had a turn (pass is always fine).

SAY *“Relay time! I show a card and ask for a volunteer. That person answers, then chooses a friend who has not had a turn. You can always say pass. A different answer gets “let us test that”, never a groan.”*

DO Keep the game to about two and a half minutes (6 to 8.5): stop it on the clock. Then the quick questions (8.5 to 10.5): ask questions 1 and 3 only (these are the two on the slide); questions 2, 4 and 5 are spare. Children answer by thumbs, a whisper to a partner. Do not name a child.

Ask	Expect

Share 40 stickers equally among 5 children. How many each?	8
It is 10:55 am. What time is it 10 minutes later?	11:05 am
Which is bigger: 1/3 or 1/5 of the same cake?	1/3
Half of a circle is called a …	Half circle
I think of a number and add 30. I get 100. What is my number?	70

10.5–23.5 min Explore (in pairs, with objects)

REC *Pause now. Try this with your objects, and a grown-up if you can, for about 13 minutes. Press play when you have an answer.*

DO Timing (13 minutes): set up and predict, 10.5 to 12. One round of breakout rooms with an auto-timer: 60 seconds to get in, 4.5 minutes of pair work (13 to 17.5), 60 seconds to come back. Then the group debrief, 18.5 to 23.5.

SAY *"Look at the four station cards. Which one will be hardest? Which way will you start it: draw, list or table? Write your guess."*

DO 1. Your pair takes two station cards (for example 1 and 3, or 2 and 4).

2. Read the card aloud. Before you solve it, write how you will start: draw, list or table.

3. Solve it and check it (about 2 minutes a card). Then take the other card.

4. Be ready to explain how you started and how you know.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: "Can you draw it? Can you list the ways in order?"

Graph card: "Look at the key first. What is one picture worth?"

Finished early: "Is there another answer? How do you know there is not?"

DO Station cards (all four are on ONE slide in the deck). Each pair takes two cards, about 2 minutes each (4.5 minutes of pair work in all):

Station card	Answer
Card 1: picture graph. Each picture stands for 5 books. Monday has 3 pictures, Tuesday 5 pictures, Wednesday 2 pictures. How many books were read on these 3 days?	50 books (10 pictures × 5)
Card 2: logic grid. Ali, Ben and Cai each have one pet: a cat, a dog or a fish. Ben has the fish. Ali does not have the dog. Who has the cat?	Ali has the cat (Ben has the fish, so Ali and Cai share the cat and the dog; Ali does not have the dog, so Ali has the cat and Cai has the dog)

Card 3: coin puzzle. Make 70¢ with exactly 4 coins. You can use 5¢, 10¢, 20¢ and 50¢ coins. Find all the ways.	2 ways: 50¢ + 10¢ + 5¢ + 5¢, and 20¢ + 20¢ + 20¢ + 10¢
Card 4: fraction strip. A strip is cut into 8 equal parts. 3 parts are red and 2 parts are blue. What fraction is NOT coloured?	3/8

DO Go further (a second what-if, so nobody waits): it is the box at the foot of the Explore slide; post it in the chat as well when pairs go to their rooms. Any child may try it any time, as a choice. Never name who does. Question: what if the coin puzzle is 70¢ with exactly 5 coins (using 5¢, 10¢, 20¢ and 50¢ coins)? Can it be done? How? Answer: Yes: 50¢ + 5¢ + 5¢ + 5¢ + 5¢; 20¢ + 20¢ + 20¢ + 5¢ + 5¢; 20¢ + 20¢ + 10¢ + 10¢ + 10¢.

DO At about minute 17.5, say "Pencils down" and bring the group back together (60 seconds). Take the three questions below, one at a time (about 5 minutes in all). Ask "How do you know?" on question 2 only; take about 15 seconds of talk on the others.

Ask	Expect
How did you start each card?	Open answers. Draw it, list it, or make a table.
Which card was hardest? Why?	Open answers.
What helped when you were stuck?	Drawing it, going in order, trying small numbers first, checking with the clues.

23.5–33.5 min Mini-lesson (you teach, they join in)

REC *Keep watching and copy the examples onto your paper. Missed a step? Rewind. You will try some yourself soon.*

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Mei had some stickers. She gave half to Ben. Then she gave 6 to Siti and had 4 left. I cannot hold that in my head, so I draw a bar cut into two halves. One half went to Ben. The other half is the 6 for Siti and the 4 left: 6 + 4 = 10. Both halves are the same, so Mei had 10 + 10 = 20 stickers. Check: half of 20 is 10, 10 − 6 = 4 ✓."*

"Digit cards 2, 0 and 5. How many 3-digit numbers? I list them in order, smallest hundreds digit first. 0 cannot go first. 2: 205, 250. 5: 502, 520. Four numbers, and I know I have them all because I went in order."

"Chickens and goats: 7 heads and 20 legs. I make a table. 0 goats: 14 legs. 1 goat: 16. 2 goats: 18. 3 goats: 20 ✓. So 3 goats and 4 chickens. Each goat I swap in adds 2 legs."

Worked example on the slides	Answer

Draw it: a before-and-after story. Mei had some stickers. She gave half of them to Ben. Then she gave 6 to Siti. Now she has 4 left. How many stickers did she have at first?	20. The half she kept is 6 + 4 = 10, so she had 10 + 10 = 20. Check: 20 → 10 → 4.
Make a table: heads and legs. A farm has chickens and goats. There are 7 heads and 20 legs. How many goats are there?	3 goats and 4 chickens. Table: 0 goats 14 legs, 1 goat 16, 2 goats 18, 3 goats 20 ✓. Check: 12 + 8 = 20.

SAY *"The key idea today is: how to start a puzzle. Stuck? Draw it, list it in order, or make a table. Say it with me."*

DO Show the key idea slide at about minutes 32.5 to 33.5. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Picture graph: counting the pictures instead of using the key (writing 6 instead of 30 for Tuesday).
- Listing: putting 0 first (038 is not a 3-digit number) or missing a number because the list is not in order.
- Logic grid: putting a tick without crossing out the rest of that row and column, so the next clue cannot be used.

33.5–36.5 min Brain break

REC *Movement break! Stand up and join in. The video keeps going.*

DO Freeze with maths. Dance until the music stops, then make the shape the tutor calls with your arms: a half circle, a quarter circle, a cone, a cuboid.

Water. Toilet break for anyone who needs it. Keep it to three minutes: two of movement, one for water.

36.5–71 min Worksheet (pairs, then alone, then stretch)

REC *Pause and do your workbook, starting at Start here on page 1, for about 34 and a half minutes. Press play to check your work.*

DO Minute 36.5: give out the workbook. Read the title of Lesson 10 aloud. Say "Pairs first. Talk before you write."

Minutes 36.5 to 39: a short teach slide first (this lesson). Minutes 39 to 51: pairs work together on the first page (the 2 minutes at the start are for finding the page and setting up; pairs stay in the main room). Minutes 51 to 53.5: check page 1 together (read only the answer line, children mark their own, one "How do you know?" for the page). Minutes 53.5 to 57: short teaching for page 2. Minutes 57 to 68.5: each child works alone on the second page. Minutes 68.5 to 71: check page 2 together (answer lines only, children mark their own). Think harder, Go further and the Investigation are open to any child at any time after question 3.

DO Take-home stars: Page 1: the whole of Q3 (the coins table, 6 coins worth $1.80). The how-to-start teaching slide is not shown separately: the key idea slide says the same. Each is marked on the page with a ★ and the words take-home, stays Core (not Think harder), and keeps its answer on the check-together slide so a child can mark it at home. Do not spend class time on them. Late rule, printed here on purpose: at minute 57 (the start of page 2), if the clock is more than 2 minutes

late, drop the last Core question of page 2 to take-home (children put a ★ beside it). Never shorten talk time, the exit ticket, looking back or the closing.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "How did you start this one? Draw, list or table?"
- Ask: "How do you know there is only one answer?"
- Ask: "Show me your check."

IF STUCK Hint 1: "Which lesson does this remind you of?"

Hint 2: "Draw it, list it in order, or make a table."

Hint 3: "Try a small number first, then change it one step at a time."

DO A choice, not a track: after question 3 any child may carry on or try Think harder (the neutral line is printed on page 1). Invite the WHOLE group at once, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group, and do not split a pair unless both children agree. Children may move either way at any time. The Investigation is printed on the last core page, so every child sees it.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: 30 children chose a fruit. In the picture graph each picture stands for 5 children. Apple has 2 pictures and banana has 3 pictures. The mango row is missing. How many pictures should the mango row have?	1 picture (30 − 10 − 15 = 5 children)
Workbook Think harder 2: Mei saves $3 on day 1, $6 on day 2, $9 on day 3, and so on ($3 more each day). Ali saves $10 every day. Make a table of how much each child has saved altogether after each day. On which day does Mei FIRST have more than Ali?	Day 6. Mei's totals: $3, $9, $18, $30, $45, $63. Ali's totals: $10, $20, $30, $40, $50, $60. On day 6 Mei has $63 and Ali has $60. (On day 5 Mei still has less: $45 and $50.)
Workbook Think harder 3: A bus had some people on it. At the first stop half of the people got off. At the second stop 6 people got on. Now there are 14 people on the bus. How many people were on the bus at the start? Draw a bar to start.	16 (14 − 6 = 8 is half, 8 + 8 = 16).
Workbook Think harder 4: 5, 9, 13, 17, ... Will 49 be in this pattern? Will 50? How do you know?	49 will be in the pattern and 50 will not. The pattern adds 4 each time, so every number in it is 1 more than a number in the 4 times table: 5 = 4 + 1, 9 = 8 + 1, 13 = 12 + 1. 49 = 48 + 1, but 50 is 2 more than 48.

Workbook Go further 5: A picture graph has lost its key. Football has 6 pictures and swimming has 4 pictures. 10 more children chose football than swimming. How many children chose swimming? Football?	Swimming 20, football 30 (2 pictures stand for 10 children, so 1 picture stands for 5)
Workbook Go further 6: Ali, Ben, Mei and Siti each have one coin: 10¢, 20¢, 50¢ or $1. Mei has more money than Ali but less than Ben. Ben does not have the $1 coin. Who has which coin?	Ali 10¢, Mei 20¢, Ben 50¢, Siti $1. Ben is not $1 and two people have less than him, so Ben has 50¢; then Ali 10¢ and Mei 20¢, and Siti has the $1.
Workbook Go further 7: At a hawker centre, chicken rice costs $4 a plate and noodles cost $3 a bowl. Ravi's family buys 7 dishes for $25. How many plates of chicken rice do they buy? What is the smallest total 7 dishes can cost? The biggest? Can 7 dishes cost $30? Can they cost EVERY whole-dollar amount in between?	4 plates of chicken rice and 3 bowls of noodles ($16 + $9 = $25). Table: 0 plates → $21, 1 → $22, 2 → $23, 3 → $24, 4 → $25 ✓. Smallest $21 (all noodles), biggest $28 (all chicken rice), so $30 is impossible. Each swap of a bowl for a plate adds exactly $1, so every whole dollar from $21 to $28 is possible.

These are the questions on the two Think harder pages in the workbook (Think harder, then Go further). More are in the separate Question Bank.

Investigation (several answers are right)	Model answer and what to look for
Workbook Investigation (on the "Before you go" page): Staircase. Mei climbs a staircase. She can take 1 step or 2 steps at a time. For 3 steps there are 3 ways: 1+1+1, 1+2 and 2+1. (a) Find ALL the ways for 4 steps. (b) Find ALL the ways for 5 steps. (c) Fill in a table for 1, 2, 3, 4 and 5 steps. What pattern do you see? Can you guess the number of ways for 6 steps? Easy first step: draw 3 steps and show the 3 ways: 1+1+1, 1+2 and 2+1. Needs: paper and a pencil.	Number of ways: 1 step: 1, 2 steps: 2, 3 steps: 3, 4 steps: 5, 5 steps: 8 (brute-forced; 6 steps: 13). 4 steps: 1+1+1+1, 1+1+2, 1+2+1, 2+1+1, 2+2. Pattern: each number is the sum of the two before it (3 + 5 = 8, 5 + 8 = 13). Look for: a list in order (all 1s first, then swap in a 2), a table, and the pattern spotted from the table. Do not name Fibonacci numbers unless a child asks.

71–75 min Flex (spare time)

REC *Pause or rewind whenever you need to.*

DO Spare time, kept on purpose (4 minutes). Use it to absorb a late start, a camera problem or a slow page 2. If you are on time, spend it on the Investigation share or a second talk-time volunteer, or an early start on talk time. Do not add new content. Never take time from the exit ticket, looking back or the stamp.

75–80 min Talk time

REC *Pause. Tell a grown-up your answer to the talk question, and why you think so. Then press play.*

DO Bring the group back to a circle (or camera on, mics on). Pairs talk for 90 seconds, then two volunteers share (about 5 minutes in all). One question only. You choose volunteers first: "Who would like to share?" If nobody offers, ask the pairs to share what their partner said. Never cold-call a child by name. Any child may say "pass" at any time and that is always fine. Do not keep public tallies, scores or rankings (no counting who got most right, no "first to finish"). Invite the whole group to think about whether they would like a turn, never one child.

SAY *"Which puzzle was hardest today? How did you start it?"*

Discussion question	Listen for
Pick one puzzle and teach it to the group in two sentences: how you started it, and how you know your answer is right.	Open answers. Listen for a clear start (I drew bars, I listed in order, I made a table) and a check (I put my answer back into the puzzle).

80–85 min Exit ticket

REC *Pause and do the exit ticket on your own, without looking back. Then press play to check it.*

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide. About 2 minutes of child work; then children hold their slips up one at a time (never all together) and the answer is on the next click.

Exit ticket	Answer
5 bicycles and tricycles have 13 wheels altogether. How many tricycles are there? How did you start?	3 tricycles and 2 bicycles (9 + 4 = 13). A table of 0, 1, 2, 3 tricycles works well.

85–87 min Looking back (2 minutes)

REC *Pause for about 2 minutes and fill in your looking-back page: what was hardest, and what will you try next time?*

SAY *"Looking back takes 2 minutes. Find the box called Looking back on the page "Before you go: Investigation and Looking back". Finish one line, two lines or all three, and tick what helped you. You may pass on any line. If you tick "I am still stuck", your tutor will check in during the session."*

DO Nothing is collected in front of the group and nobody has to read theirs out. The tutor may read a box while stamping journals (never by private message), and checks in on a child who ticked still stuck during the open session, for example by a quick visit to the shared breakout room or a hand signal with a parent nearby, for example "I saw you drew it when you got stuck". A child may volunteer one sentence, never by name unless they choose to. If a child ticks "I am still stuck", check in during the next open session (a quick visit to the shared breakout room or a hand signal, with a parent nearby). Praise the strategy, not the speed.

87–90 min Certificates and closing

REC *Well done! Show your journal and workbook to a grown-up and tell them one thing you found out.*

SAY *"You are real inquirers now: you notice it, test it and explain why. And when you are stuck, you know how to start."*

"Give each child a certificate and say one specific thing they did well in the program."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (this is for you only, never shared with the children). If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Think harder and Go further

Write a challenge puzzle for next year's group, with the answer and a hint that says how to start it. Check it has exactly one answer.

Answer key (workbook)

Q1 (a) 30 (b) Thursday sold 25 ice-creams: draw 5 pictures (90 − 20 − 30 − 15 = 25) (c) $30 (15 ice-creams at $2 each)

Q2 (a) 308, 380, 803, 830 (b) Greatest 830, smallest 308 (c) 308, 380, 803

Q3 Two 50¢ coins and four 20¢ coins (100¢ + 80¢ = 180¢ = $1.80). Table: 0 → $1.20, 1 → $1.50, 2 → $1.80 ✓

Q4 Ali: car. Ben: MRT. Mei: bus. Siti: walk.

Q5 (a) 5/10 (b) 2/10 (c) Mei, Ali, Ben

Q6 (a) 16, 20, 24 cubes in Towers 4, 5, 6; Tower 8 has 32 cubes (each tower adds 4: 4, 8, 12, 16, 20, 24, 28, 32) (b) Tower 9 (36 cubes: 4 × 9, or keep adding 4: 32, 36)

Investigation (Staircase). Number of ways: 1 step: 1, 2 steps: 2, 3 steps: 3, 4 steps: 5, 5 steps: 8 (brute-forced; 6 steps: 13). 4 steps: 1+1+1+1, 1+1+2, 1+2+1, 2+1+1, 2+2. Pattern: each number is the sum of the two before it (3 + 5 = 8, 5 + 8 = 13). Look for: a list in order (all 1s first, then swap in a 2), a table, and the pattern spotted from the table. Do not name Fibonacci numbers unless a child asks.

Recording chapters and what a child watching later does

Time	Chapter marker	A child watching later
0–1.5 min	Opening	Gather your workbook, pencil, paper and the materials on the slide. Press play when ready.
1.5–6 min	Sneak-peek puzzles (4.5 minutes)	Pause. Try the puzzles on your own paper (about 4 and a half minutes). Guessing is fine. Press play for the answers.
6–10.5 min	Warm-up	Play along. Say or write each answer before the video does. Pause if you need longer.
10.5–23.5 min	Explore (in pairs, with objects)	Pause. Try the explore task with your objects, with someone at home if you can (about 13 minutes). Write down what you notice.
23.5–33.5 min	Mini-lesson (you teach, they join in)	Keep watching. Copy the worked examples. Rewind any step you missed.

33.5–36.5 min	Brain break	Stand up and join the brain break.
36.5–71 min	Worksheet (pairs, then alone, then stretch)	Pause. Do the workbook, starting with Start here on page 1 (about 34 and a half minutes). Press play for each checking part and mark in a different colour.
71–75 min	Flex (spare time)	Pause or rewind whenever you need to.
75–80 min	Talk time	Pause. Explain your answer to someone at home, with a reason. Then play.
80–85 min	Exit ticket	Pause. Do the exit ticket on your own, then play to check it.
85–87 min	Looking back (2 minutes)	Pause. Fill in the looking-back page.
87–90 min	Certificates and closing	Show your journal and workbook to a grown-up. Tell them one thing you found out.

Progress tracker

After each exit ticket, write S (secure: correct and can explain), D (developing: correct but cannot explain yet) or H (needs help). Use the notes for what to revisit next lesson.

Student	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	Notes

Ending the program

- In Lesson 10, give each child a certificate and their completed inquiry journal.
- Send parents a short message through the approved channel: one thing their child did well, and one thing to keep practising.
- Delete or hand over all Student Data within 7 days after the last lesson (this is in the tutor agreement).

Tutor checklist

Before each lesson

- Read the lesson plan and solve the worksheet yourself without looking at the key.
- Check materials are ready, or the online set-up has been sent.
- Test the platform, screen share and annotation tools 10 minutes early (online).
- Print or open the slide deck for the lesson.

During the lesson

- Circulate and ask questions rather than giving answers.
- Use only the approved channels to contact parents. No private messages to children. Invite the whole group, never one child. If a child ticks “I am still stuck”, check in during the open session (a quick visit to the shared breakout room or a hand signal, with a parent nearby). Never use personal accounts.
- Do not take your own photos or recordings. The only recording is the lesson recording made on the approved platform under “Recording the lesson”.
- If you have a concern about a child’s safety or welfare, tell the Company straight away.

After the lesson

- Stamp inquiry journals and fill in the tracker.
- Collect worksheets and answer keys (face to face) and keep them secure.
- Note any child who needs help, and plan a short catch-up at the start of the next lesson.

Math Inquirers Holiday Program

Notice it. Test it. Explain why.

A 10-lesson holiday maths program for children finishing Primary 2

Your child will spend the holidays as a young mathematician: noticing patterns, testing ideas with real objects and explaining their thinking. Lessons are live online in small groups, hands-on, and built around thinking, not speed.

What your child will do

- Lesson 1, Digit Detectives: numbers to 1000
- Lesson 2, Missing Digit Mysteries: addition and subtraction within 1000
- Lesson 3, Legs, Wheels and Arrays: multiplication and division
- Lesson 4, Pattern Machines: patterns with multiplication
- Lesson 5, Mystery Number Machines: working backwards
- Lesson 6, Market Day: money
- Lesson 7, Fair Shares: fractions
- Lesson 8, Shape Builders: 2d and 3d shapes
- Lesson 9, Measure Masters: length, mass, volume and time
- Lesson 10, Challenge Day: mixed puzzles

How you can help

- Ask “How do you know?” when your child tells you an answer.
- Let your child try first. Please sit nearby, but let them find the answer.
- Print the workbook and cut-outs, and keep counters, scissors and a pencil ready.

Practical details

Dates	[start date] to [end date], two lessons a week
Time	[day] and [day], [start time] to [end time]
Where	Live online video class (link sent after registration)
Group size	[number] children
Fee	S$500 introductory (regular S$650), workbook included, payable in 3 instalments
Contact	[name, phone or email]

Math Inquirers Holiday Program is an independent enrichment program. It is inspired by enrichment-style thinking skills but is not affiliated with, run by or endorsed by the Ministry of Education or any school.